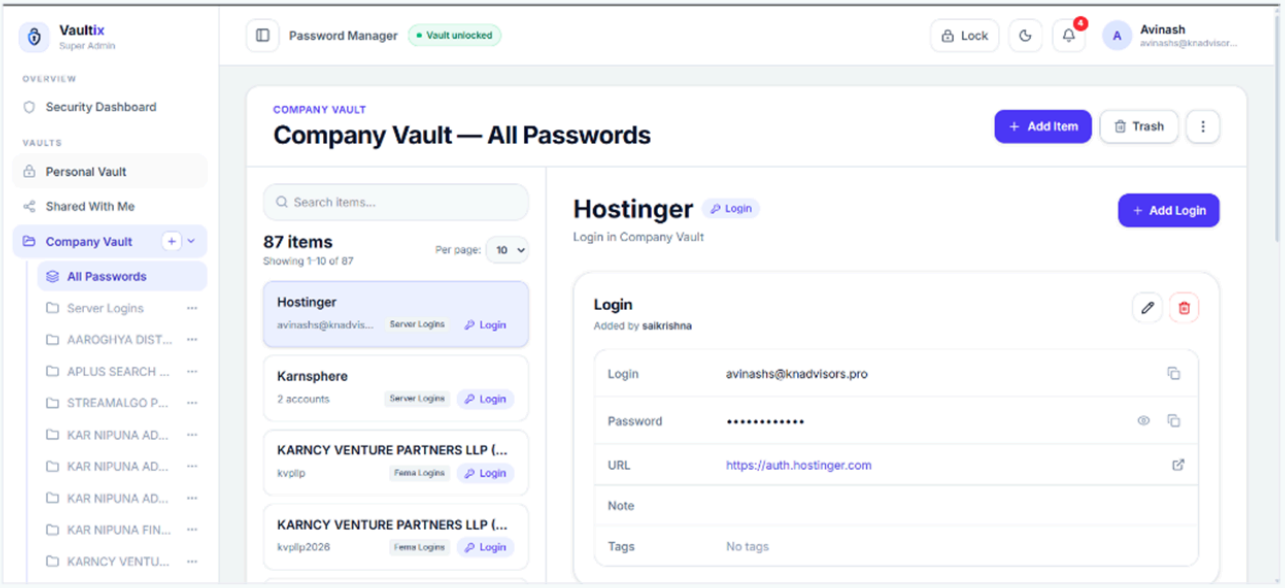
VAULTIX

DOC-VX-2026-V3.4

Vaultix Password Manager

User Guide & Technical Reference Manual

This manual covers the system architecture, zero-knowledge encryption model, administrative access controls, browser extension (v3.4.0), and day-to-day operational workflows for Vaultix.



APPLICATION URL	SECURITY MODEL
https://vaultix.knadvisors.pro	Zero-Knowledge (AES-256-GCM, PBKDF2, RSA-2048)
EXTENSION RELEASE	PUBLICATION DATE
Manifest V3 — Version 3.4.0	November 2026

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1. Platform Overview

Vaultix is a zero-knowledge password management and credential governance platform designed for organizations. It centralizes credential storage, allows secure credential sharing across teams, and provides item-level two-factor authentication (TOTP) without exposing plaintext secrets to the central server.

All cryptographic operations occur on the user's workstation using standard Web Crypto APIs. Sensitive payloads are encrypted with 256-bit AES keys before transmission. The central API server and PostgreSQL database store only encrypted ciphertext and authentication hashes. Plaintext passwords and private keys are never stored on or transmitted to the backend.

2. Security Architecture & Encryption Model

Vaultix uses standard, audited cryptographic algorithms for authentication, key derivation, and data encryption:

- **PBKDF2-SHA256 (600,000 iterations):** Derives a 256-bit symmetric AES Master Key from the user's Master Password and a unique 16-byte random salt.
- **AES-256-GCM:** Encrypts passwords, secure notes, custom fields, and private keys with a unique 12-byte initialization vector (IV) per record. GCM authentication tags detect data tampering.
- **RSA-OAEP 2048-bit Key Wrapping:** Enables shared folder access. Each user has an asymmetric RSA keypair. Shared credentials are encrypted with an AES item key, which is then wrapped individually with the public RSA key of each authorized member.
- **In-Memory Key Lifecycle:** Decrypted private keys and credentials reside in browser memory only while the vault is unlocked. They are never written to disk, localStorage, or cookies, and are purged on logout or session timeout.
- **Bcrypt (Cost 12):** Validates account login passwords on the backend server.

2.1. Authentication & Master Password Separation

Vaultix separates account authentication from vault decryption. The account password authenticates the user with the backend API and generates a short-lived JWT session token. The Master Password is used locally in the browser to derive the AES key that decrypts the user's private RSA key. The server never receives or stores the Master Password.

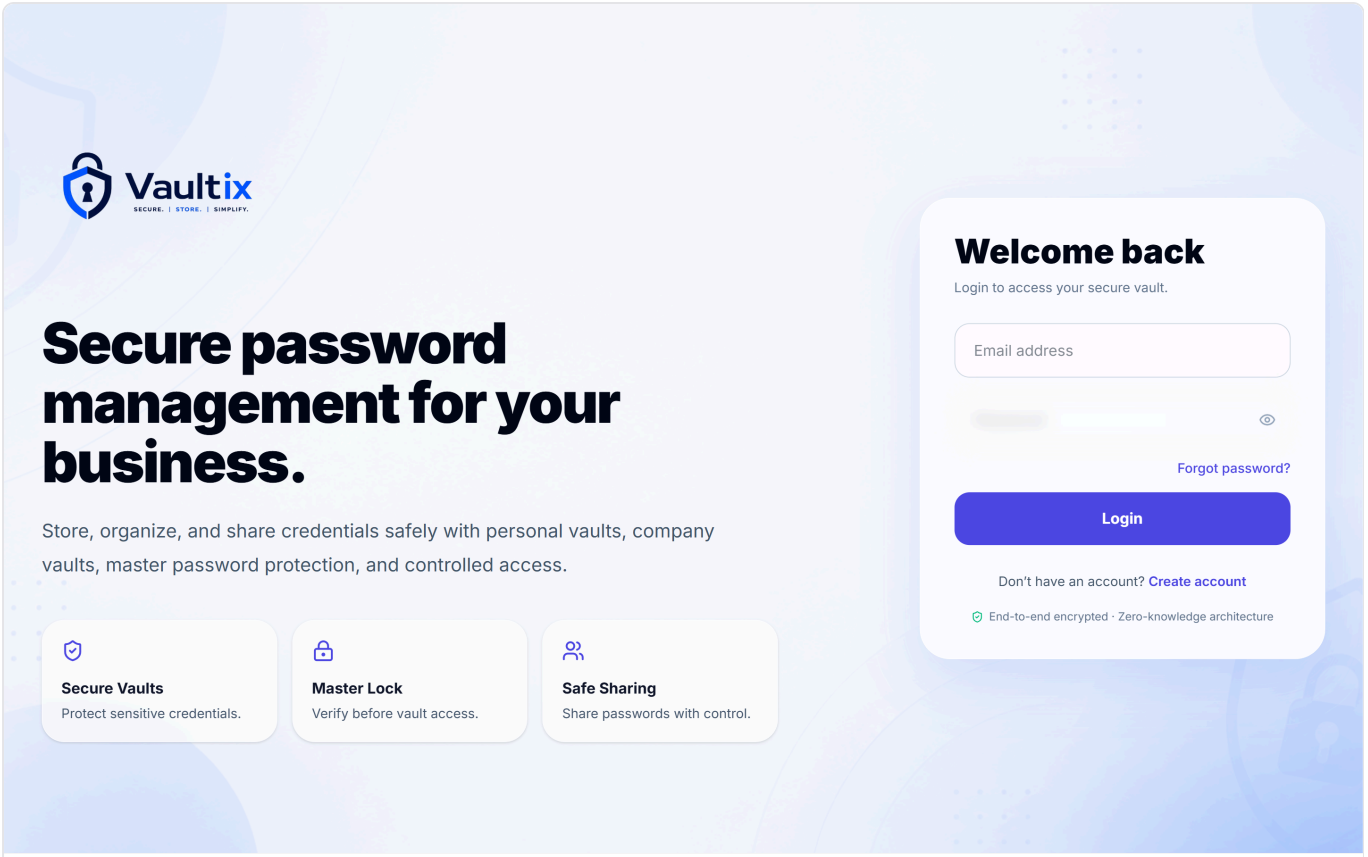

Figure 1: Account Login (`/login`). Primary authentication screen. Users log in with their registered email address and account password. The backend validates credentials using bcrypt and issues a session JWT.

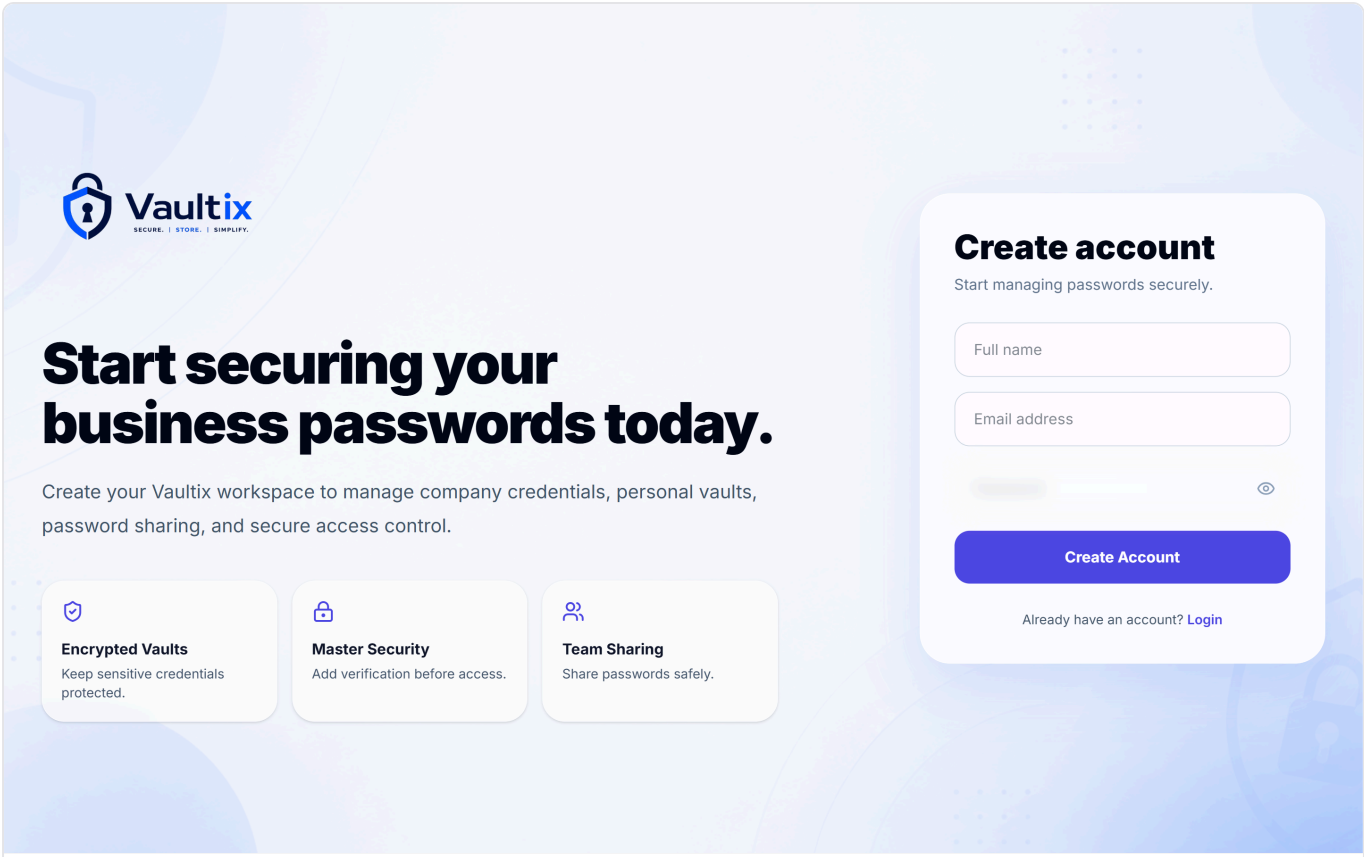

Figure 2: User Registration (`/register`). Account creation form for new team members. Registration initializes the user profile prior to administrator approval and role assignment.

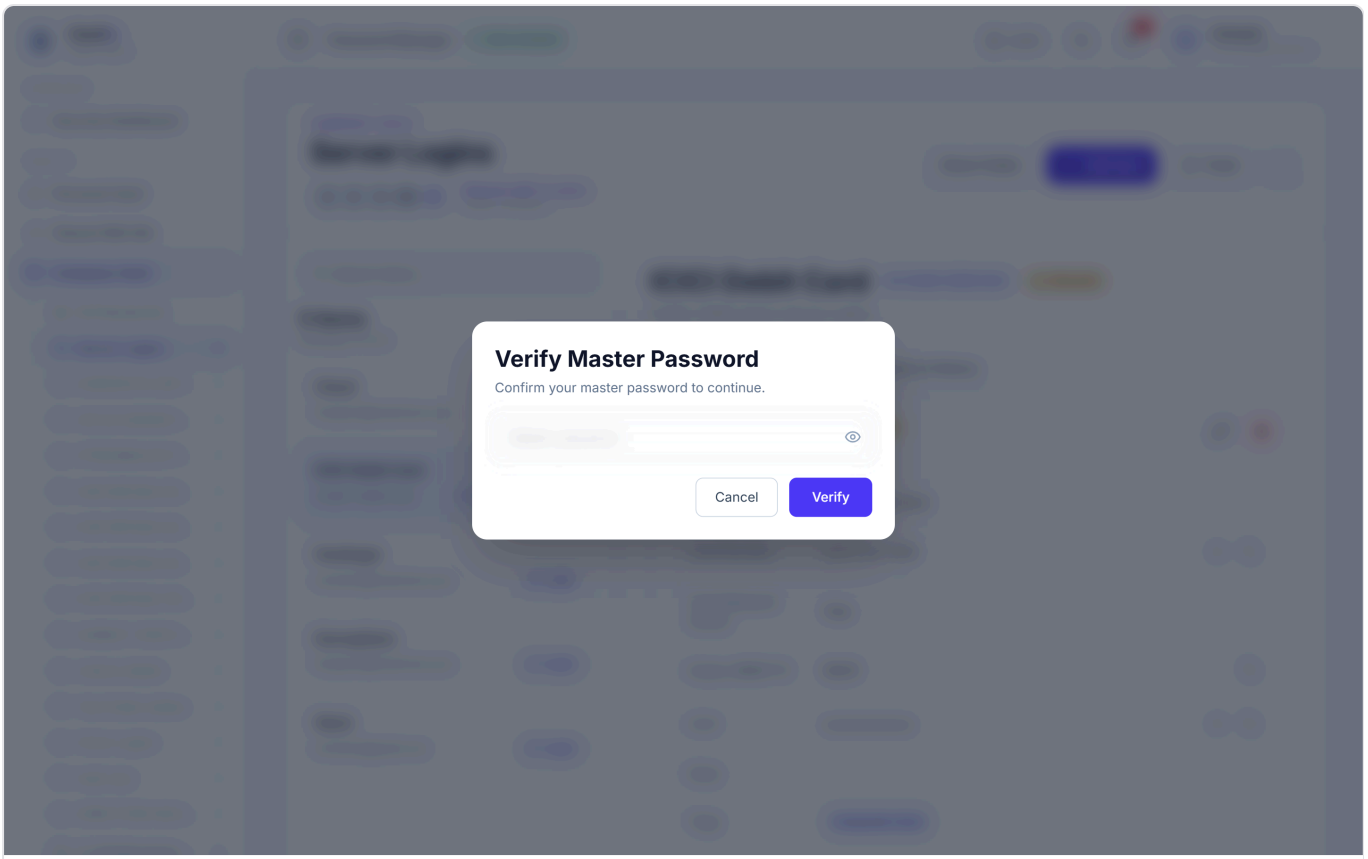

Figure 5: Master Password Verification & Vault Unlock (`Modal`). Vault unlock prompt. The Master Password derives a 256-bit AES key in browser memory via PBKDF2 (600,000 iterations) to decrypt the user's private RSA key.

2.2. Master Password Rotation

When a user updates their Master Password in Profile settings, the client decrypts the user's private RSA key and personal vault data using the current master key, derives a new key from the new Master Password, re-encrypts the data, and updates the server record in an atomic transaction.

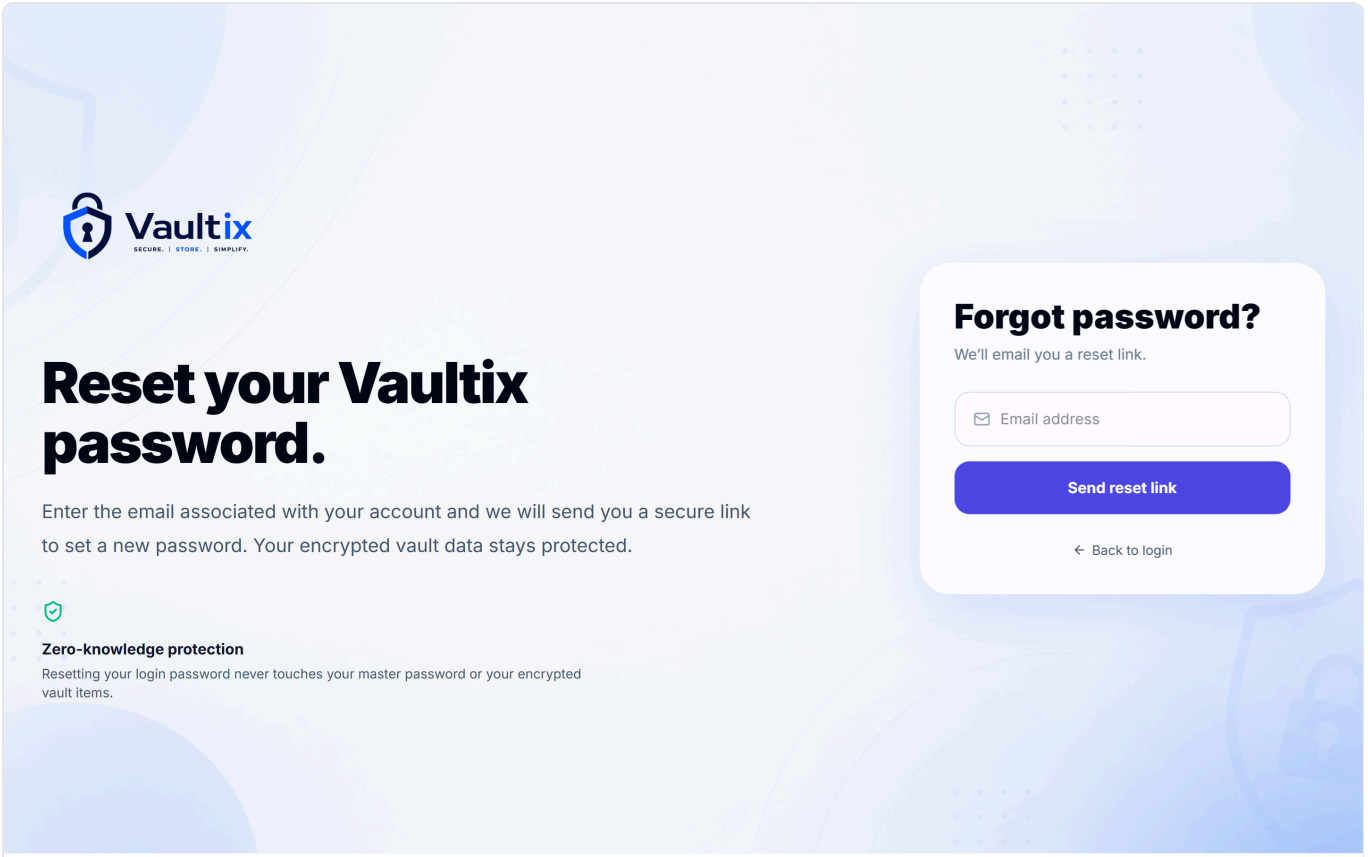

Figure 3: Account Password Reset (`/forgot-password`). Password reset request screen. Resetting the account password updates server login credentials but does not affect the client-side master encryption key.

2.3. Emergency Recovery Key

During initial account setup, Vaultix generates a 12-character emergency recovery key (e.g. `XK7P-9M2Q-4TRH`). The user's private RSA key is encrypted with an AES key derived from this recovery key and stored on the server. If a user forgets their Master Password, they can use this key to restore vault access.

> **Security Note**
>
> Because Vaultix uses zero-knowledge encryption, administrators cannot recover a user's vault without this emergency recovery key.

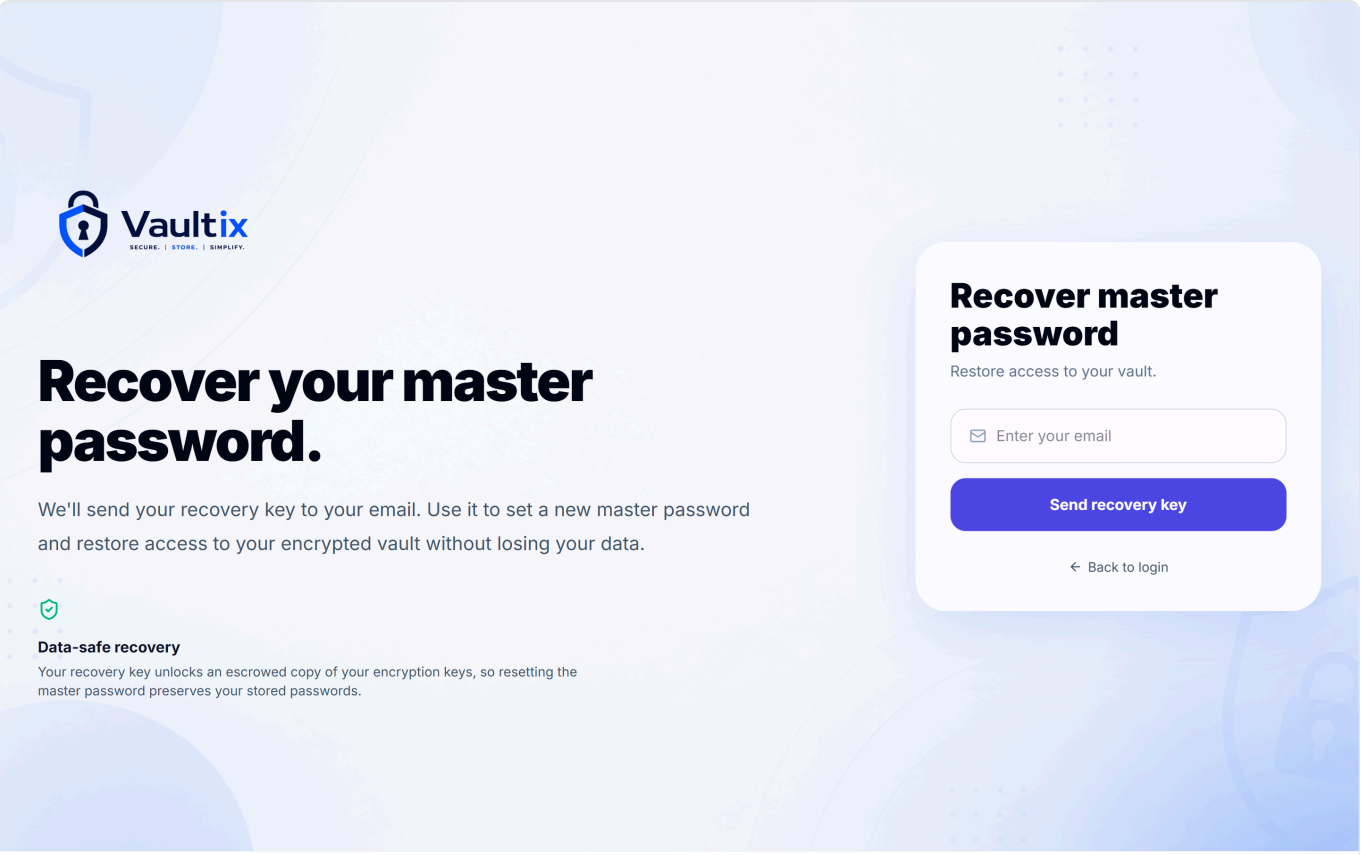

Figure 4: Master Password Recovery (`/forgot-master-password`). Master password recovery screen. Users input their 12-character emergency recovery key to re-derive their master key and regain access to their encrypted vault.

3. Vault Structure: Personal & Shared

Vaultix provides two isolated storage scopes: a private Personal Vault and shared Company Vaults.

3.1. Personal Vault ("My Vault")

- **Private Ownership:** Automatically created for each user. Inaccessible to system administrators or managers.
- **Custom Folders:** Users can create custom subfolders to organize personal services, banking, and notes.
- **Direct Symmetric Encryption:** Personal items are encrypted directly with the user's AES Master Key without RSA key-wrapping overhead.
- **Audit Privacy:** Access logs for personal vault items are visible only to the individual user.

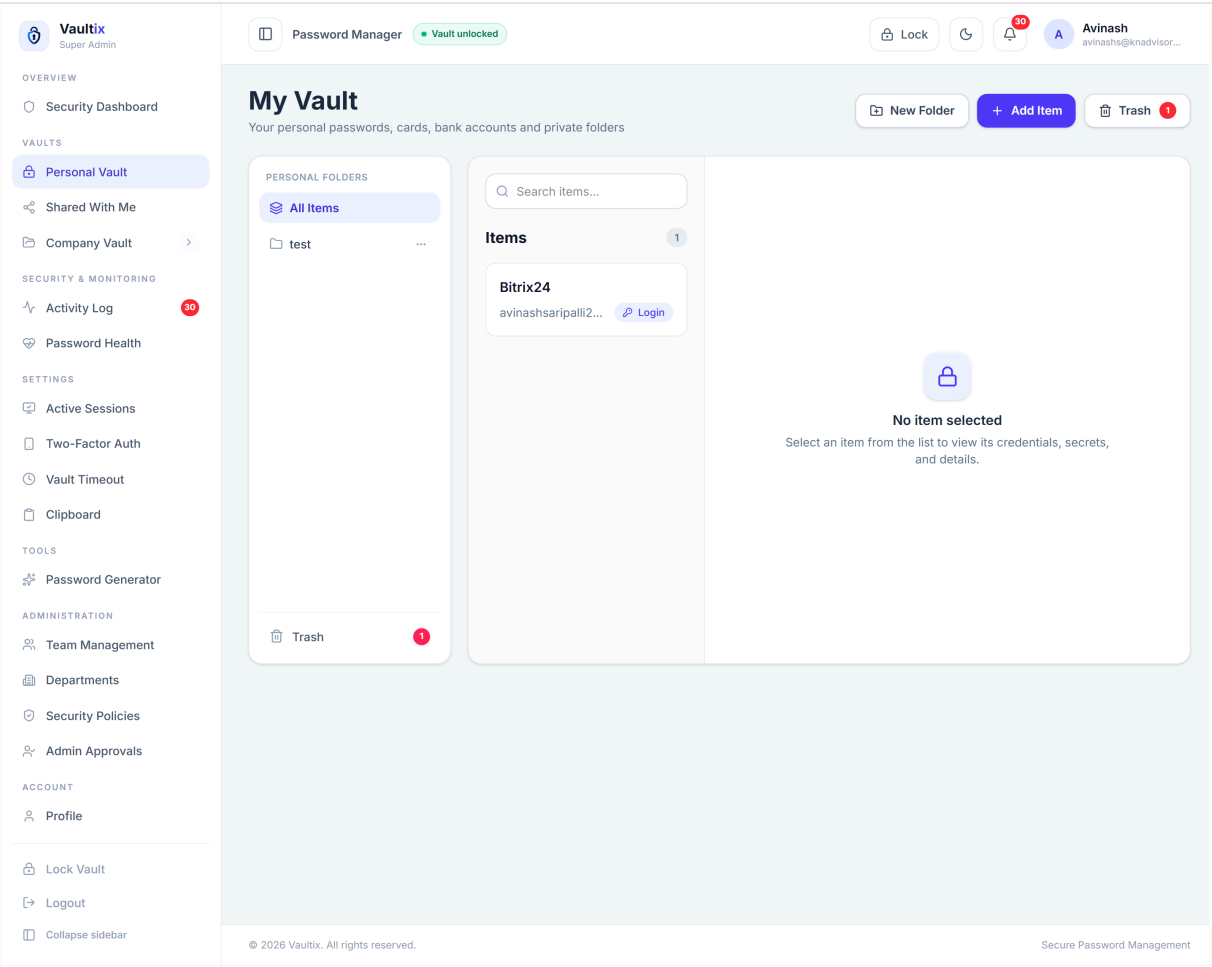

Figure 8: Personal Vault ("My Vault") (`/vault/personal`). Personal vault inventory. Contains private credentials, payment cards, bank accounts, and notes encrypted with the user's master key. Inaccessible to administrators.

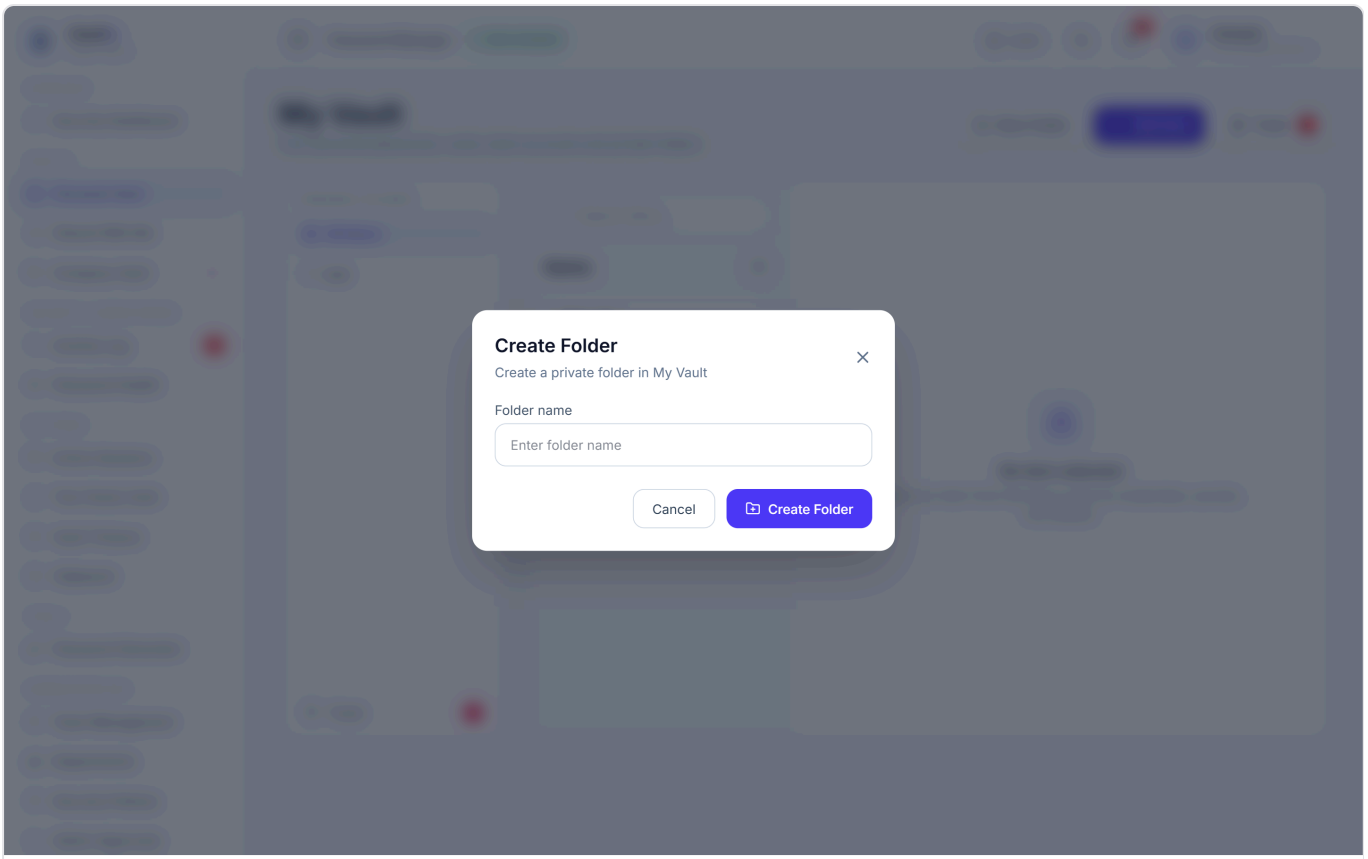

Figure 10: Create Personal Folder Modal (`Modal`). Folder creation dialog for organizing personal vault credentials into hierarchical folders.

3.2. Company & Team Vaults

- **Collaborative Vaults:** Administrators can create multiple company vaults for departments, teams, or client engagements.
- **Hybrid Multi-Recipient Encryption:** Items in shared vaults are encrypted with single-use AES item keys and RSA-wrapped for authorized users.
- **Folder Hierarchy:** Company vaults contain structured folders with granular user permissions.

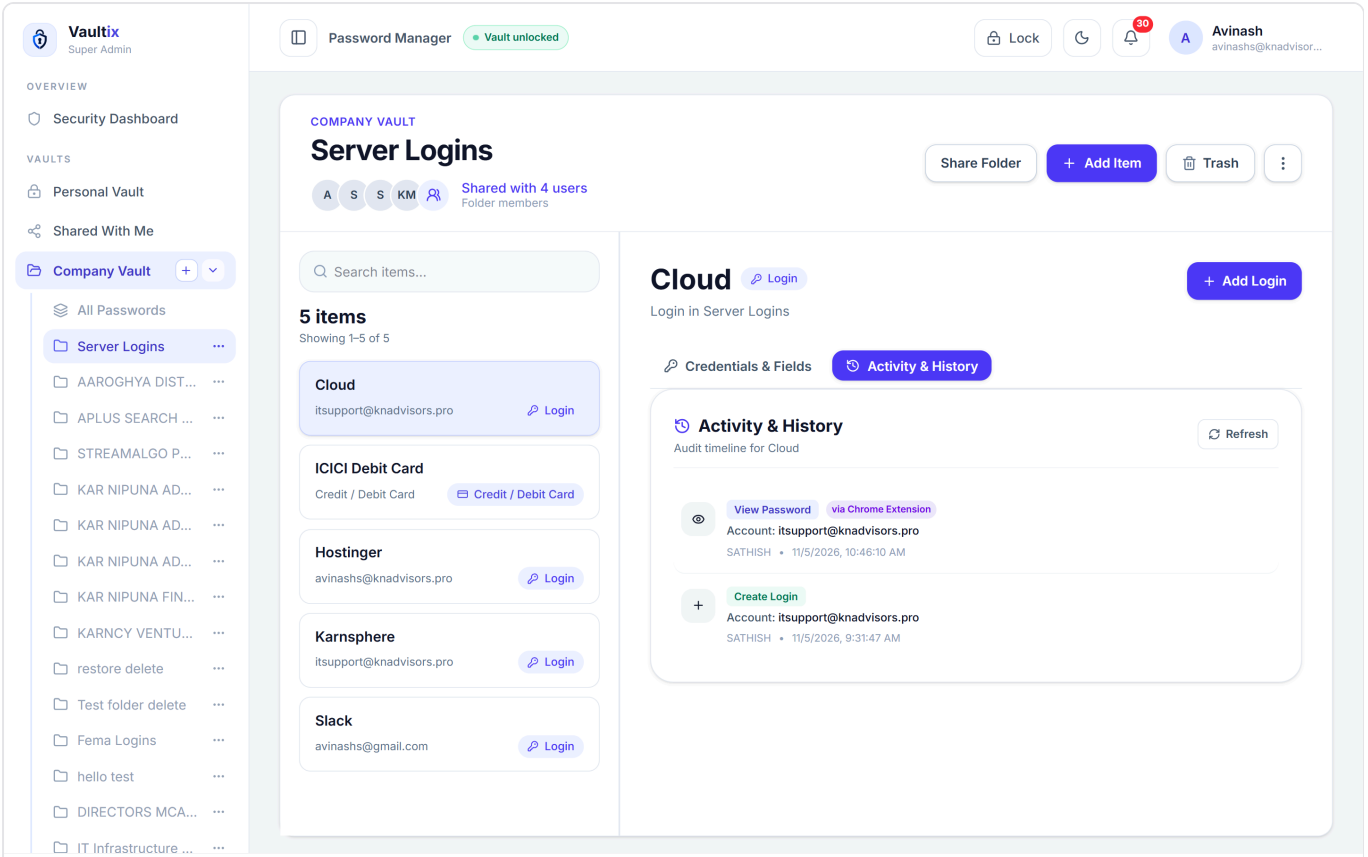

Figure 14: Company Vault Collaborative Workspace (`/vault/company`). Shared organizational vault. Holds shared team folders and credentials encrypted with hybrid RSA-OAEP / AES-256-GCM encryption.

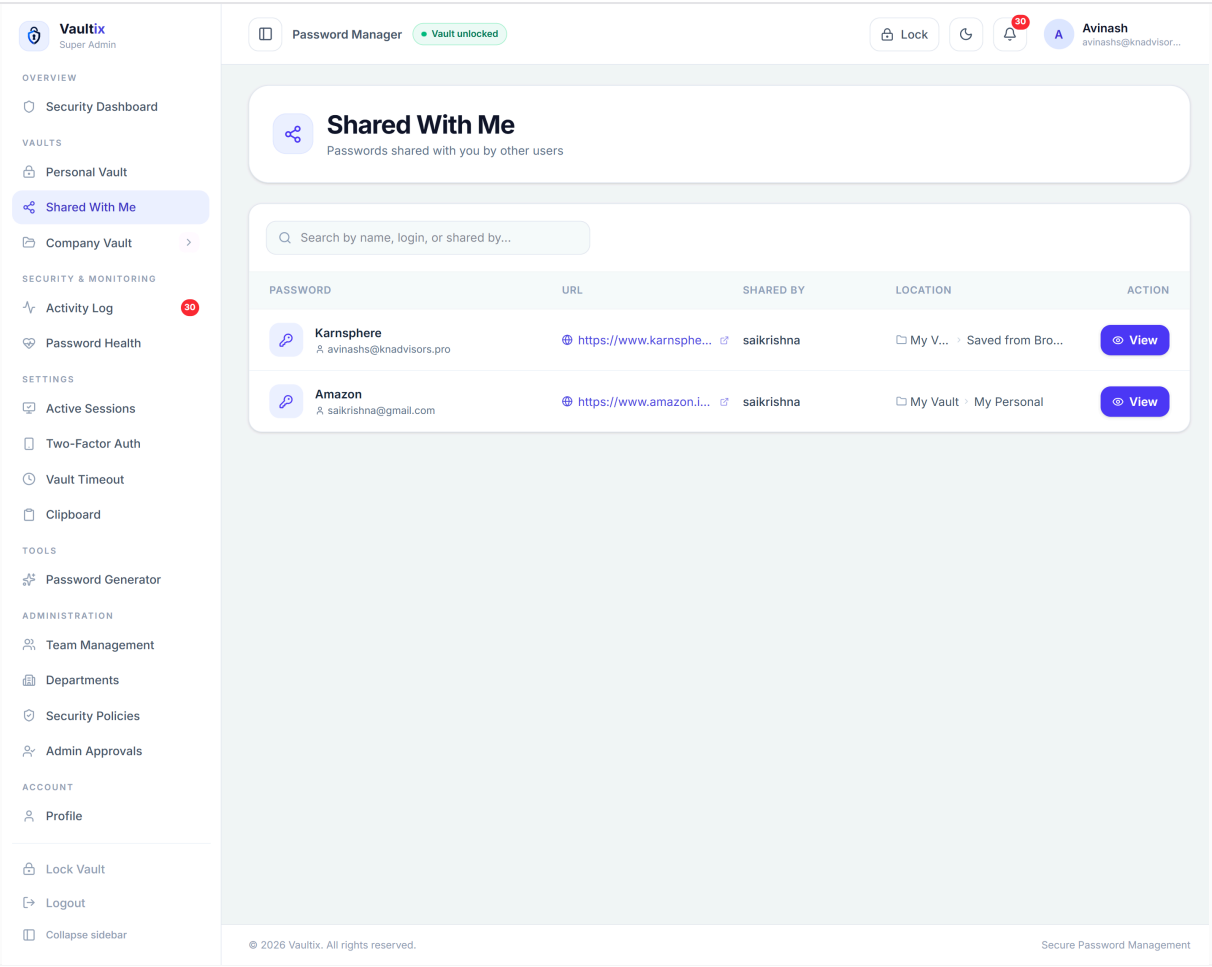

Figure 12: Shared with Me Directory (`/vault/shared`). Directory displaying all folders and credentials that other team members or administrators have shared with the logged-in user.

4. Item Schemas & Custom Fields

Vaultix provides specialized schemas and validation rules for 11 distinct credential types:

Schema Type	Category	Built-in Fields
LOGIN	Web & App Logins	Username, Password, Website URL, Built-in TOTP Authenticator Key.
CARD	Payment Cards	Cardholder Name, Card Number (Luhn validated), Expiration Month/Year, CVV, PIN, Brand Detection.
BANK_ACCOUNT	Banking & NetBanking	Bank Name, Customer ID / Username, Account Number, NetBanking Password, Transaction Password, URL.
IDENTITY	Personal Identification	Full Name, National ID / SSN, Passport Number, Driver's License Number, Date of Birth.
SSH_KEY	DevOps & Infrastructure	Hostname, Port, Username, Private Key (PEM format), Public Key, Passphrase.
API_TOKEN	Developer & API	Service Name, API Key ID, Secret Token, Environment (Dev/Staging/Prod).
DATABASE	Database Credentials	Host Endpoint, Port, Database Name, Database Engine (PostgreSQL, MySQL, MongoDB, Redis), Username, Password.
WIFI	Network Access	Network SSID, Security Protocol (WPA2, WPA3, Enterprise), Password, Hidden Network flag.
LICENSE	Software Licenses	Software Name, Licensed Email/Org, License Key / Activation Code, Expiration Date.
RECOVERY_CODE	Account Recovery	Primary Service, Account Identifier, Single-use emergency recovery codes.
SECURE_NOTE	Encrypted Notes	Title, Encrypted Markdown notes, configuration snippets, recovery instructions.

Dynamic Custom Fields: In addition to default schema fields, any item can include arbitrary custom fields (Text, Masked, or Hidden). Custom fields are encrypted with the item's AES key before transmission.

> **Banking & NetBanking Schema**
>
> The `BANK_ACCOUNT` schema combines bank account records with NetBanking credentials. Dedicated fields for NetBanking Password and Transaction Password include individual copy buttons, eliminating the need to create separate login items.

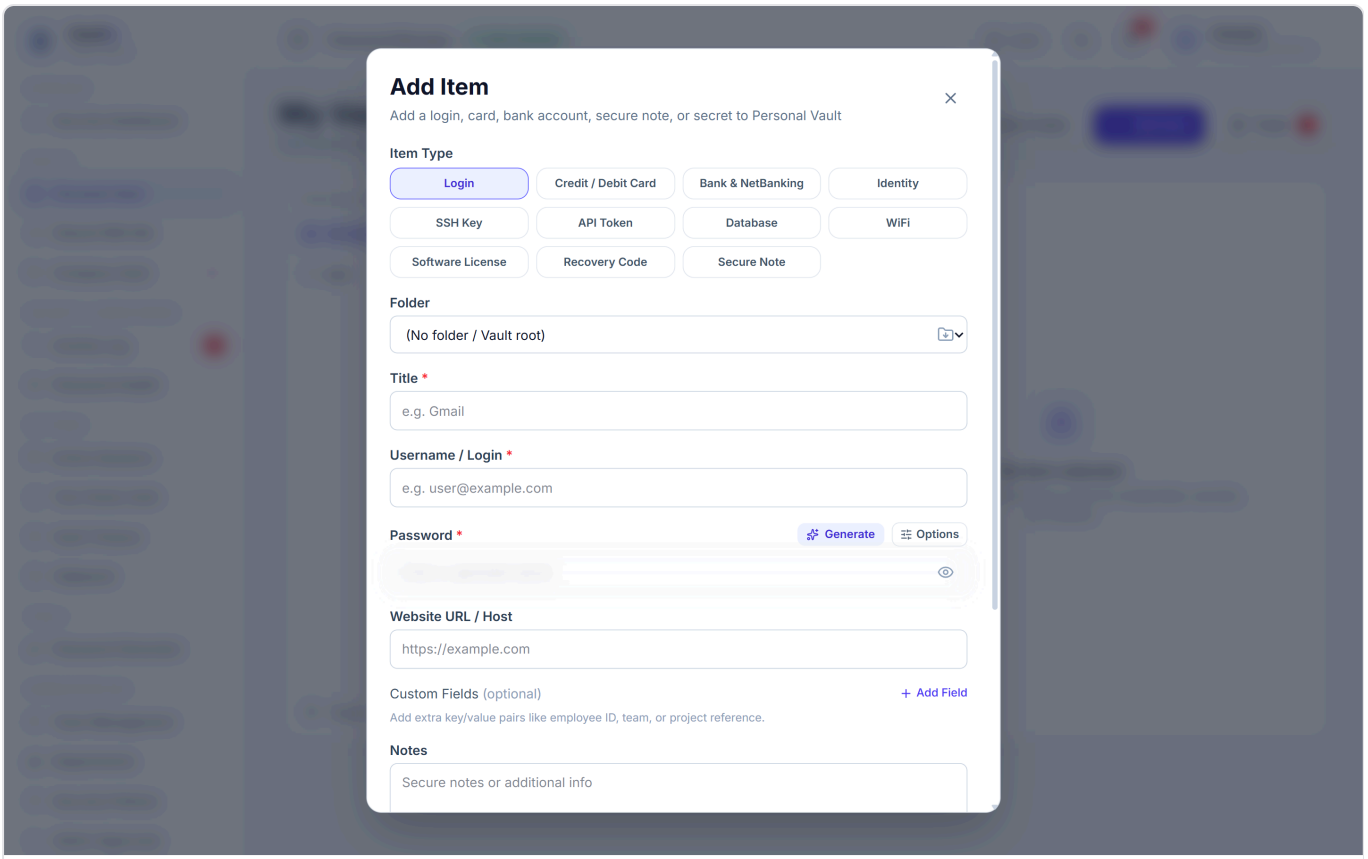

Figure 9: Add Credential Modal (`Modal`). Item creation modal supporting 11 schemas, custom fields (text, masked, hidden), built-in TOTP secret keys, and color tags.

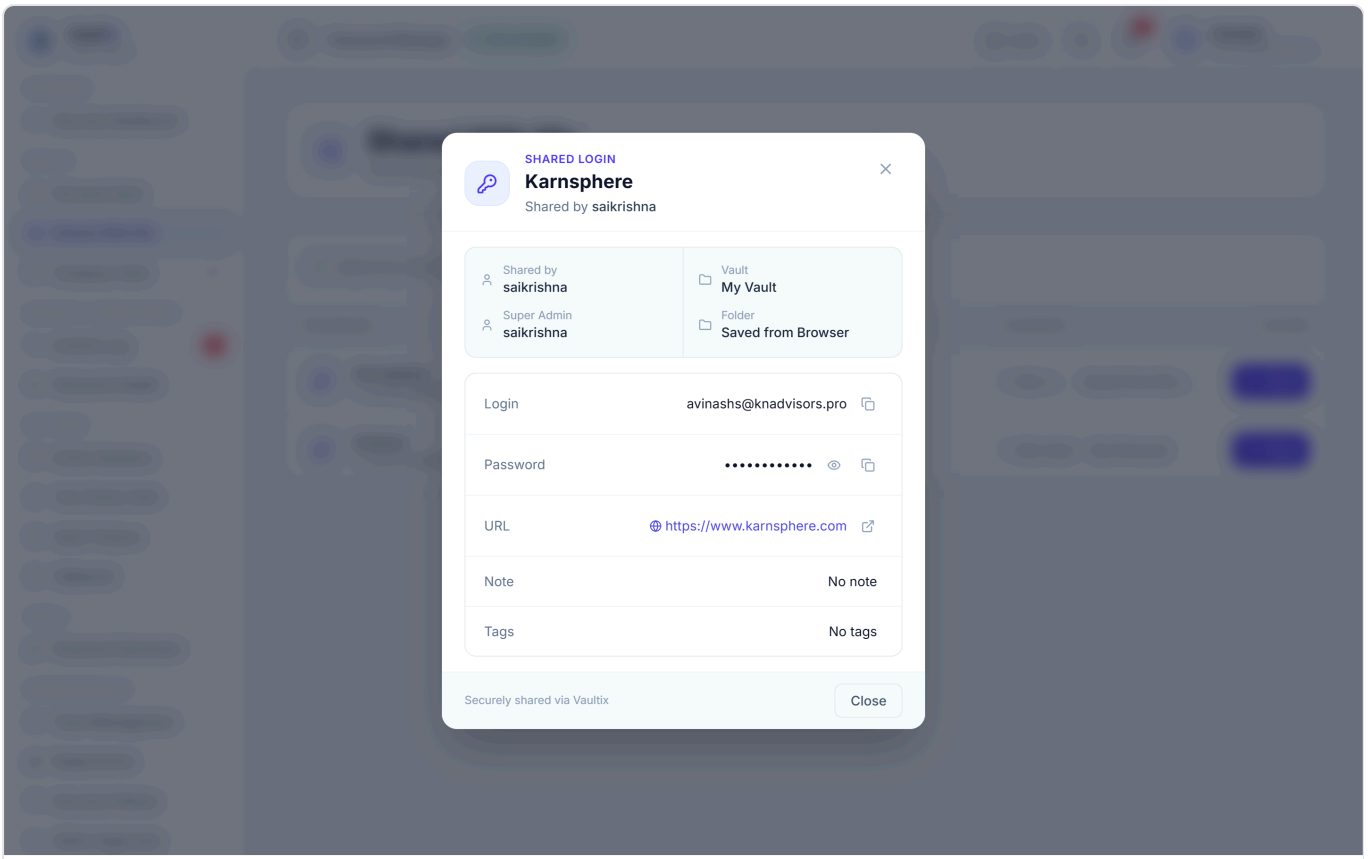

Figure 13: Credential Details & TOTP Modal (`Modal`). Decrypted item view displaying usernames, masked passwords, custom fields, and an active RFC 6238 TOTP circular countdown timer.

5. Built-in TOTP Authenticator (2FA)

Vaultix includes an integrated RFC 6238 TOTP engine within credential items:

- **Secret Storage:** Base32 secret keys can be saved directly in the TOTP field of Login items.
- **Circular Countdown:** Displays an active 30-second circular countdown timer with real-time 6-digit codes.
- **One-Click Copy:** Users can copy 2FA verification codes directly to the clipboard.
- **Shared Team 2FA:** When an item with TOTP is stored in a shared folder, all members with access generate identical OTP codes.

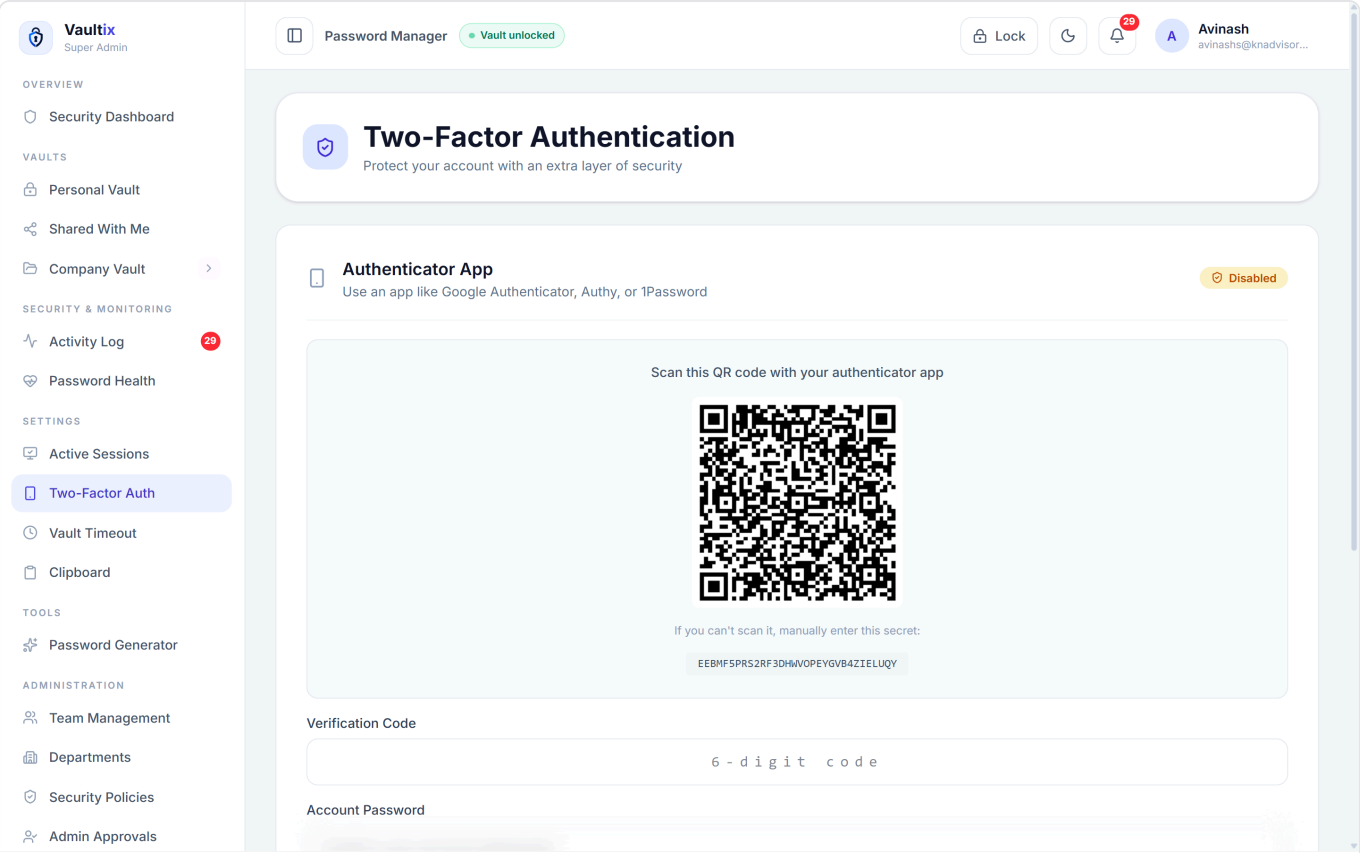

Figure 29: Authenticator App Setup Modal (`Modal`). TOTP enrollment modal displaying an RFC 6238 QR code, manual base32 secret key, and code verification field.

6. User Roles & Permissions

Vaultix separates administrative privileges into system roles and folder permission levels.

6.1. System Roles

Role	Scope	Permissions
SUPER_ADMIN	Global Organization	Full control over all vaults, security policies, departments, user accounts, and system logs.
ADMIN	Administration	Can manage company vaults, team members, folder permissions, and access approval requests.
USER	Standard Member	Access to private Personal Vault and shared folders explicitly granted to them.

6.2. Folder Permission Levels

Level	Operations	Description
FULL_ACCESS	Create, Edit, View, Copy, Soft-Delete	Full control over items within the specific folder, including moving items to trash.
READ_WRITE	Create, Edit, View, Copy	Users can add new items, edit existing fields, and copy passwords. Cannot delete items.
READ	View & Copy Only	Users can view credential details and copy passwords. All actions are logged.

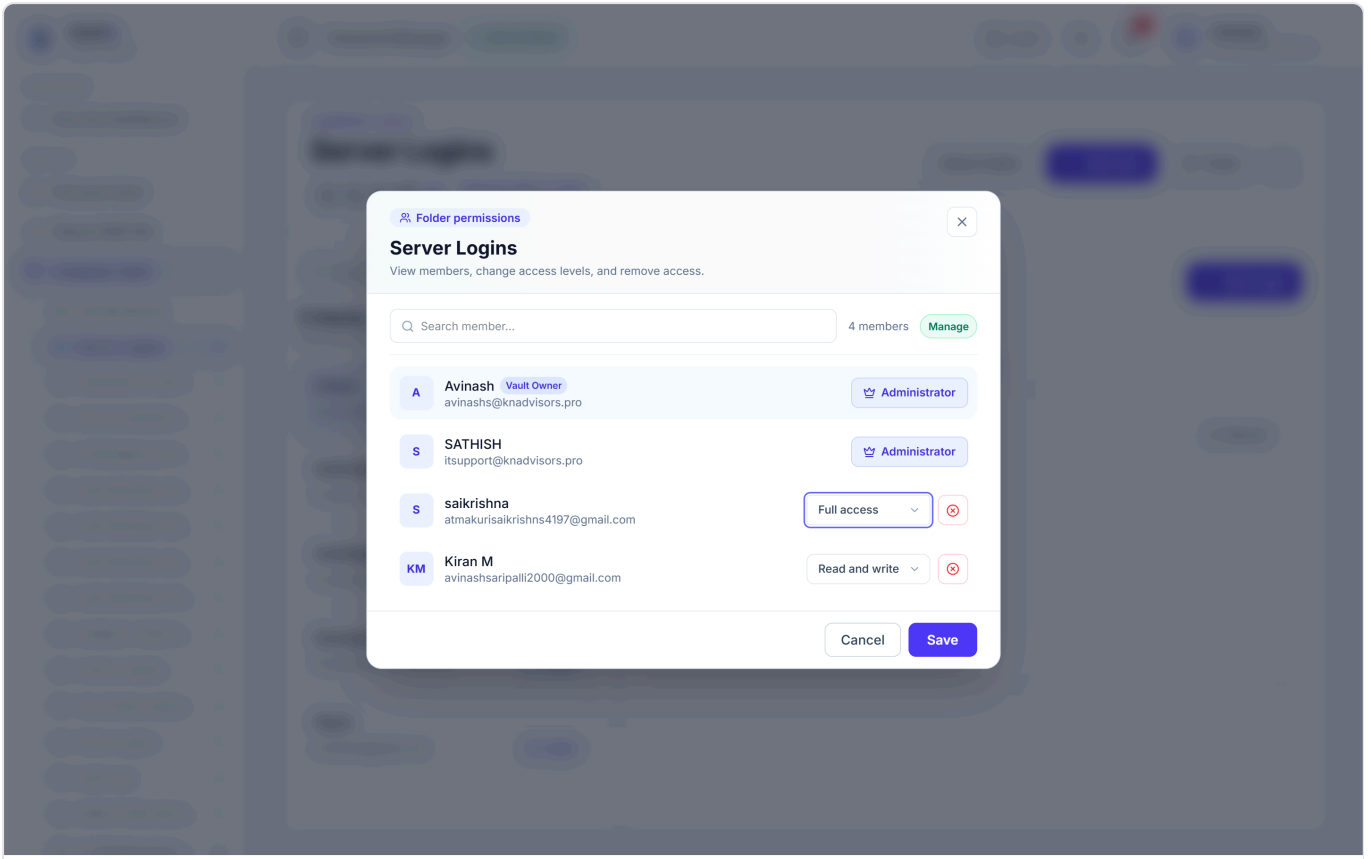

Figure 15: Folder Permissions Management (`Modal`). Permission assignment interface for granting direct user access (Full Access, Read/Write, Read Only) with client-side RSA key wrapping.

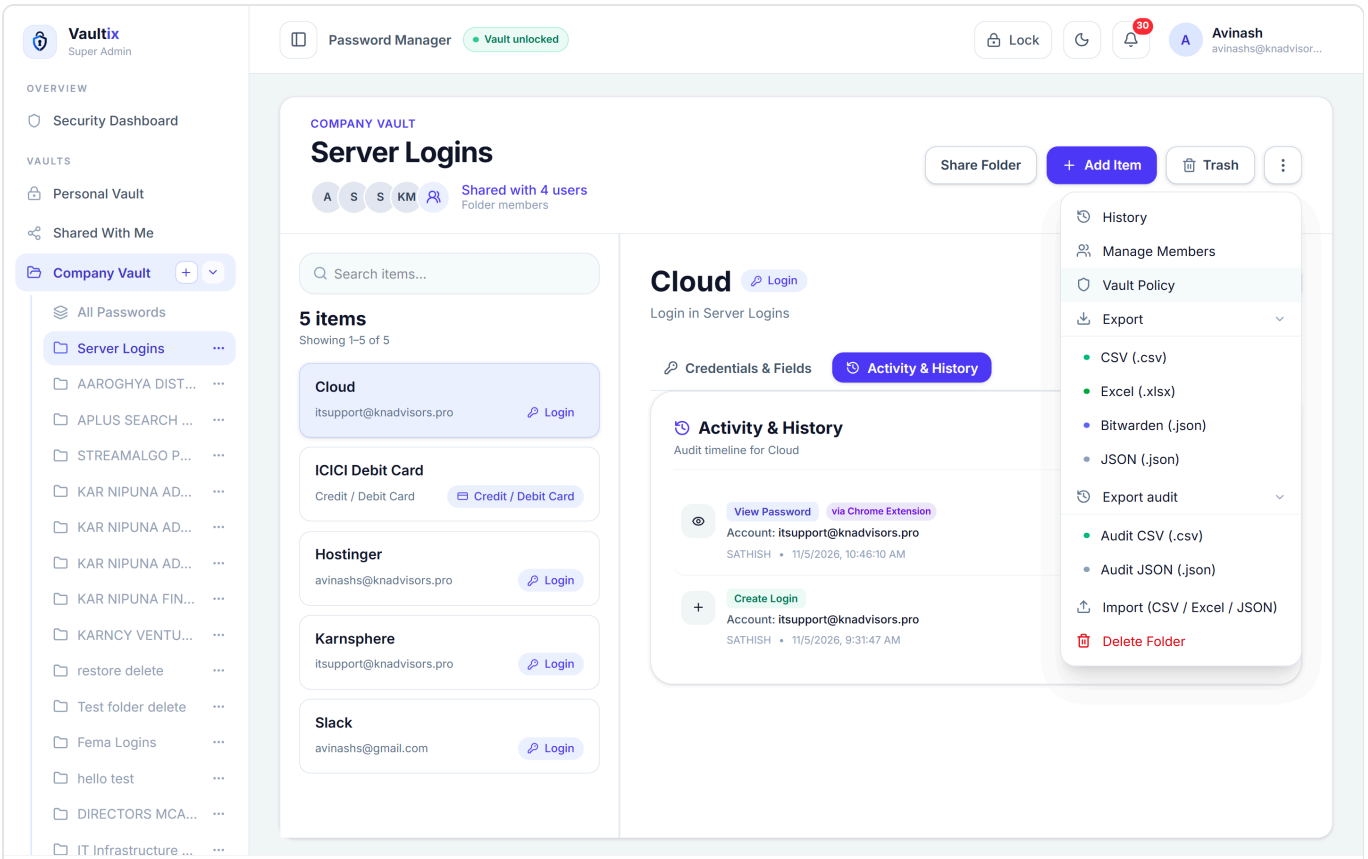

Figure 18: Folder Action Menu (`Context Menu`). Context menu for renaming folders, managing user access, reviewing audit history, configuring policies, and moving folders to trash.

7. Department Management

Vaultix separates corporate organizational hierarchy from cryptographic folder access grants:

- **Department Structure:** Supports multi-level department hierarchies (e.g. IT → Systems Engineering → Cloud Operations).
- **Department Roles:** Employees are assigned as Members or Managers within departments.
- **Direct User Grants:** Shared folder permissions are assigned directly to individual users. Each user receives their own RSA-wrapped folder key, ensuring least-privilege access.
- **My Departments Page:** Employees can view their assigned departments, managers, and corporate contacts.

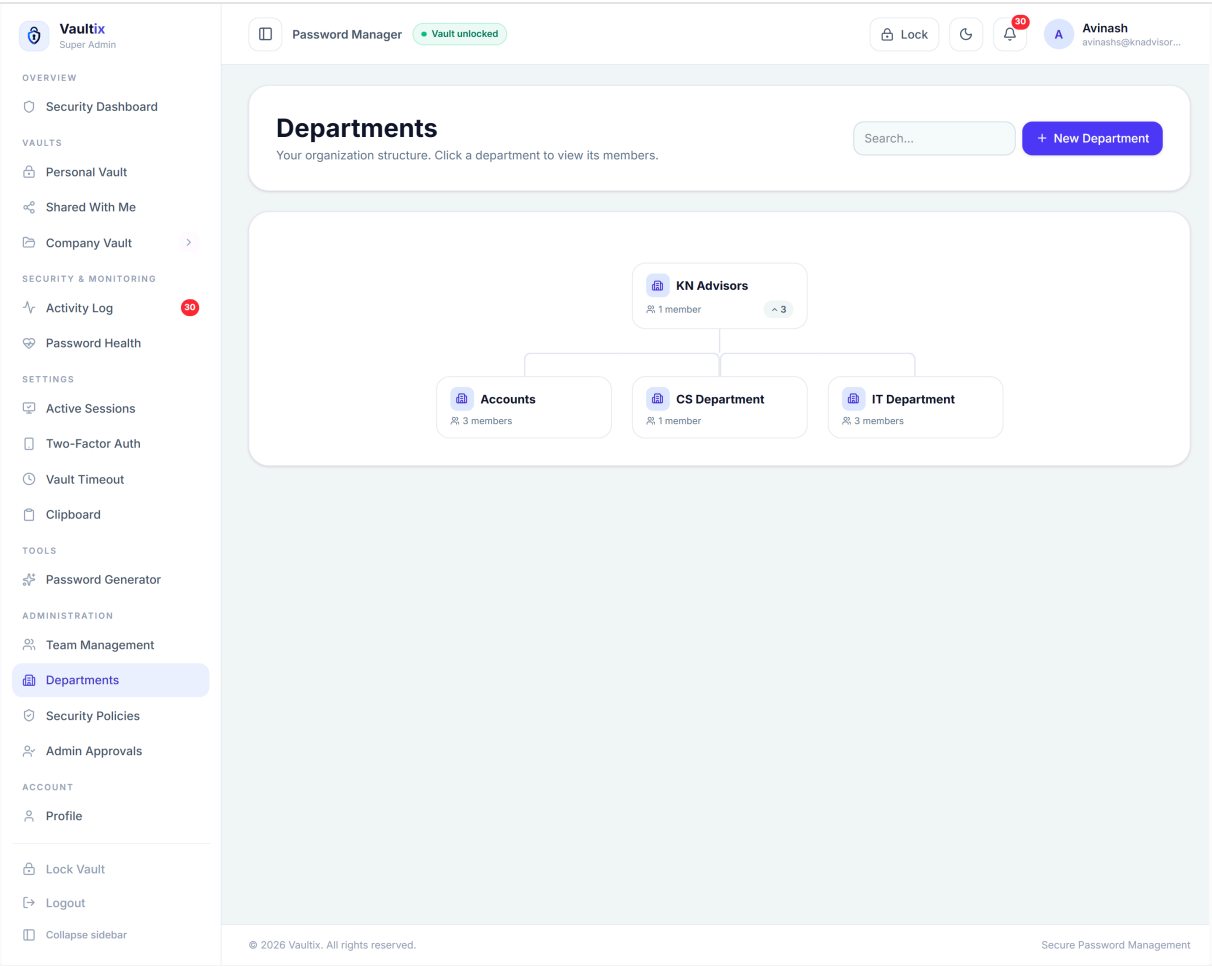

Figure 39: Departments Directory (`/departments`). Overview of organizational units showing department hierarchy, member counts, and assigned managers.

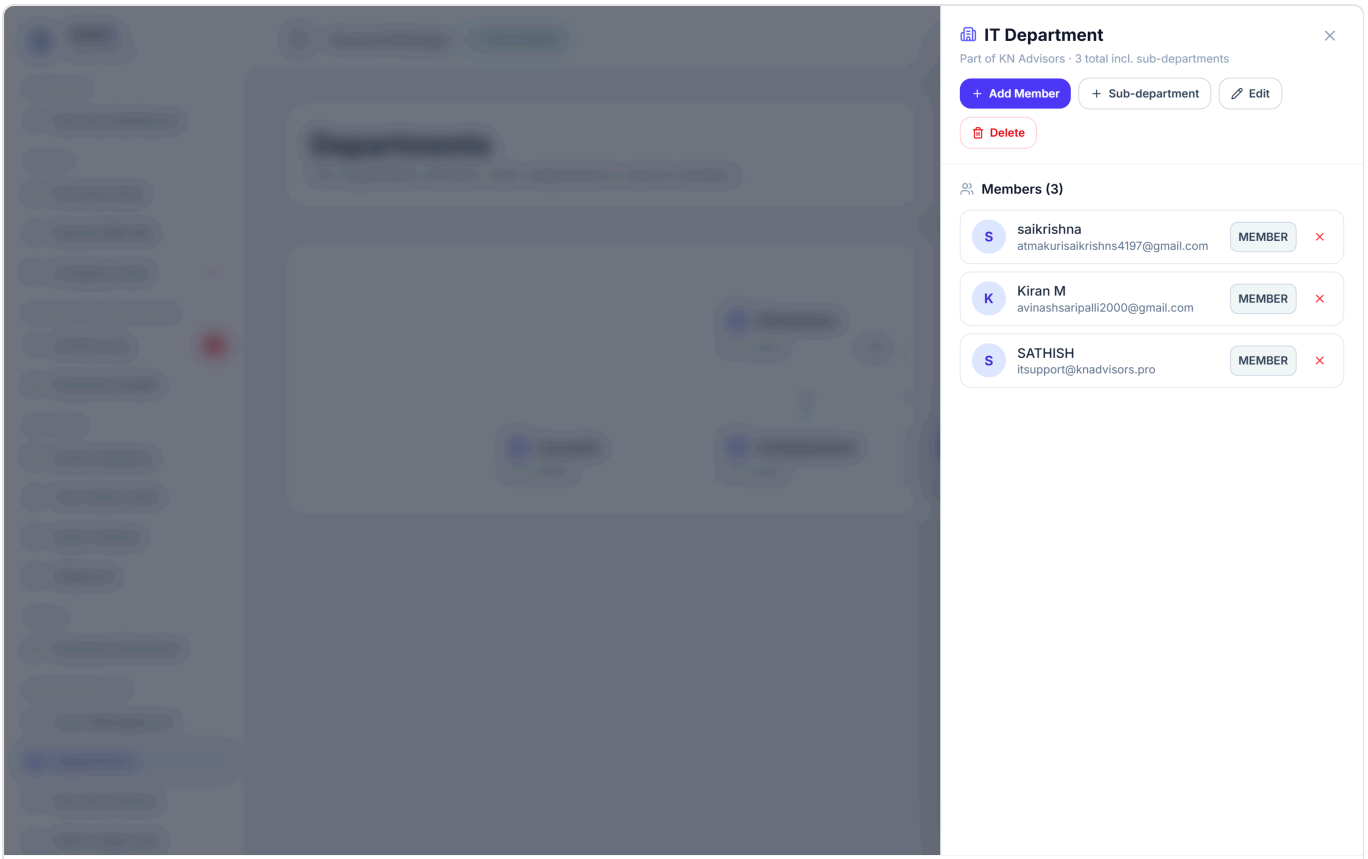

Figure 40: Department Members Drawer (`Drawer`). Slide-out panel displaying department leadership, member roster, and member roles (Manager vs Member).

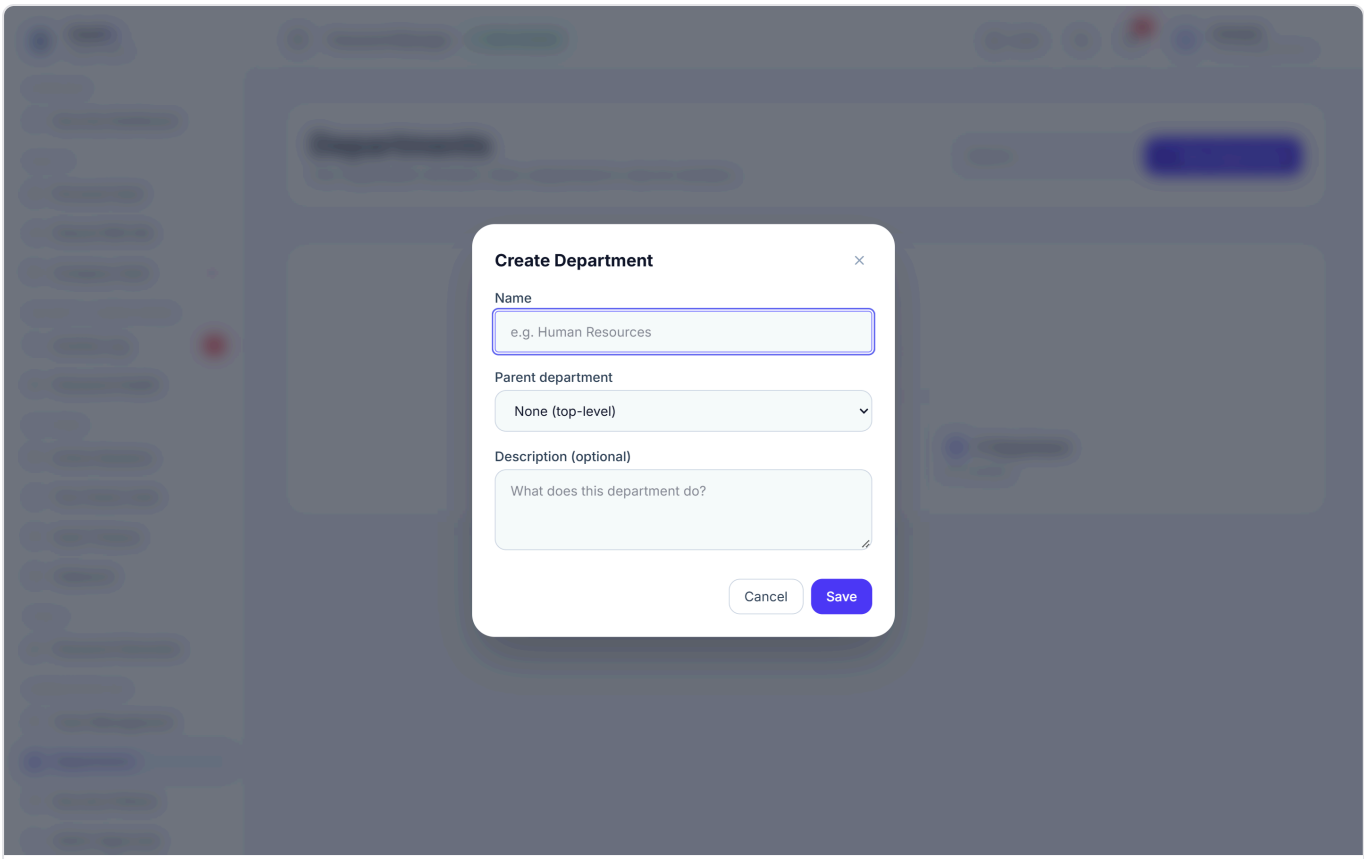

Figure 41: Create Department Modal (`Modal`). Modal for creating top-level departments with custom descriptions and designated managers.

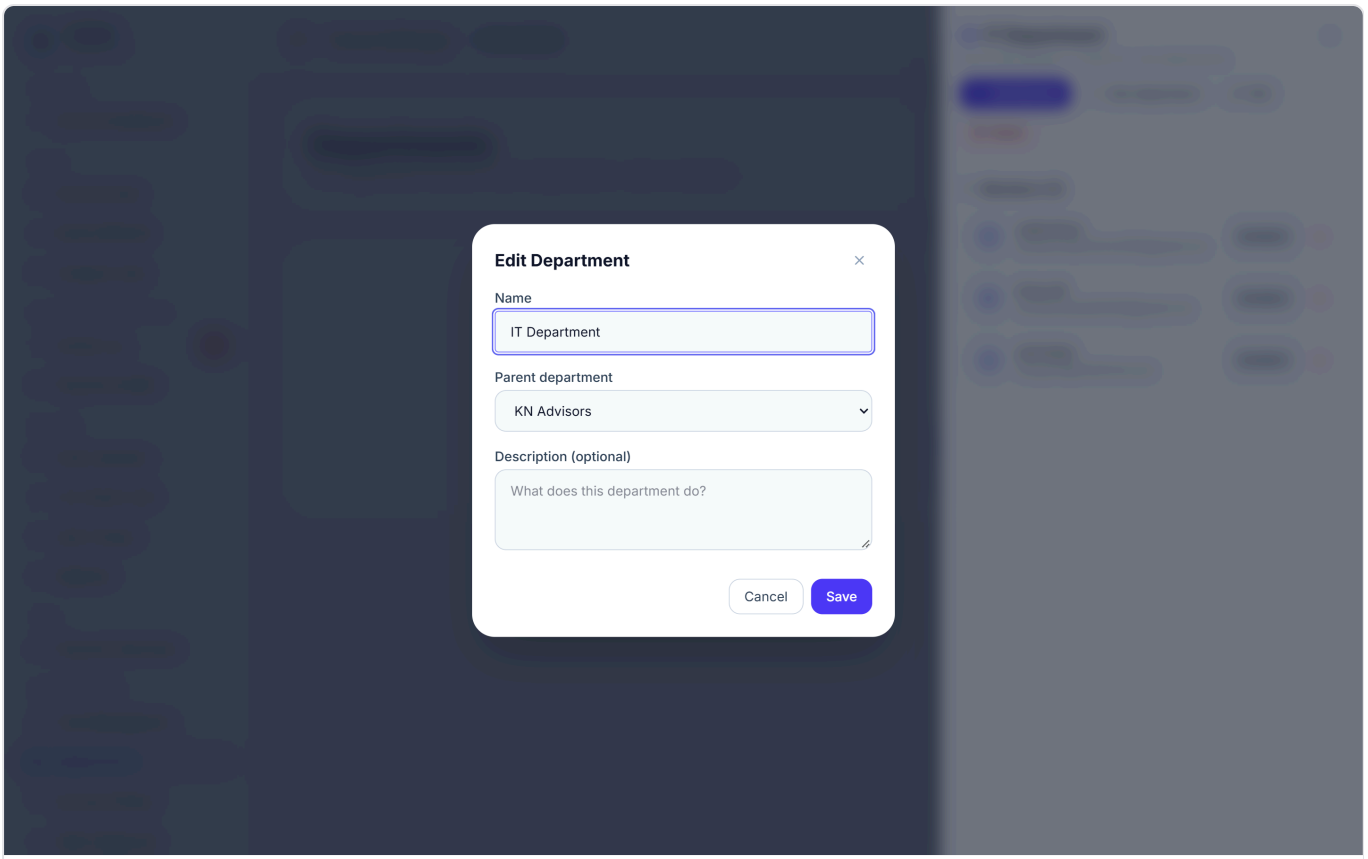

Figure 42: Edit Department Modal (`Modal`). Dialog for editing department details, updating descriptions, and reassigning managers.

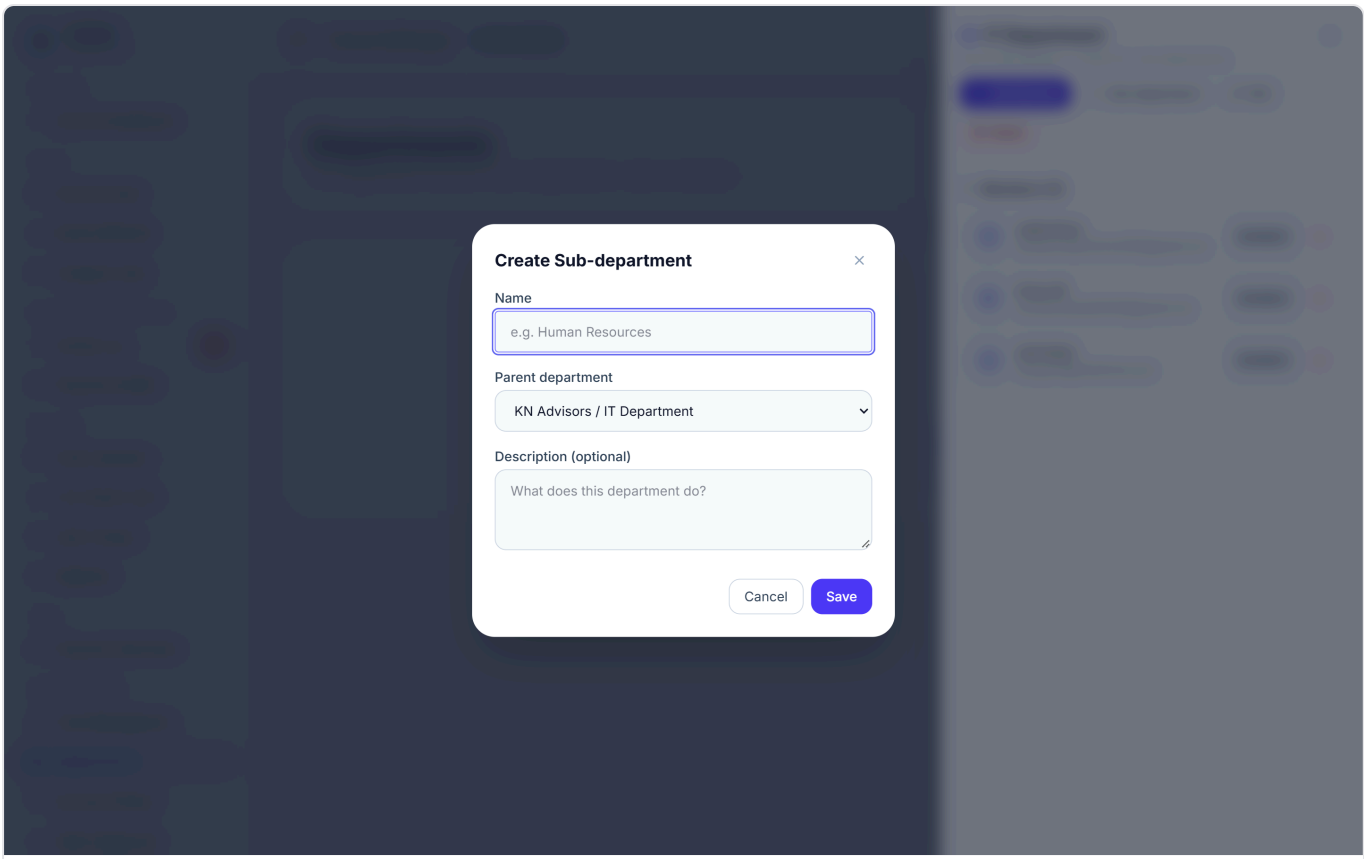

Figure 43: Create Sub-Department Modal (`Modal`). Dialog for adding nested sub-departments within existing department structures.

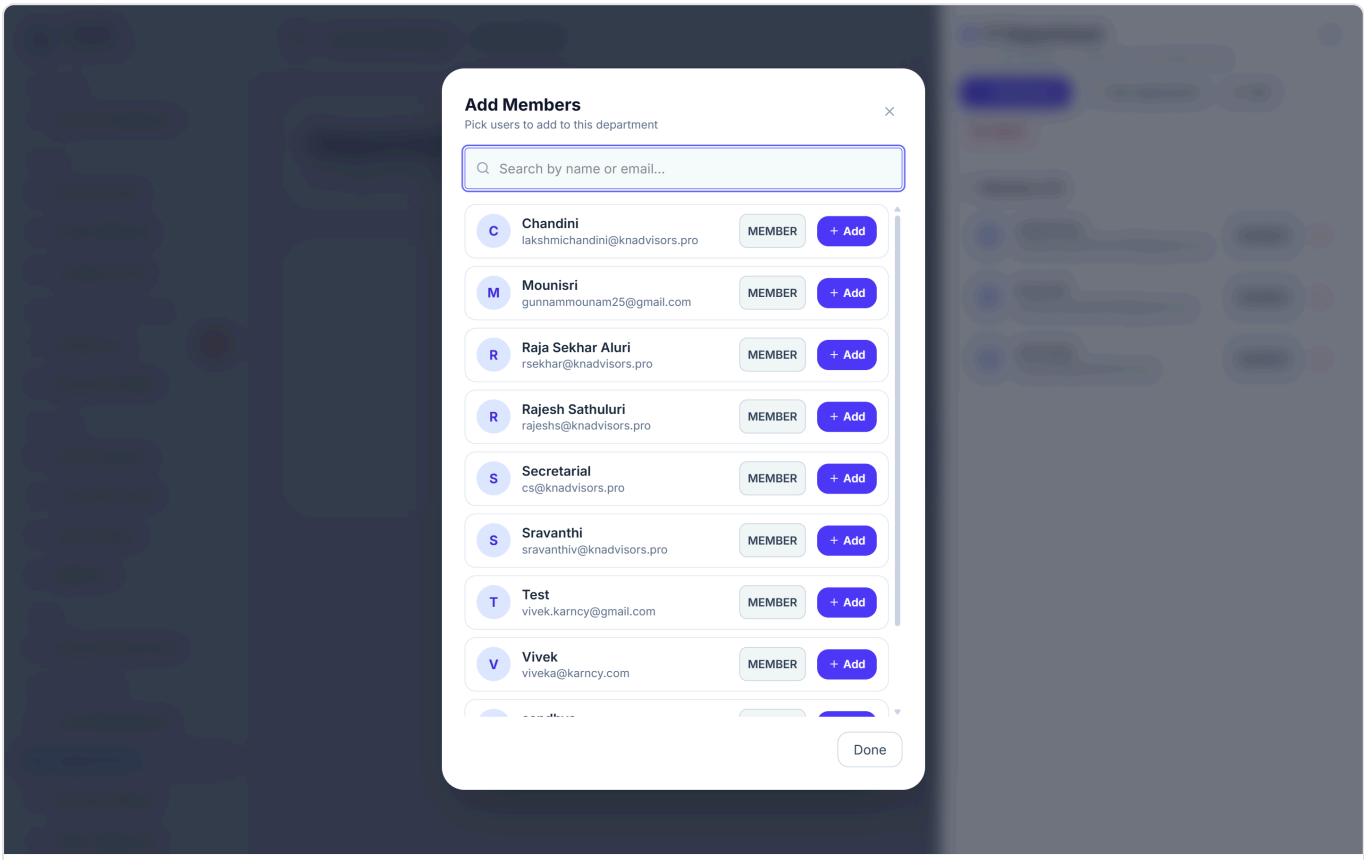

Figure 44: Add Department Members Modal (`Modal`). Search and multi-select dialog for adding employees to a department.

8. User Access Auditing & Grants

In the User Access modal, administrators can audit and configure folder permissions for any employee:

- **Accessible Folders Tab:** Displays all shared folders accessible to the employee with current permission badges and a Revoke Access button.
- **Grant New Access Tab:** Allows administrators to select a vault and folder, set the permission level, and wrap the folder key with the user's RSA public key.
- **Immediate Revocation:** Revoking access removes the user's wrapped key from the database, preventing future item access.

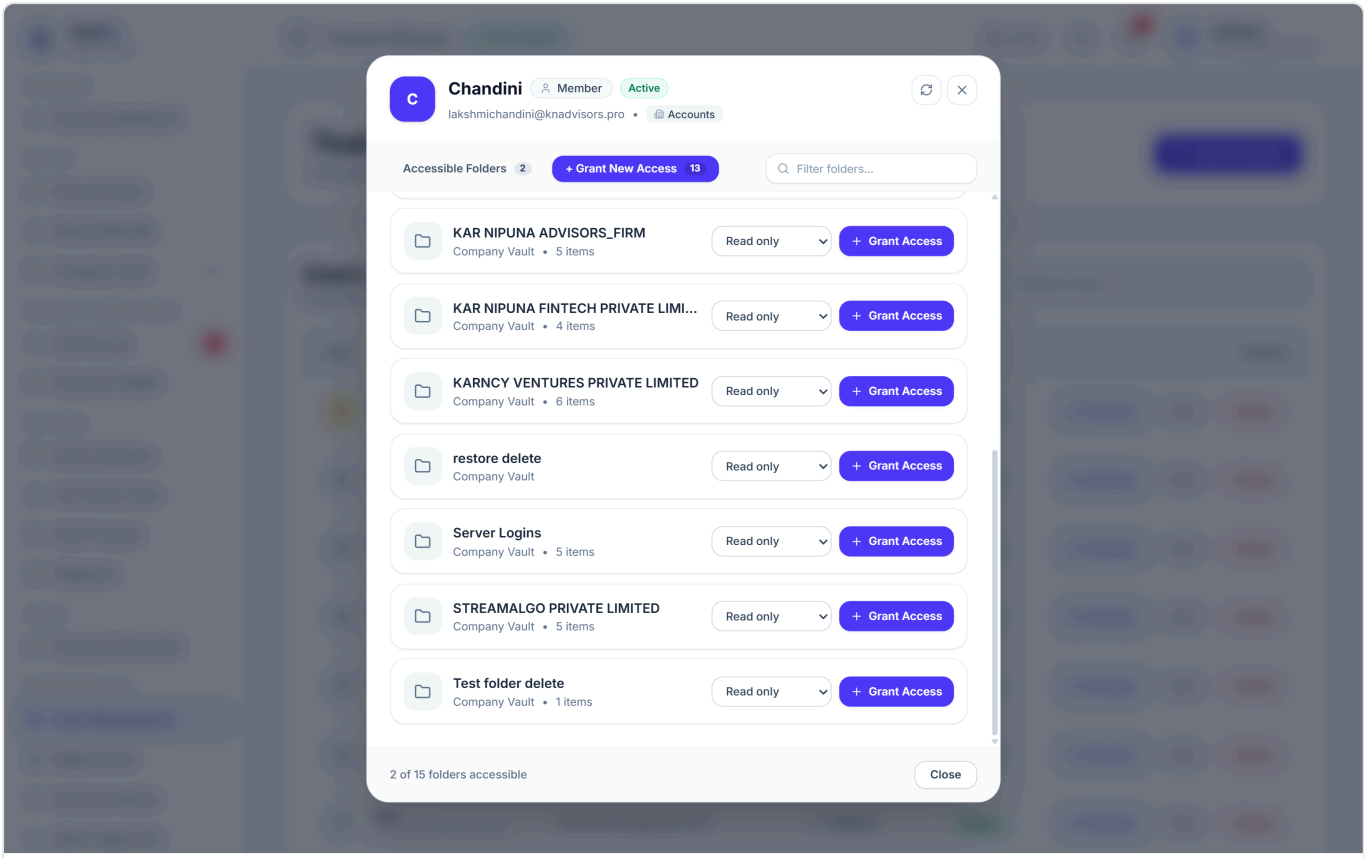

Figure 34: Grant Folder Access Modal (`Modal`). Interface for assigning shared folder permissions directly to a specific user with automated RSA key wrapping.

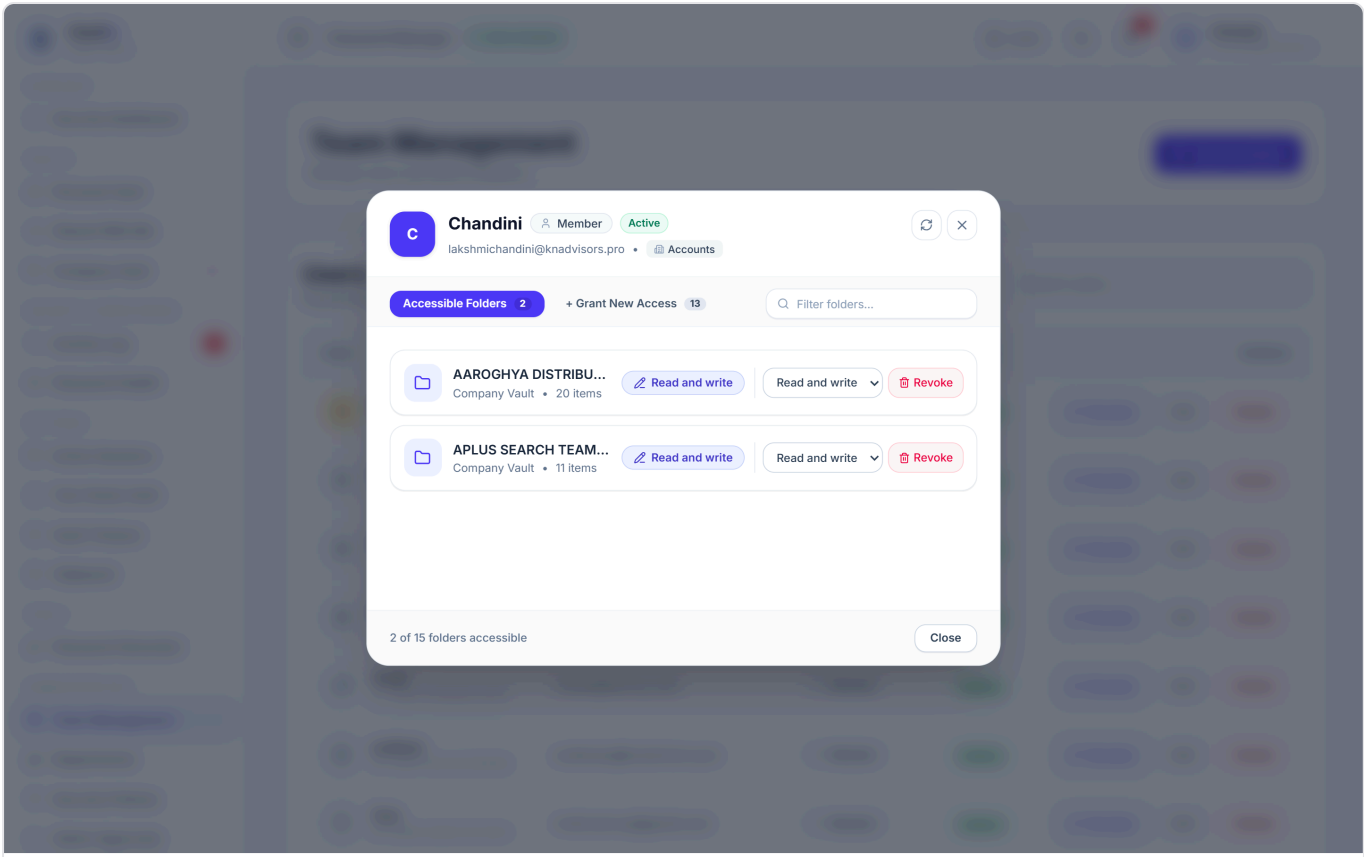

Figure 35: User Accessible Folders Modal (`Modal`). Audit view listing all shared folders accessible by a specific employee, with quick-revoke controls.

9. Sensitive Items & Admin Approvals

For high-value credentials (e.g. production database passwords, root credentials), Vaultix supports an access approval workflow:

- **Sensitive Flag (isSensitive):** Items marked as sensitive require administrator approval before plaintext passwords can be viewed or copied.
- **Access Request Form:** Users submit an access request providing a business justification.
- **Admin Approvals Portal:** Administrators review pending requests and approve or reject them with audit logging.
- **Status Indicator:** The top navigation bar displays a counter of pending access requests.

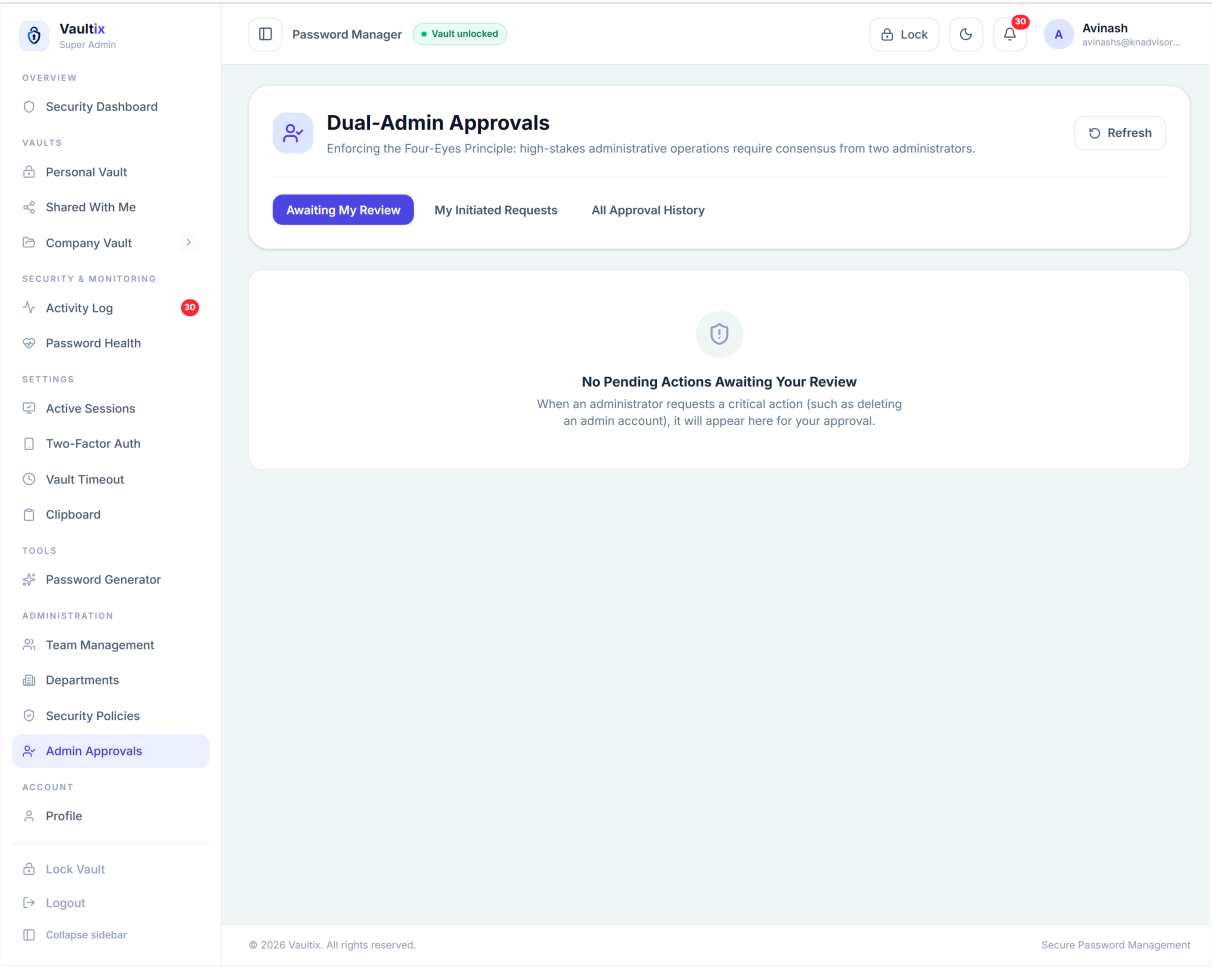

Figure 46: Pending Sensitive Access Approvals (`/approvals (Pending Tab)`). Approval queue where requests to view or copy sensitive credentials await administrator review.

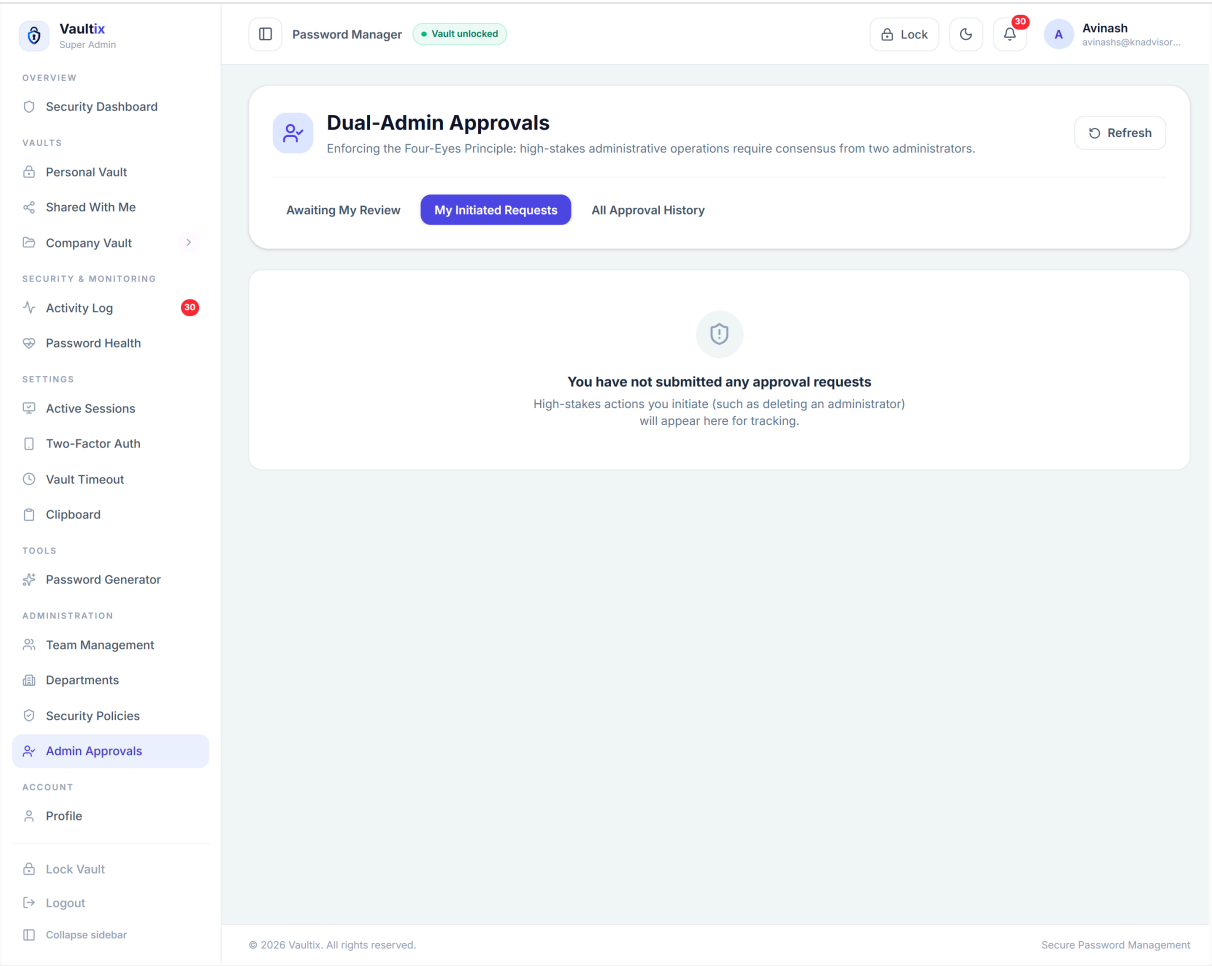

Figure 47: My Access Requests Tab (`/approvals (My Requests)`). Status view showing access requests submitted by the current user (Pending, Approved, Denied, Expired).

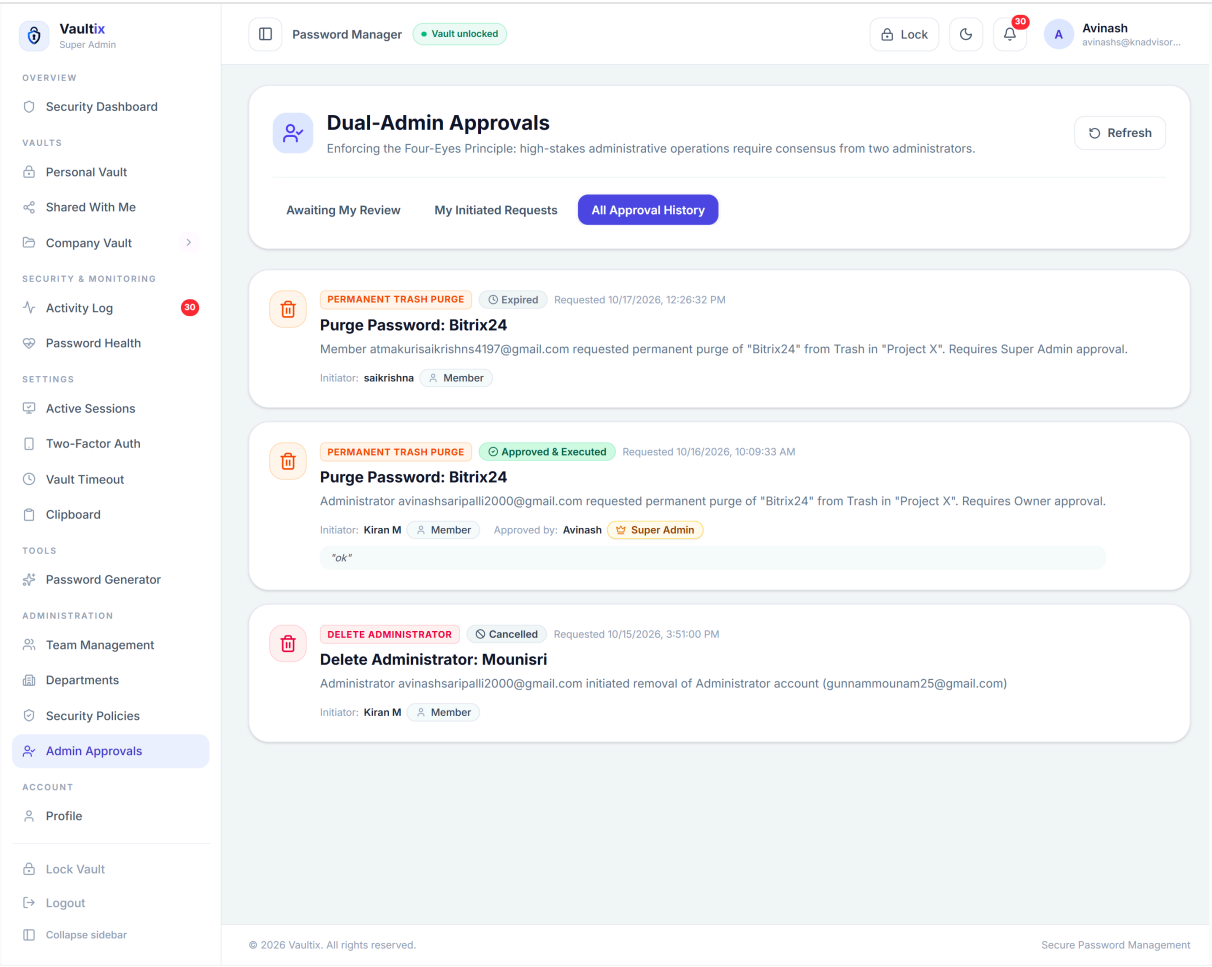

Figure 48: Access Request History Tab (`/approvals (History)`). Audit log of all sensitive credential access requests, showing reviewer decisions, timestamps, and justifications.

10. Password Health & Breach Auditing

Vaultix audits credentials stored in vaults using client-side checks and privacy-preserving breach detection:

- **Weak Passwords:** Identifies passwords under 8 characters or with low entropy.
- **Reused Passwords:** Flags credentials used across multiple accounts.
- **Aging Credentials:** Flags passwords that have remained unchanged for over 90 days.
- **Breach Detection (k-Anonymity):** Hashes passwords with SHA-1 and sends only the first 5 characters of the hash to the HaveIBeenPwned API. Full passwords or full hashes never leave the client.
- **Health Score:** Calculates an overall security hygiene score (0–100%) displayed on the dashboard.

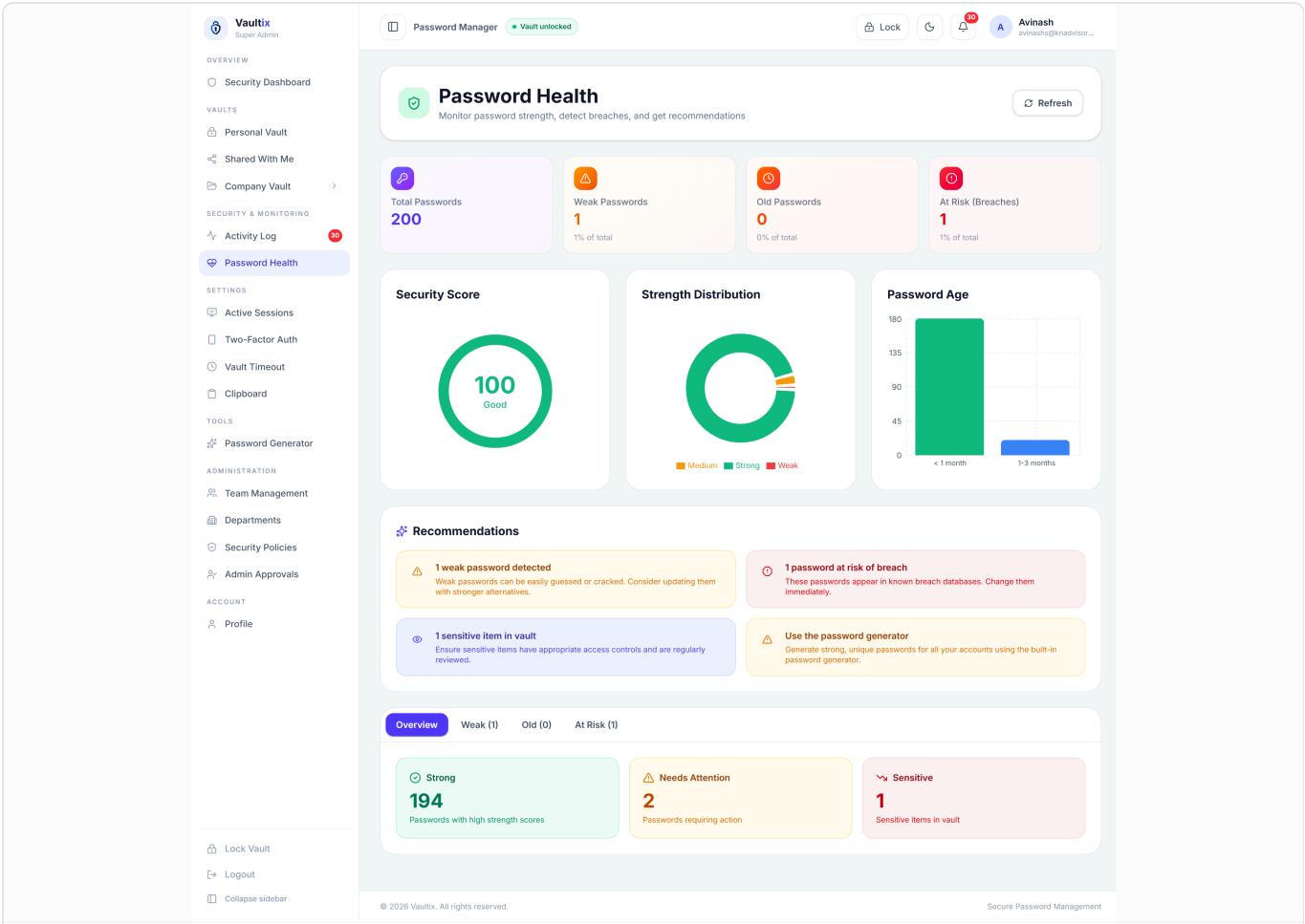

Figure 24: Password Health Audit (`/security/health`). Client-side password analysis tool evaluating password strength, detecting reused credentials, and flagging breach exposure using k-anonymity checks.

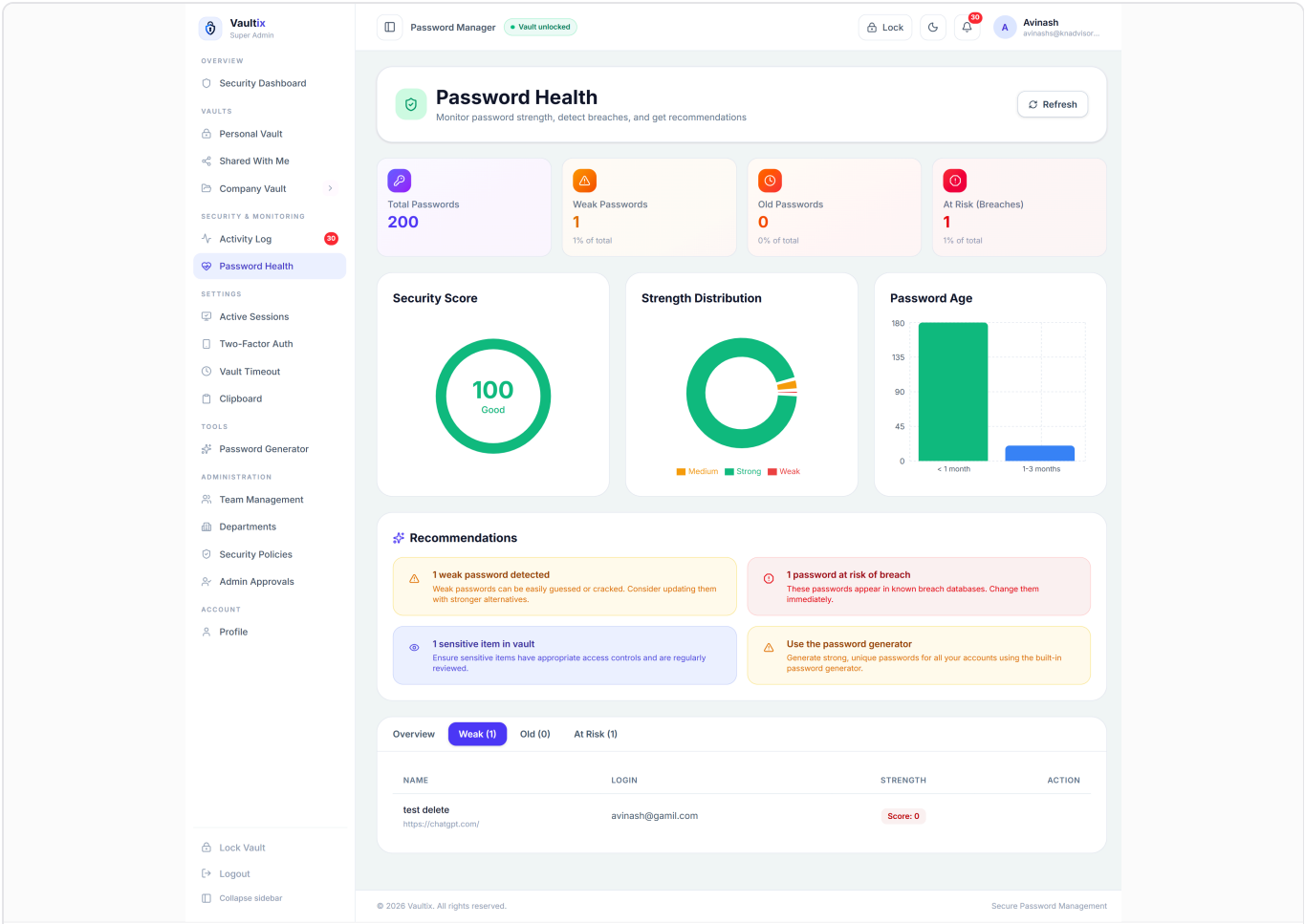

Figure 25: Weak Passwords Report (`/security/health (Weak Tab)`). Filtered list of credentials that do not meet length or complexity requirements.

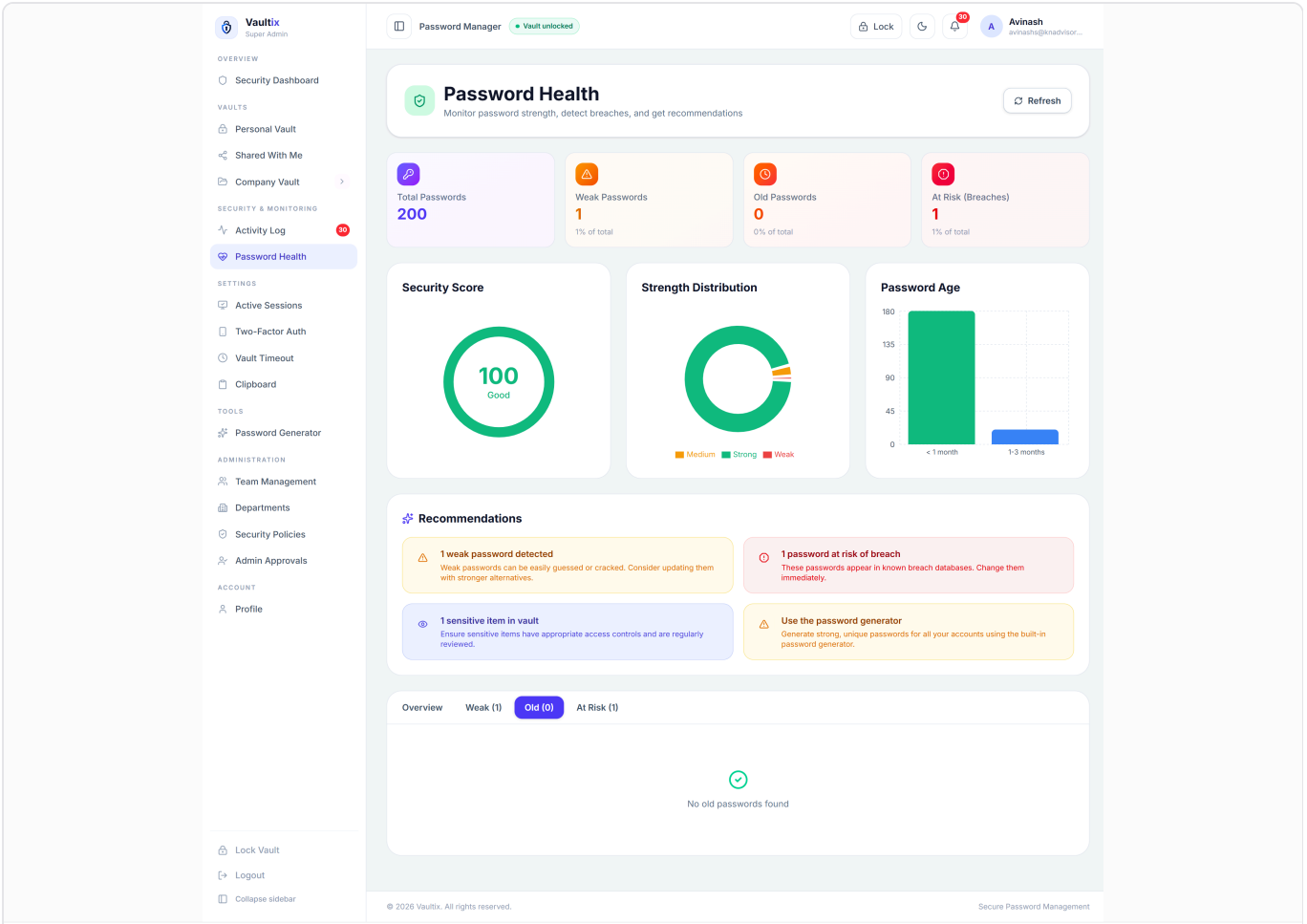

Figure 26: Aging Passwords Report (`/security/health (Old Tab)`). Filtered list of credentials that have not been rotated in over 90 days.

11. Password Generator

Vaultix includes a cryptographic pseudorandom password generator using `crypto.getRandomValues` :

- **Adjustable Length:** Configurable from 8 to 64 characters.
- **Character Sets:** Uppercase (A–Z), lowercase (a–z), numbers (0–9), and symbols (!@#$%^&*).
- **Ambiguous Characters:** Option to exclude visually similar characters (e.g. 0, O, 1, l, I).
- **Entropy Rating:** Displays real-time bit strength and quality rating (Weak to Very Strong).
- **Availability:** Accessible in item forms, the top navigation bar, and the browser extension.

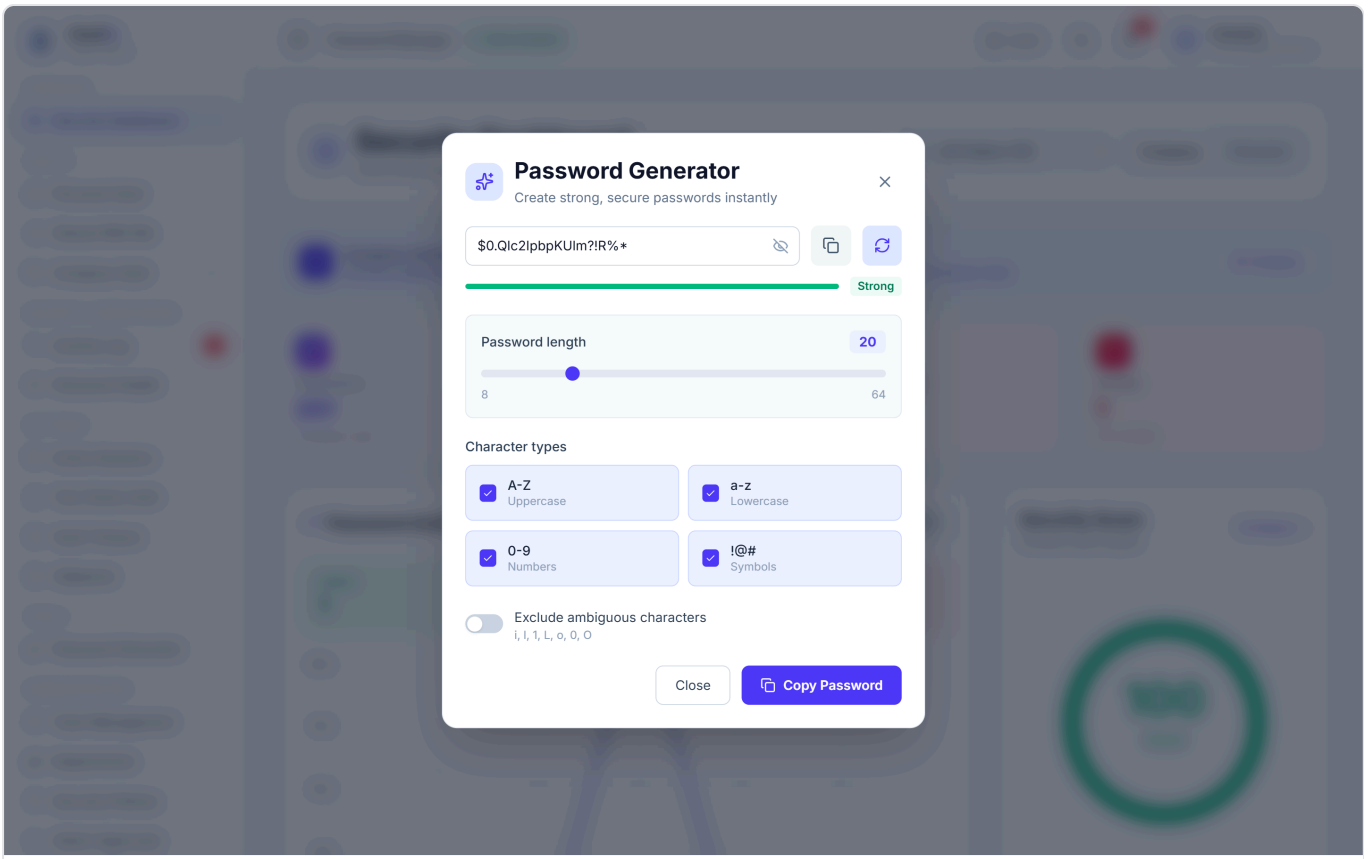

Figure 7: Password Generator (`/generator`). Cryptographic random password generator (crypto.getRandomValues). Supports configurable length (8–64 characters), character set toggles, and ambiguous character filtering.

12. Security Dashboard

The dashboard (`/dashboard`) provides high-level security metrics for team leads and administrators:

- **Security Score:** Radial score gauge reflecting overall credential hygiene.
- **Inventory Metrics:** Total password counts, company passwords, personal passwords, and items in trash.
- **Activity Charts:** Visual charts showing password creation, update, and deletion activity over 7 days, 30 days, or 6 months.
- **Recent Credentials:** Quick-access list of the 5 most recently created or modified credentials.

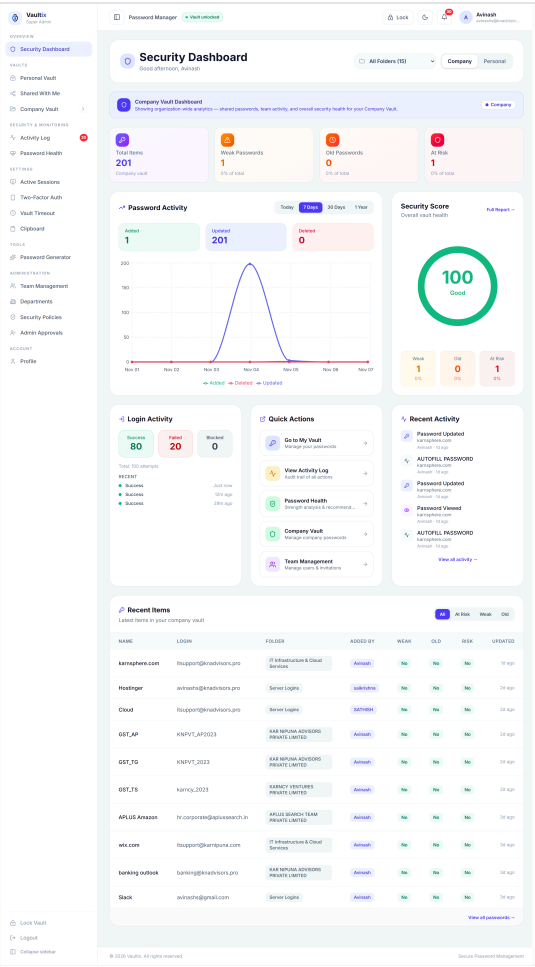

Figure 6: Security Dashboard (`/dashboard`). Overview dashboard showing overall password health score, vulnerable/reused password counts, credential totals, and recent vault activity.

13. Activity Logging & Audit Trail

Vaultix maintains an operational audit log (`/activity-log`) recording security events:

- **Event Types:** `PASSWORD_VIEW` , `PASSWORD_COPY` , `PASSWORD_CREATE` , `PASSWORD_UPDATE` , `PASSWORD_DELETE` , `FOLDER_CREATE` , `ACCESS_GRANTED` , `ACCESS_REVOKED` , `USER_LOGIN` , `POLICY_CHANGE` .
- **Audit Metadata:** User ID, email, IP address, user agent, target vault, folder ID, item title, and timestamp.
- **Filters:** Filter records by event type, specific user, vault, or date range.
- **Personal Privacy:** Events inside personal vaults are visible only to the owner and excluded from organizational logs.

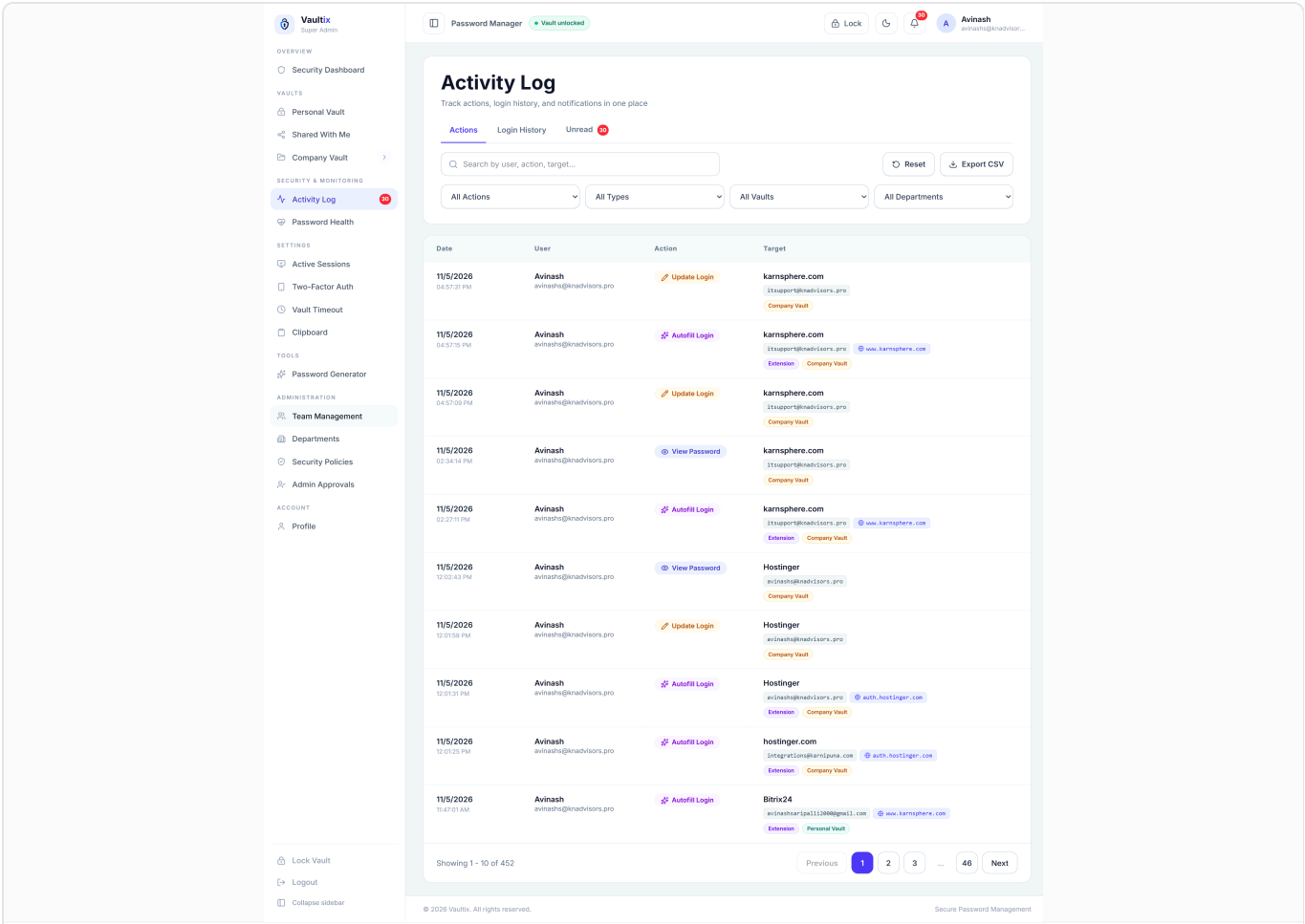

Figure 20: System Activity Log (`/activity-log`). Audit log recording vault operations including credential views, copies, exports, permission changes, and user logins with timestamps and IP addresses.

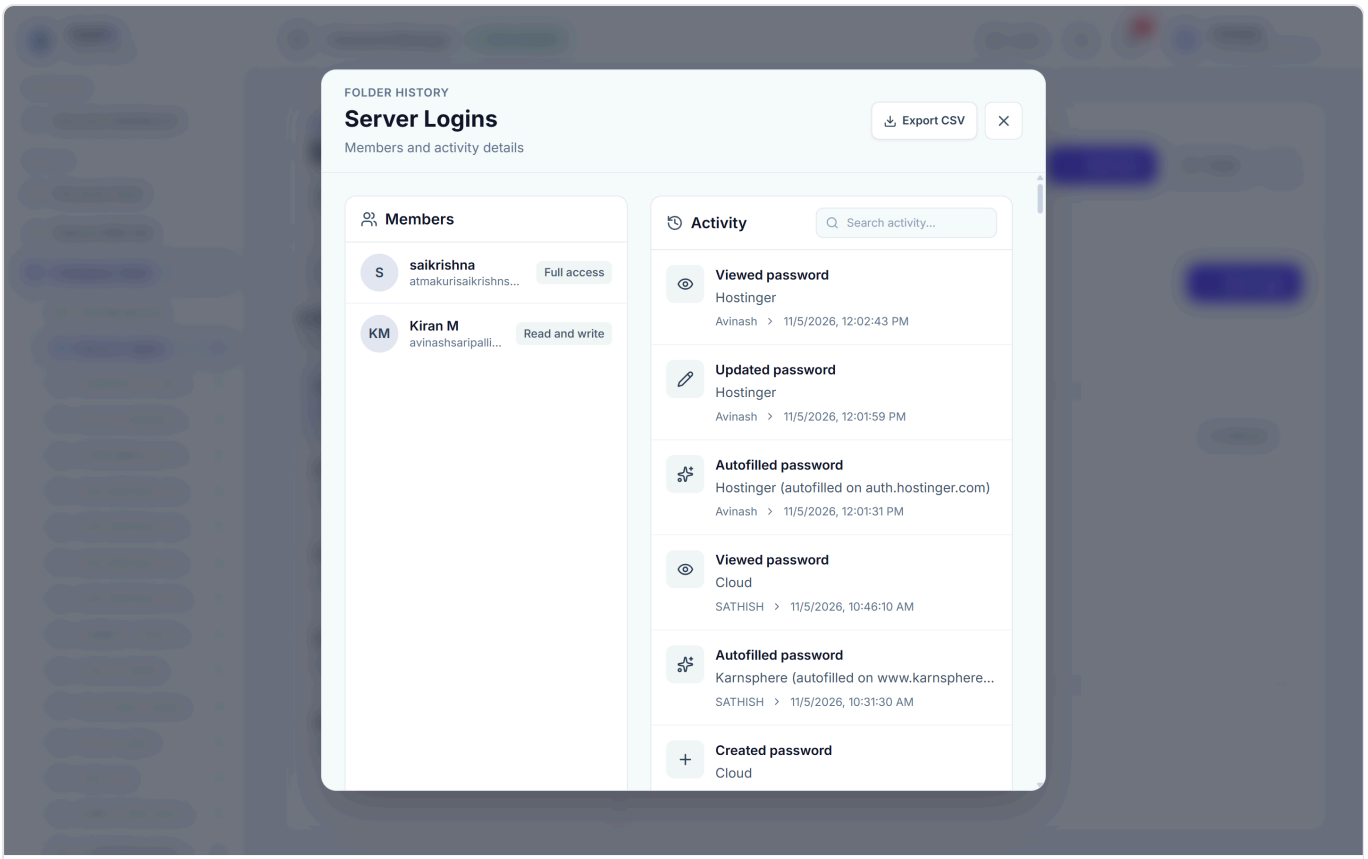

Figure 16: Folder Audit History (`Modal`). Change log recording credential updates, permission changes, additions, and deletions within the selected folder.

14. Login Activity & Device History

Tracks authentication attempts across the organization:

- **Parameters:** Records client IP address, browser user agent, operating system, timestamp, and status (`SUCCESS` , `FAILED` , `BLOCKED`).
- **Lockout Protection:** Accounts are locked for 15 minutes after 5 consecutive failed login attempts.
- **Deactivated Accounts:** Login attempts against deactivated user accounts are immediately blocked and logged.

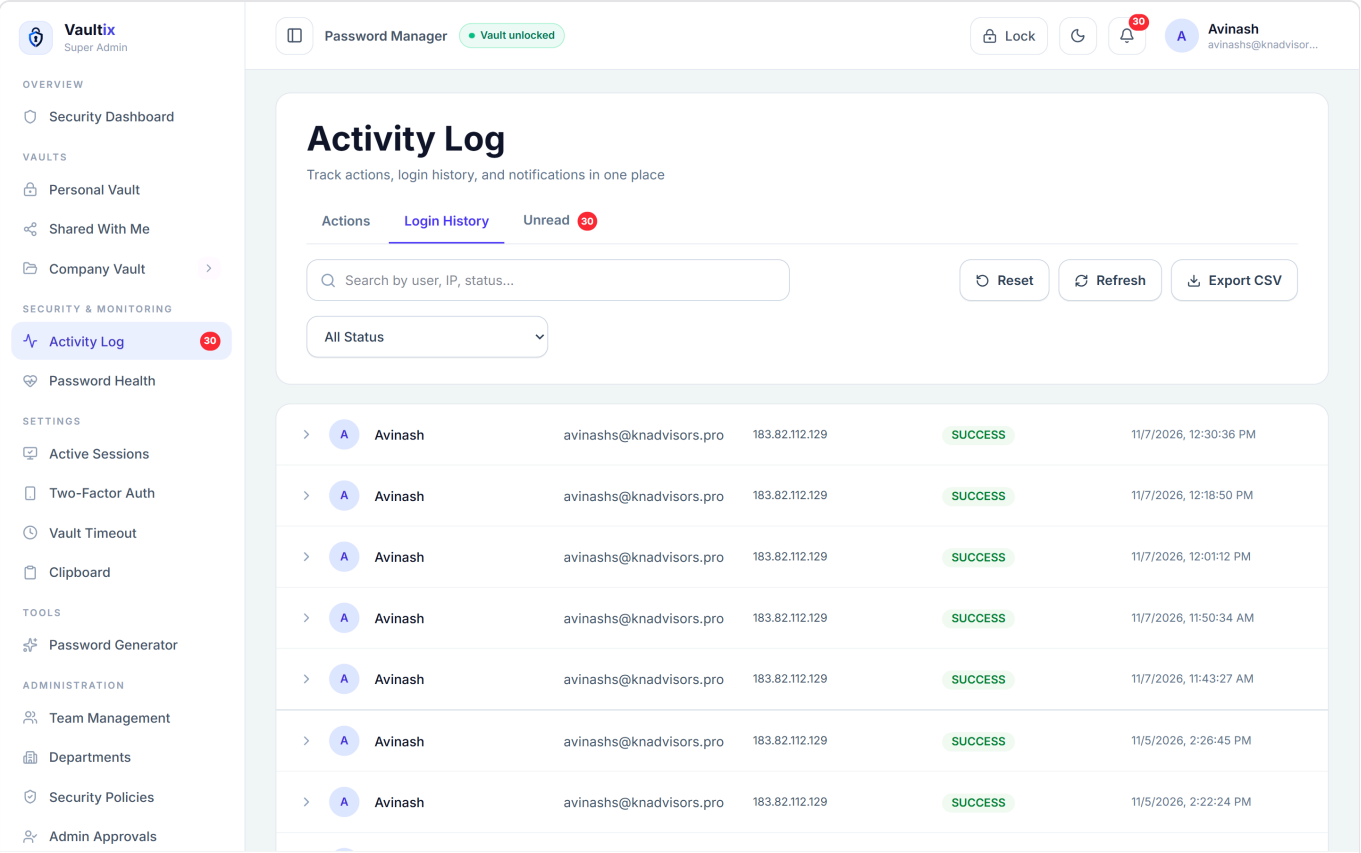

Figure 21: Login Activity History (`/activity-log (Login History)`). Authentication audit trail recording login timestamps, client IP addresses, browser user agents, operating systems, and MFA verification outcomes.

15. Two-Factor Authentication (MFA & Passkeys)

Vaultix supports two-factor authentication for user accounts:

- **TOTP Authenticator Apps:** Compatible with Google Authenticator, Microsoft Authenticator, and standard RFC 6238 apps.
- **Emergency Backup Codes:** Generates 10 single-use recovery codes for account access during phone loss.
- **FIDO2 / WebAuthn Passkeys:** Supports passwordless authentication using biometric sensors (Touch ID, Windows Hello) and physical security keys (YubiKey).
- **Passkey Management:** Register, label, and revoke individual hardware passkeys from security settings.

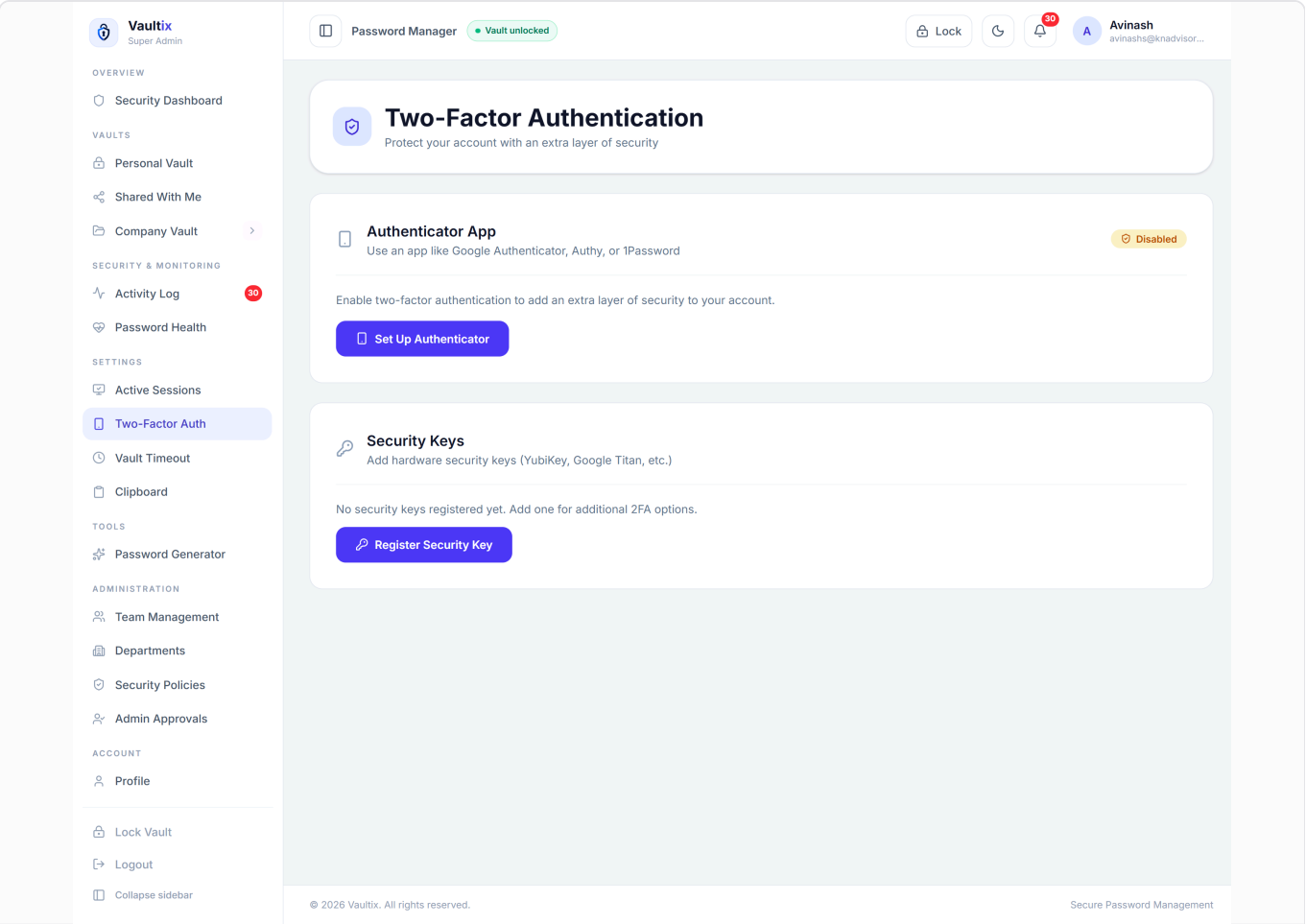

Figure 28: Two-Factor Authentication Settings (`/security/2fa`). Two-factor configuration portal supporting authenticator applications (TOTP) and FIDO2/WebAuthn hardware passkeys.

16. Session Management & Auto-Lock

Enables users and administrators to monitor and manage active sessions:

- **Active Session List:** Shows device, browser, operating system, IP address, and last active timestamp for each login.
- **Remote Revocation:** Terminate individual sessions with immediate JWT invalidation.
- **Revoke All Other Sessions:** One-click option to log out all other active sessions across devices.

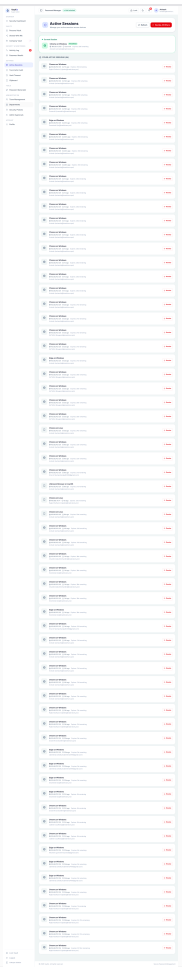

Figure 27: Active Device Sessions (`/security/sessions`). Session manager showing active logins, client browsers, IP addresses, and last active times, with options to revoke individual or all other sessions.

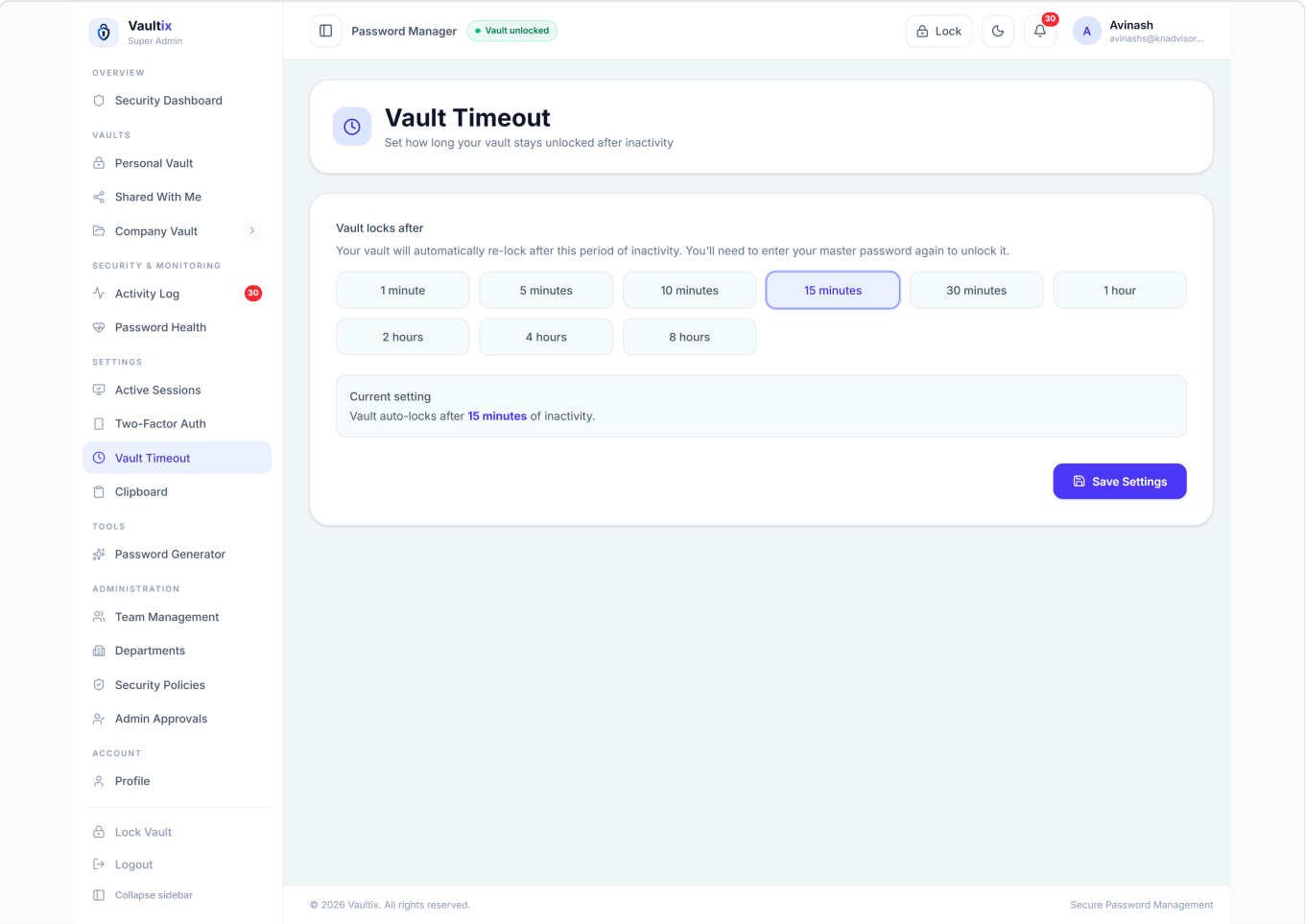

Figure 30: Vault Inactivity Timeout Settings (`/security/timeout`). Inactivity auto-lock configuration for automatically clearing decrypted keys from browser memory after 1, 5, 15, or 60 minutes, or upon browser close.

17. Security Policies

Vaultix supports two tiers of policy enforcement: Organization-Wide Policies and Vault-Specific Policies.

17.1. Organization-Wide Policies

- **Enforce Two-Factor Authentication:** Requires all members to configure 2FA before accessing company vaults.
- **Master Password Complexity:** Minimum length requirement (12–16 characters) and special character rules.
- **Default Timeout:** Organization-wide vault inactivity lock timer.

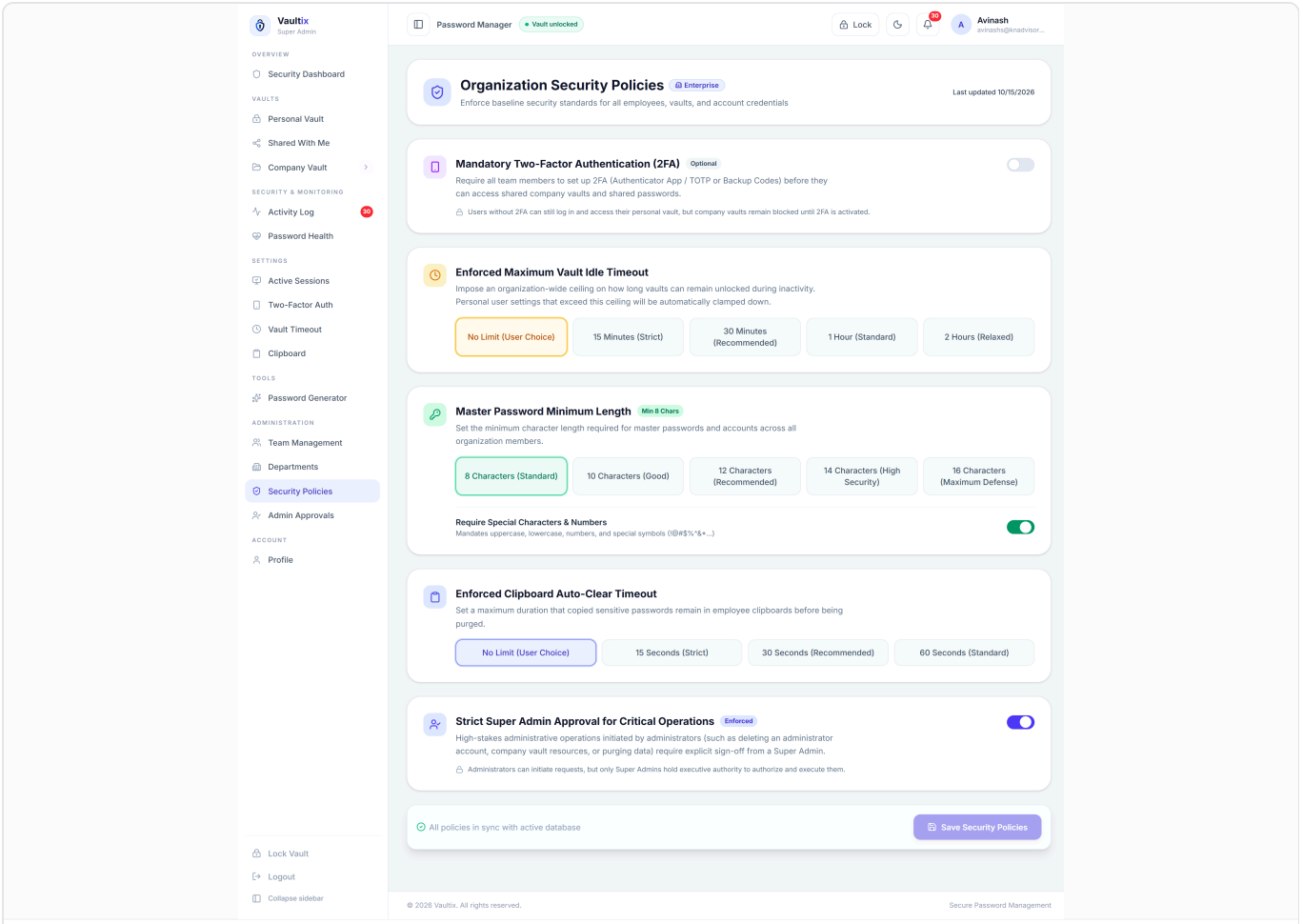

Figure 45: Organization Security Policies (`/admin/policies`). Global policy controls enforcing minimum master password length, mandatory 2FA, session timeout limits, and clipboard auto-clear intervals.

17.2. Vault-Specific Policies

- **Minimum Strength Score:** Sets minimum required password complexity score for items saved in the vault.
- **Maximum Password Age:** Sets credential rotation periods (e.g. 60 or 90 days), flagging stale credentials.
- **Common Password Filter:** Blocks known common passwords from being saved.
- **Allowed Item Types:** Restricts vaults to specific schemas (e.g. only SSH keys and database credentials).

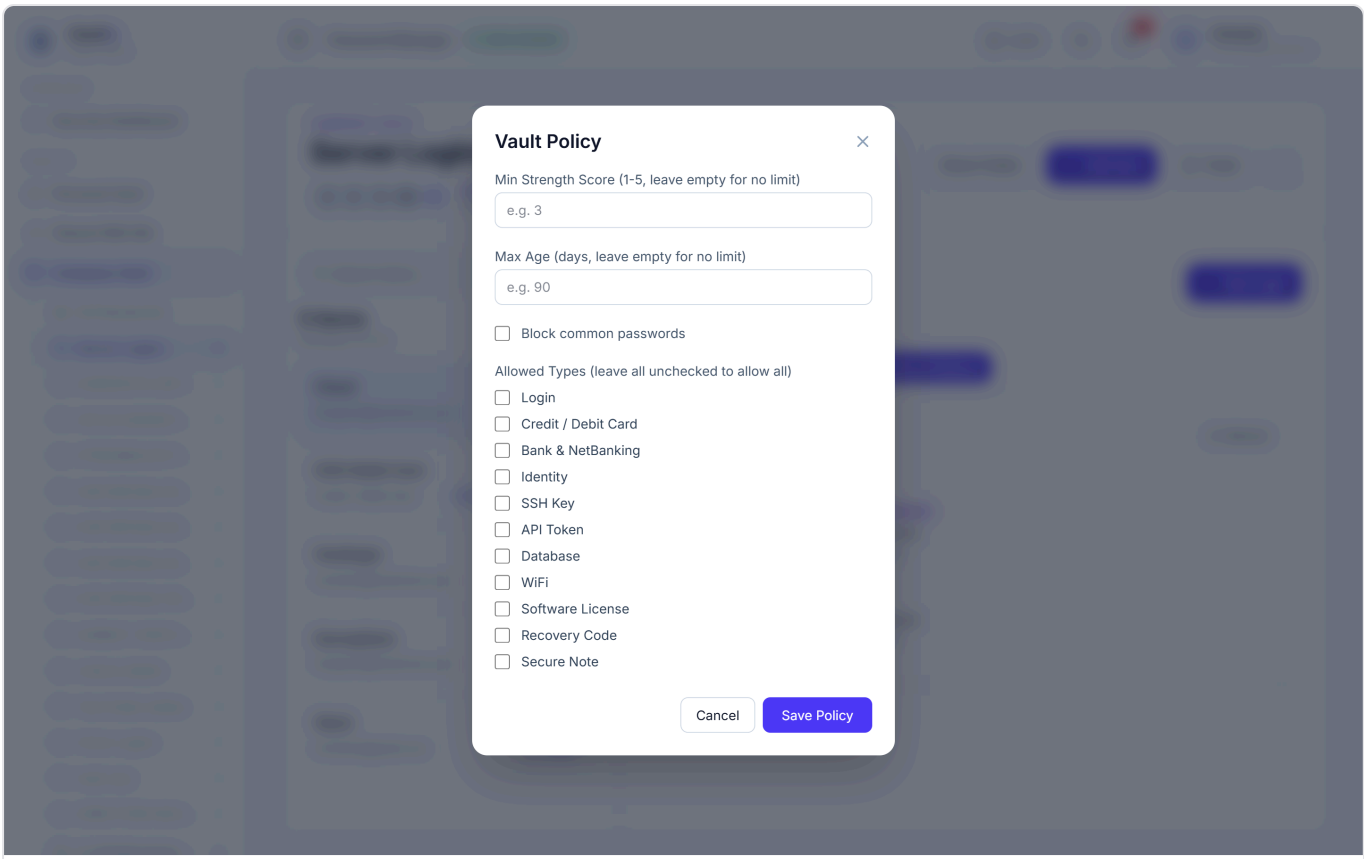

Figure 17: Vault Security Policy Settings (`Modal`). Folder-level policy configuration enforcing minimum password strength, maximum credential age, and allowed item schemas.

18. Clipboard Security

Protects copied credentials from remaining indefinitely in the operating system clipboard:

- **Configurable Auto-Clear:** Automatically clears copied passwords from the clipboard after 10, 30, 60, or 120 seconds.
- **Toast Notification:** Displays a notification when the clipboard has been cleared.

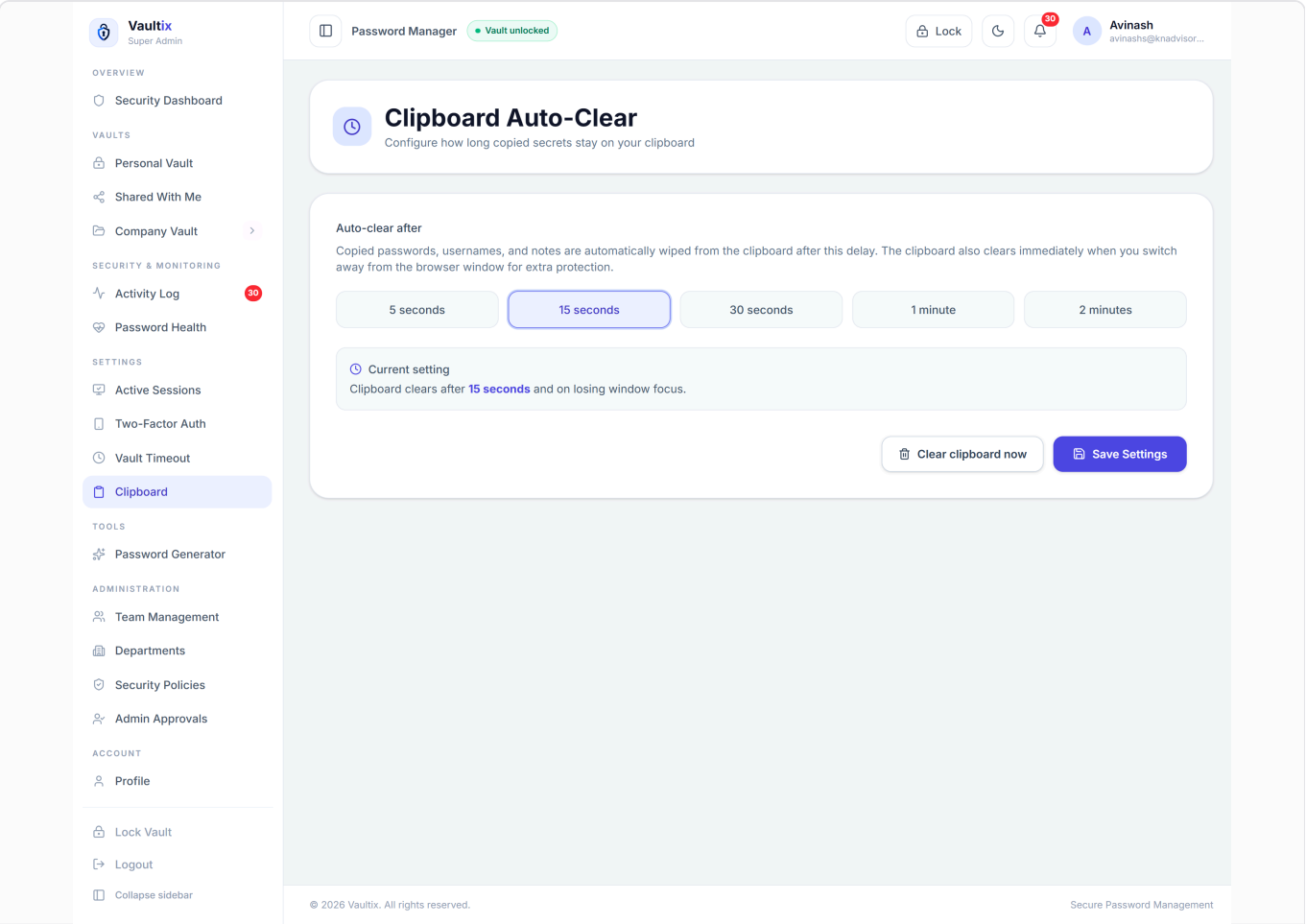

Figure 31: Clipboard Auto-Clear Settings (`/security/clipboard`). Clipboard security settings for automatically clearing copied credentials from the operating system clipboard after 10, 30, 60, or 120 seconds.

19. Bulk CSV & Excel Import/Export

Vaultix supports credential migration and backups via Excel (`.xlsx`) and CSV:

- **Import Formats:** Ingests exports from Dashlane, Bitwarden, KeePass, LastPass, and standard CSV templates.
- **Automatic Type Mapping:** Maps external credential types to native Vaultix schemas during import (e.g. server records map to Login items with host URLs).
- **Client-Side Encryption:** Imported files are parsed in browser memory. Items are encrypted client-side using AES-GCM and RSA key wrapping before sending to the backend. Plaintext files never touch the server.
- **Vault Export:** Decrypted vault records can be exported into structured spreadsheets for offline disaster recovery.

20. Trash & Deletion Lifecycle

Vaultix uses a two-stage deletion lifecycle to prevent accidental data loss:

- **Soft-Delete (Move to Trash):** Deleting a credential marks it with a deletion timestamp. The item is hidden from active views but remains in the Trash.
- **Item Restoration:** Restore soft-deleted items back to their original folder with one click.

- **Permanent Purge:** Completely deletes items from the database. Requires administrative privileges in shared vaults.

Figure 11: Personal Vault Trash (`/vault/personal/trash`). Recycle bin for soft-deleted personal credentials. Items can be restored to their original folder or permanently purged.

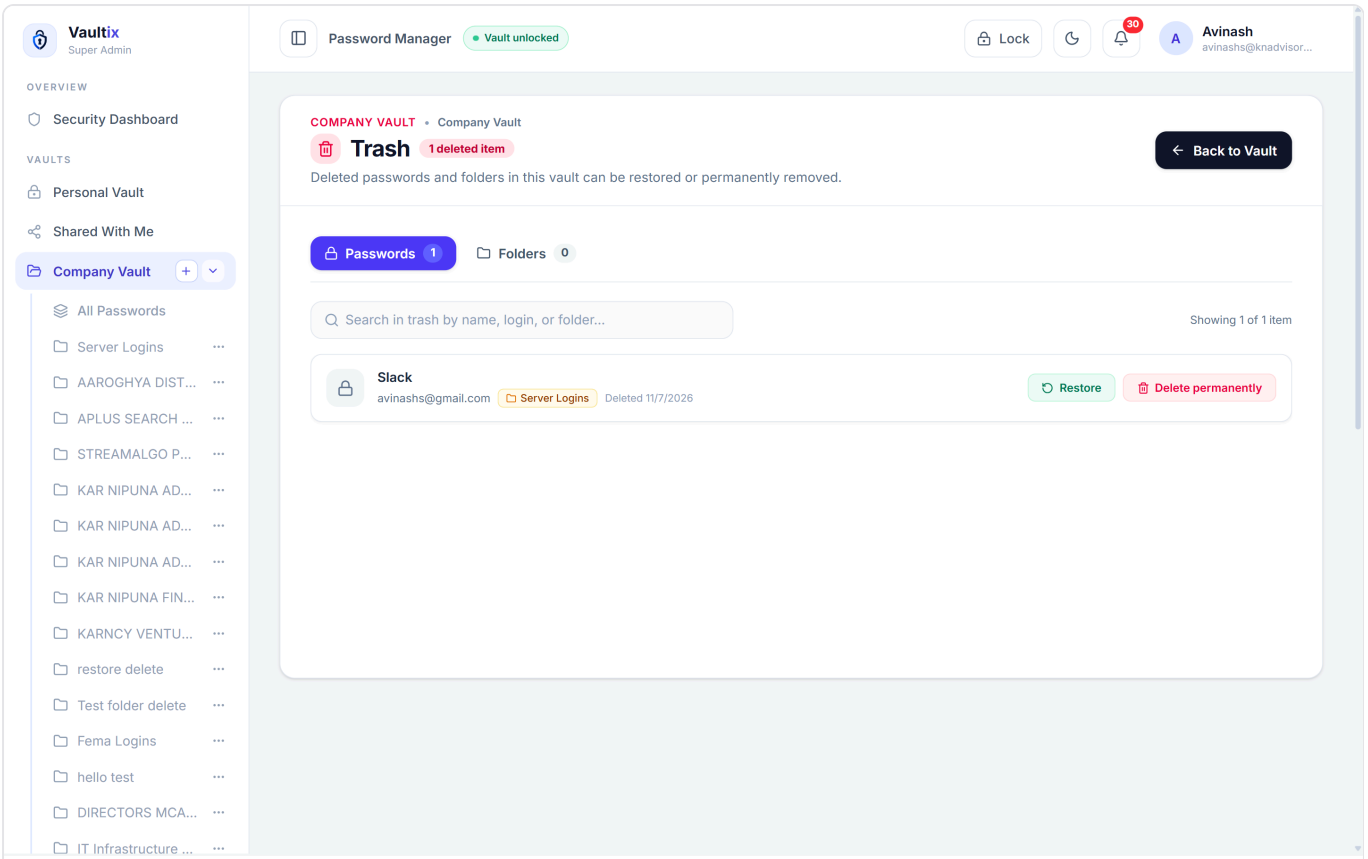

Figure 19: Company Vault Trash (`/vault/company/trash`). Shared repository for soft-deleted company credentials. Items can be restored by folder managers or permanently purged by administrators.

21. Team Administration

Administrators manage team member accounts in Team Management (`/team`):

- **User Provisioning:** Create accounts directly with assigned roles (User, Admin).
- **Email Invitations:** Send invitations with 7-day registration links.
- **Account Deactivation:** Instantly suspend user access and revoke active sessions.
- **Search & Filter:** Filter team members by name, email, role, department, or active status.

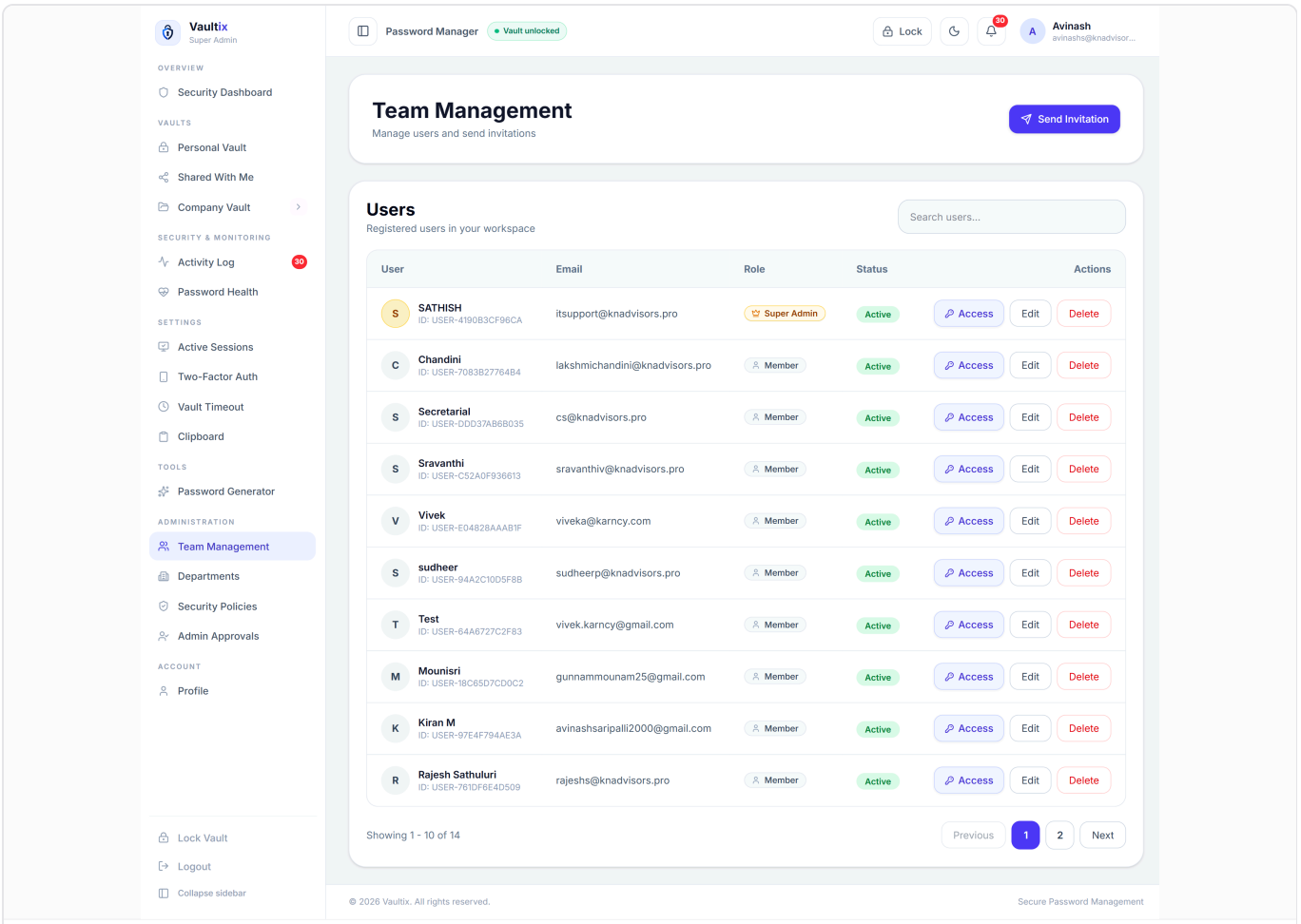

Figure 32: Team Directory & User Roles (`/team`). Administrator console for managing team members, system roles (Super Admin, Admin, User), account statuses, and department assignments.

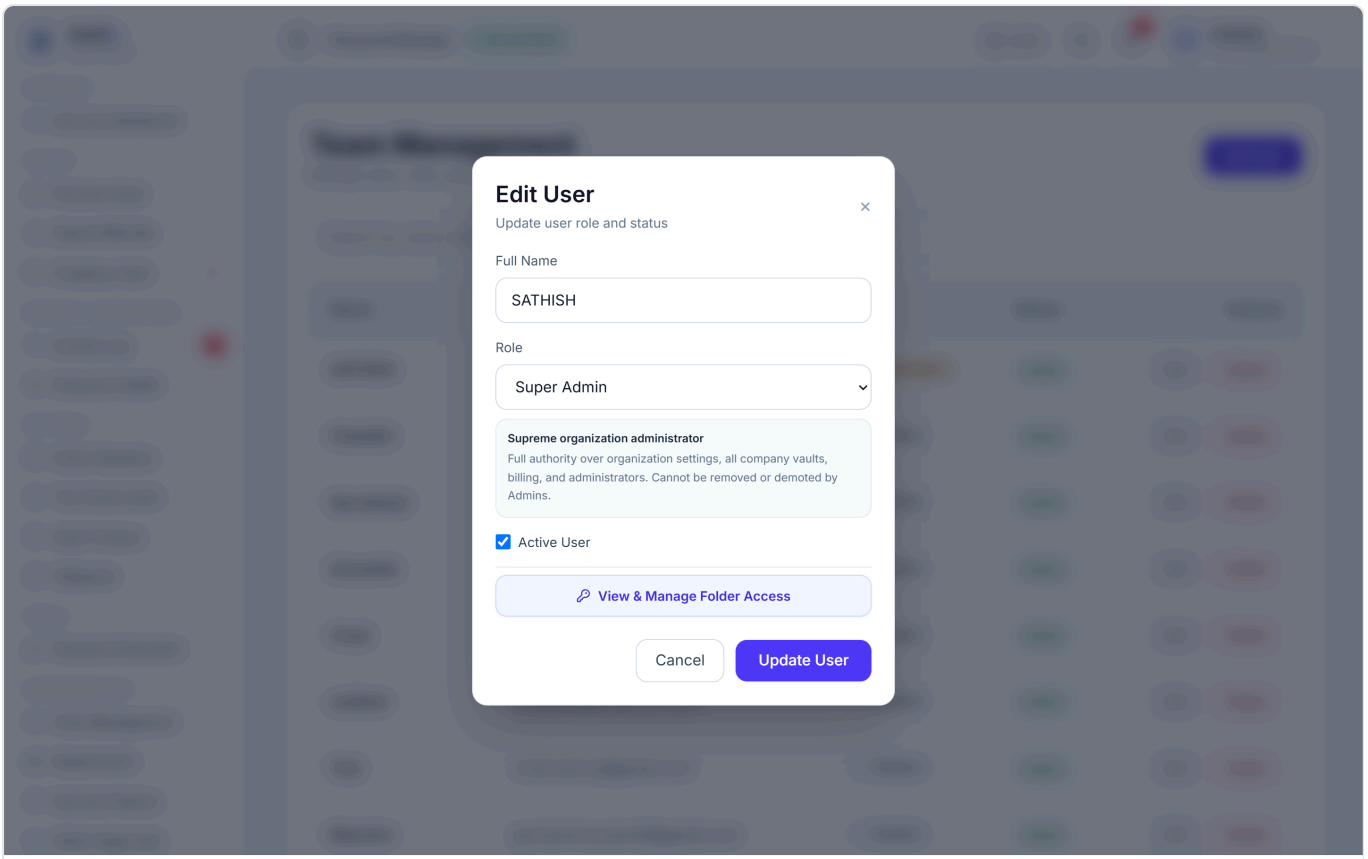

Figure 33: Edit User Modal (`Modal`). Dialog for updating team member details, changing system roles, and reassigning departments.

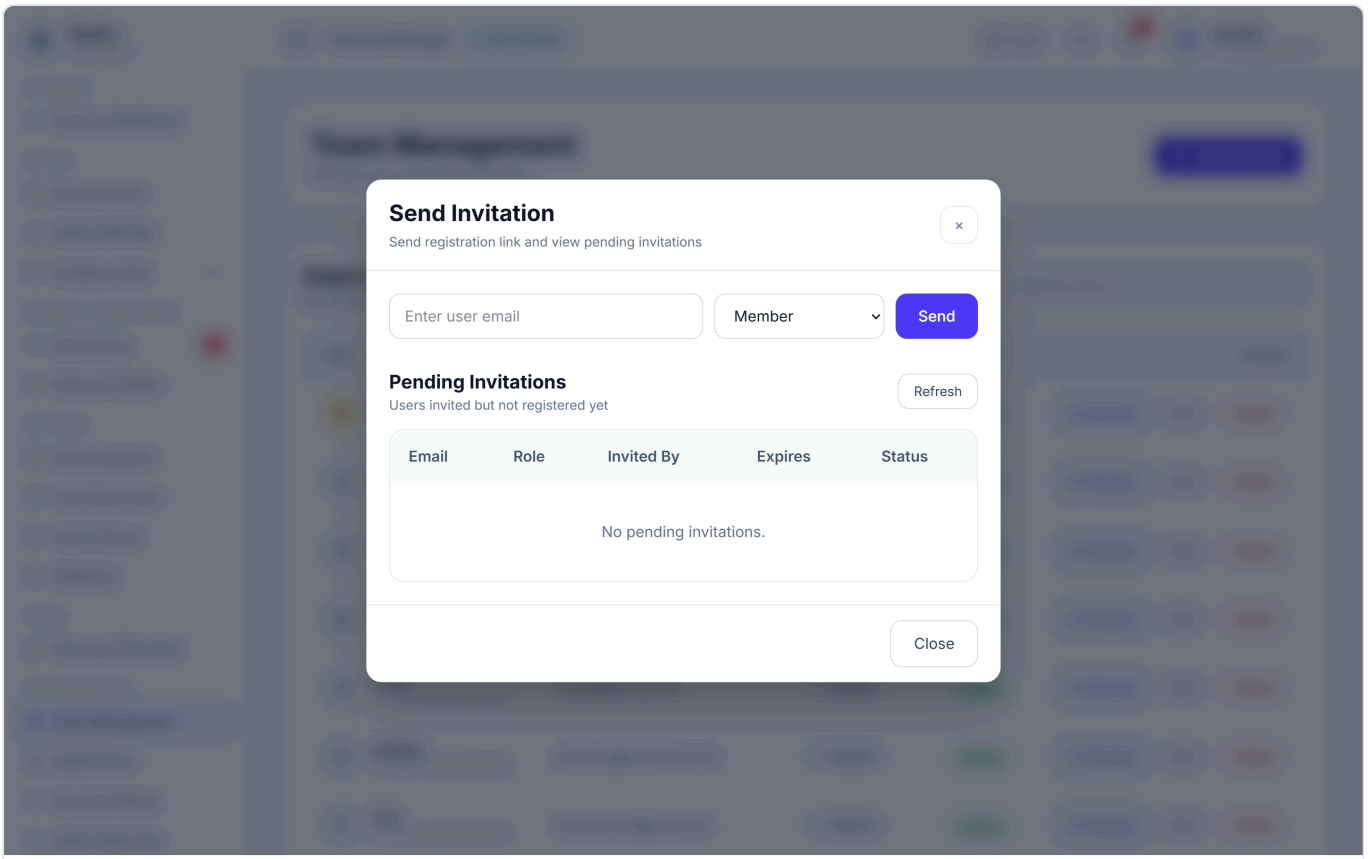

Figure 36: Send Invitation Modal (`Modal`). Dialog for sending email onboarding invitations with pre-configured roles and department assignments.

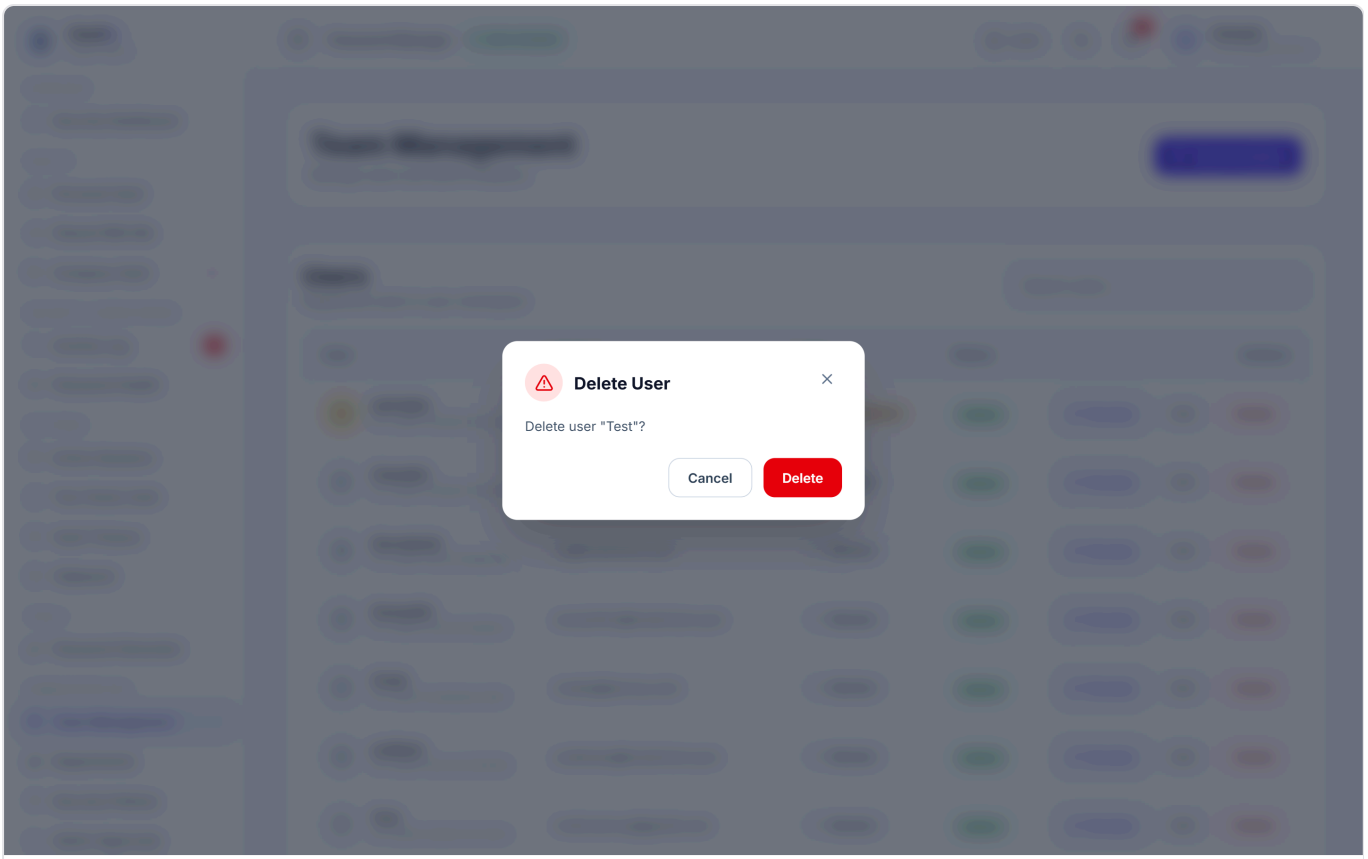

Figure 37: Delete User Confirmation Dialog (`Dialog`). Confirmation prompt for deprovisioning user accounts and revoking their access to shared folder keys.

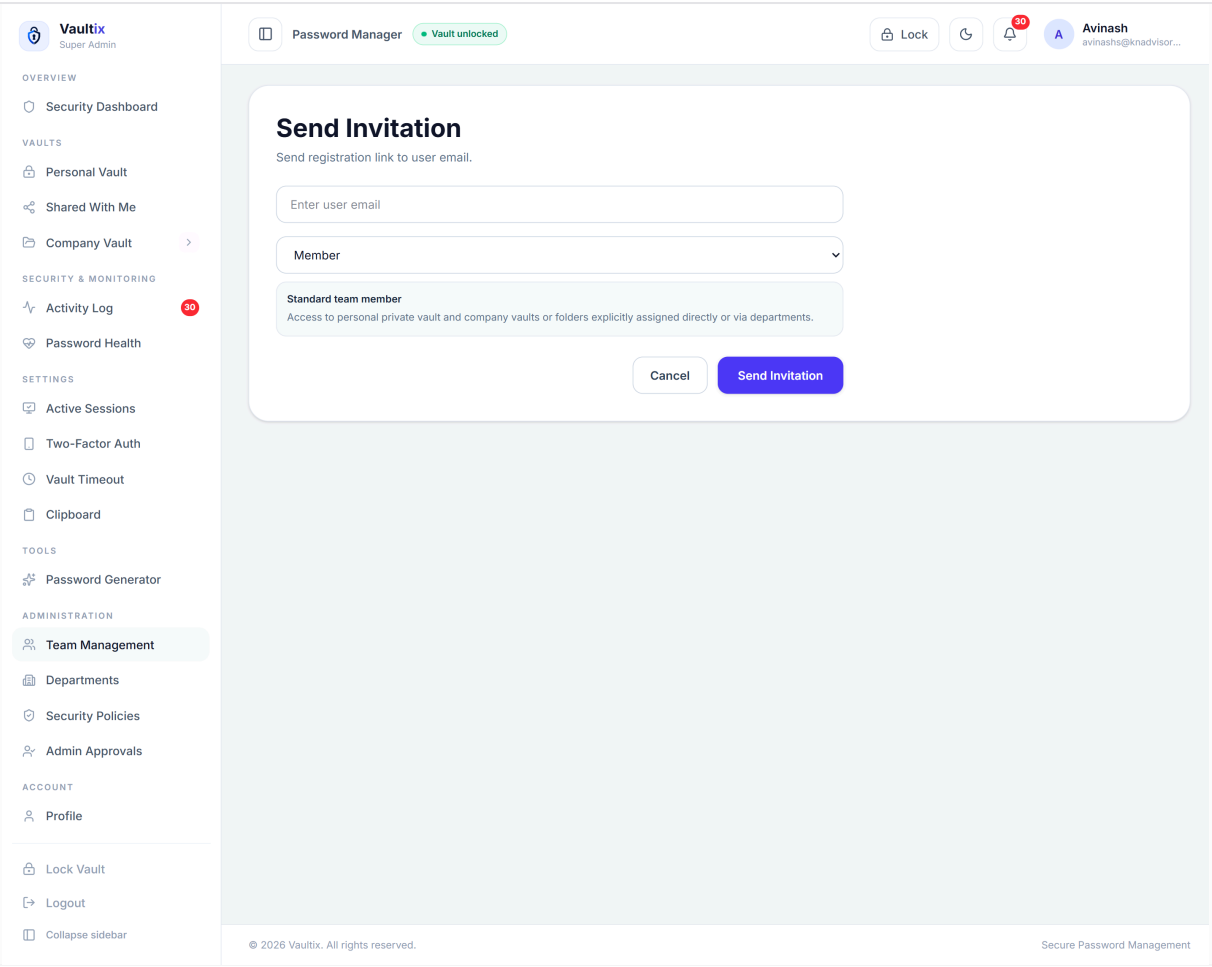

Figure 38: Member Invitation Page (`/team/invite`). Full-page form for inviting new employees, selecting their initial role, and assigning them to departments.

22. Notifications & Preferences

- **Notification Drawer:** Alerts users to shared folder invitations, policy changes, and access approvals.
- **Mark All Read:** Clears unread notification counters.
- **Dark & Light Themes:** Full theme support with preferences saved to user account settings.

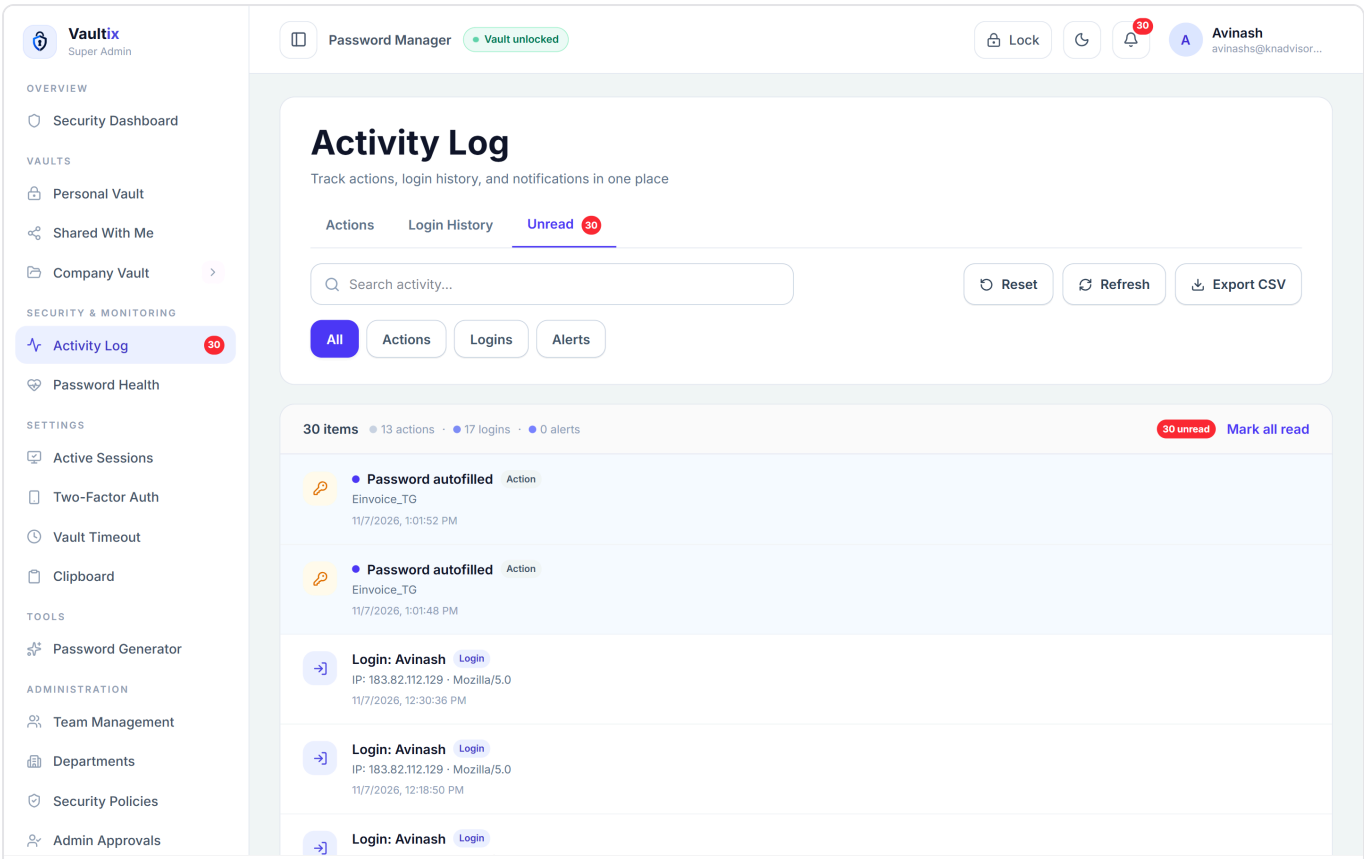

Figure 22: Notifications Center (`/activity-log (Notifications)`). Central notification feed displaying shared folder invitations, security policy changes, and pending access requests.

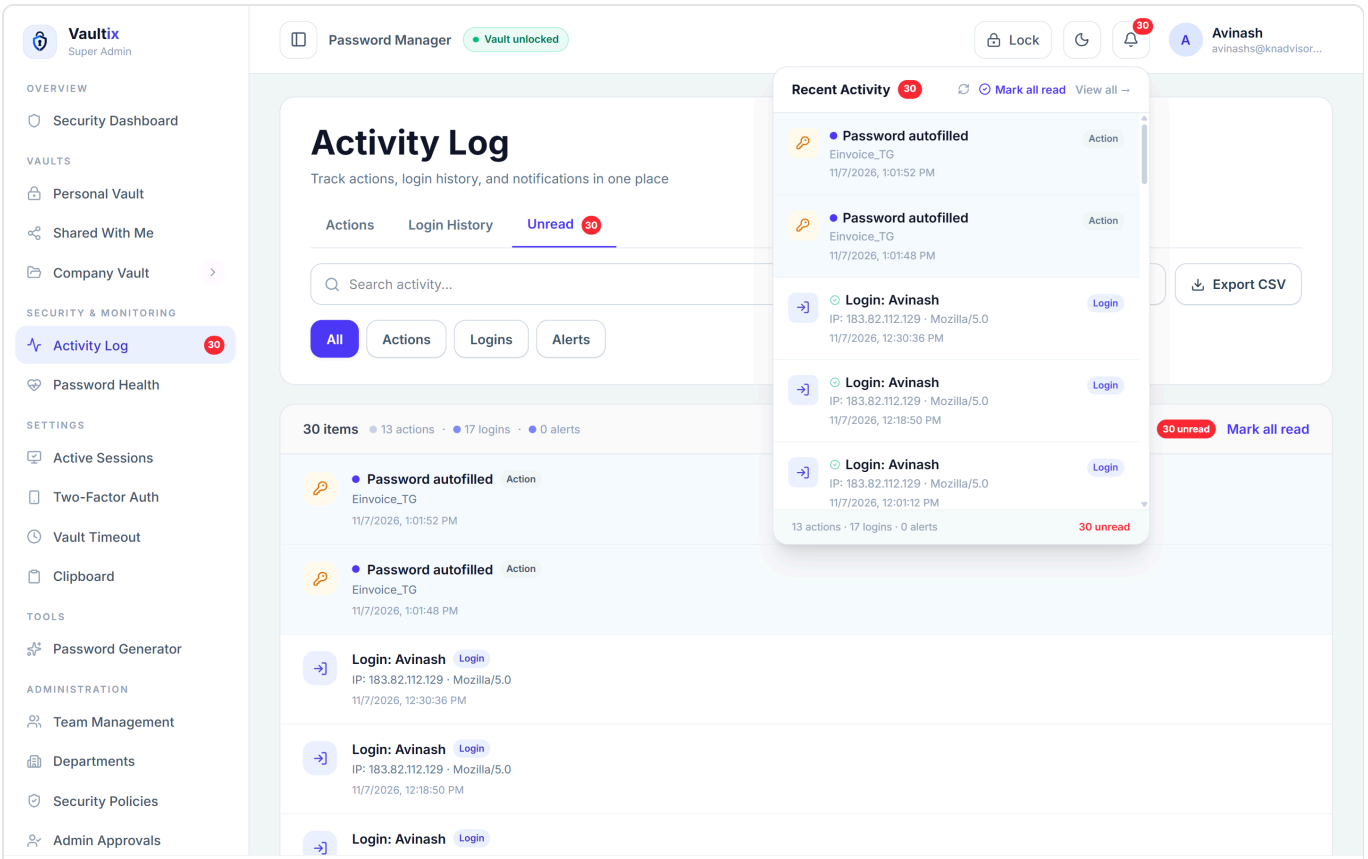

Figure 23: Header Notification Menu (`Header Widget`). Quick-access header dropdown displaying unread notifications, pending approval alerts, and system notices.

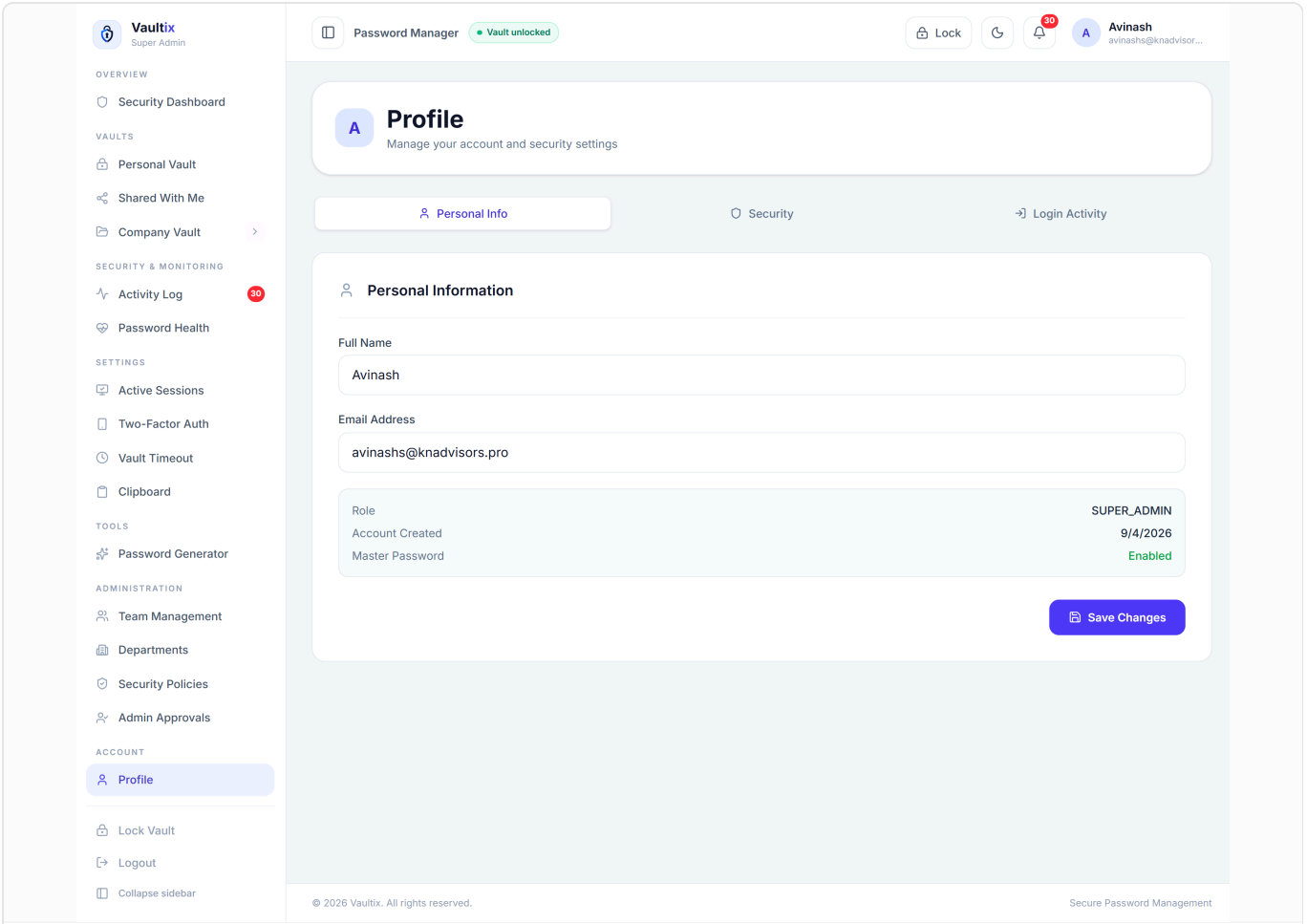

Figure 49: User Profile & Preferences (`/profile`). Profile settings page for managing personal information, rotating the Master Password, and toggling dark/light mode.

23. Browser Extension (Manifest V3 - Version 3.4.0)

The Vaultix Browser Extension is built on Chrome/Edge Manifest V3, providing credential autofill, in-field suggestions, and password capture directly on web pages:

23.1. In-Page Field Detection & Overlay

- **Field Icon:** Places an unobtrusive Vaultix shield icon inside username, email, and password fields.
- **Position Tracking:** Tracks input dimensions using `ResizeObserver` and scroll listeners to maintain icon alignment on responsive layouts.

23.2. Shadow DOM Dropdown & Generator

- **Shadow DOM Isolation:** The autofill dropdown is encapsulated in an isolated Shadow DOM container, preventing host page CSS interference and protecting against script scraping.
- **One-Click Autofill:** Clicking a matching account populates both username and password fields.
- **Keyboard Navigation:** Supports Arrow Up/Down navigation, Enter to fill, and Escape to dismiss.
- **In-Field Generator:** Includes a "+ Generate Password" shortcut directly in input fields.

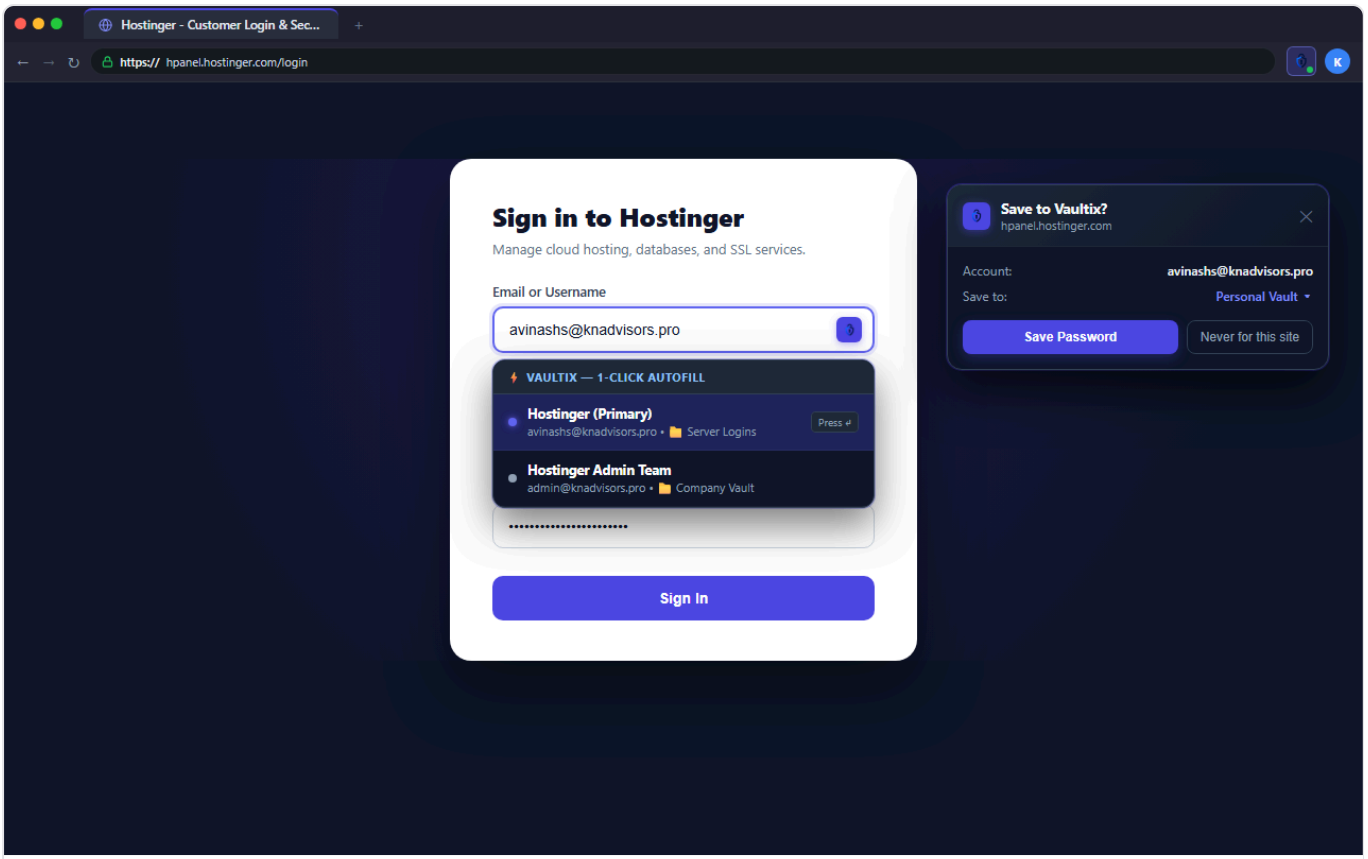

Figure 50: Browser Extension In-Page Autofill (`Extension Content Script`). Manifest V3 content script injecting an autofill dropdown into login fields using Shadow DOM isolation to prevent host page style conflicts.

23.3. SPA & React Framework Support

- **Synthetic Events:** Dispatches synthetic `input` , `change` , and `keyup` events so modern JavaScript frameworks detect autofilled text.
- **React Value Tracker:** Calls native input setters and triggers React's internal `_valueTracker` for complete compatibility with React, Vue, Angular, and Next.js applications.

23.4. Auto-Save & Auto-Update Prompts

- **Form Detection:** Monitors form submissions and AJAX login requests.
- **Auto-Save Prompt:** Displays a slide-down prompt asking to save credentials when registering or logging in with new details.
- **Auto-Update Prompt:** Detects password change forms and prompts to update existing vault entries.
- **Form Exclusions:** Skips CAPTCHA inputs, OTP boxes, search inputs, and CVV fields.

23.5. Extension Popup & Session Storage

- **Toolbar Popup:** Provides quick vault search, one-click TOTP copying, and password generation from the browser toolbar.
- **RAM Session Storage:** Unlocked keys are stored in `chrome.storage.session` with configurable auto-lock timers; master keys are never written to disk.

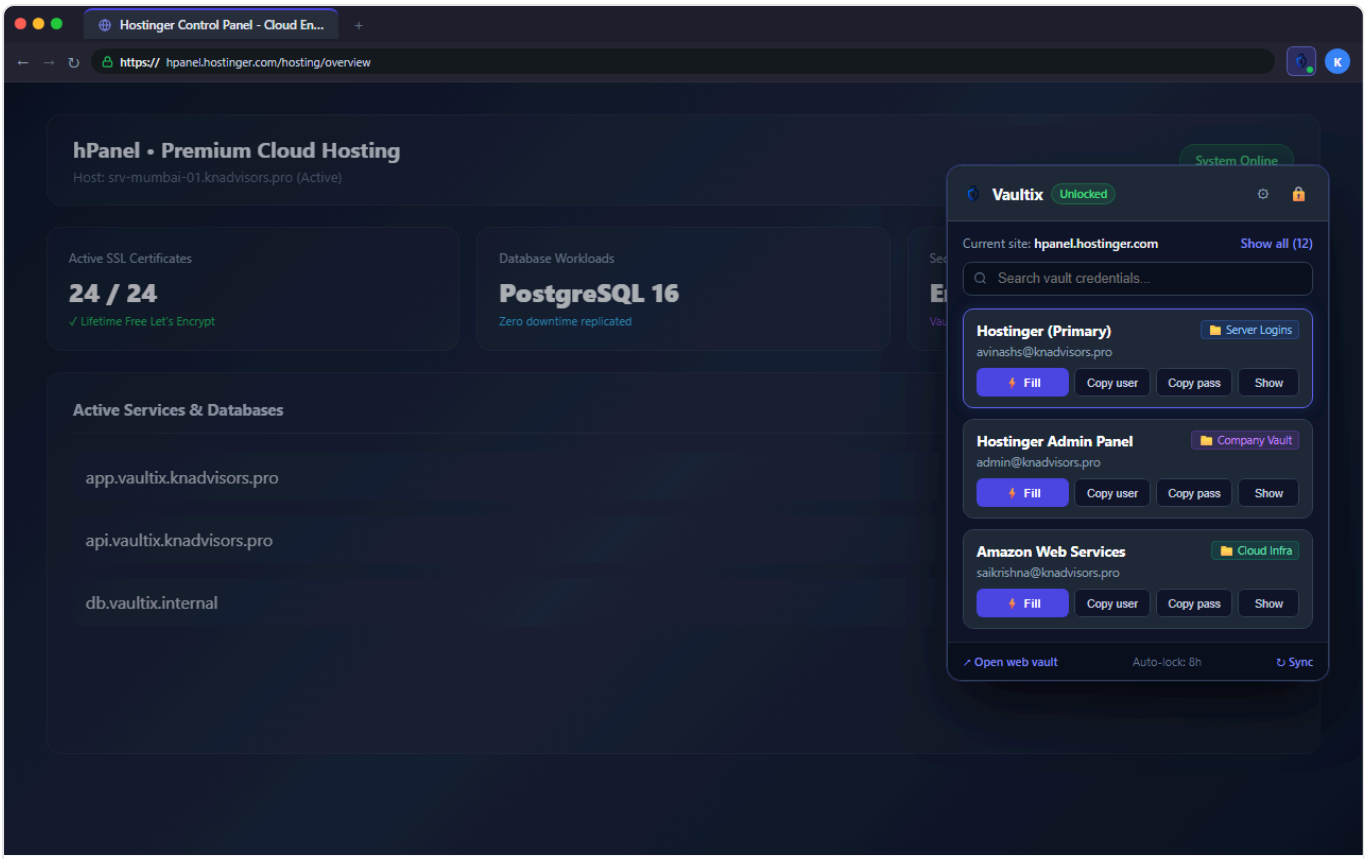

Figure 51: Browser Extension Toolbar Popup (`Extension Popup`). Extension popup providing quick vault search, one-click TOTP code copying, and inline password generation from the browser toolbar.

24. Bitrix24 Intranet Integration

Vaultix provides an integration module (`kn.vaultix`) for Bitrix24 intranets:

- **Sidepanel Frame:** Opens Vaultix inside Bitrix24's native sliding sidepanel so employees can access credentials without leaving their active task or CRM record.
- **Sidebar Link:** Adds a "Vaultix" item with brand icon to the Bitrix24 left navigation menu.
- **SSO Mapping:** Links Bitrix24 employee accounts with Vaultix profiles while preserving the client-side Master Password requirement.

25. Step-by-Step Operational Workflows

Workflow 1: User Onboarding & Keypair Generation

Step 1: User opens the registration link or invitation email.

Step 2: User inputs Full Name, Email, and Account Password (hashed with bcrypt on the server).

Step 3: User creates their Master Password. The browser derives an AES master key (PBKDF2-SHA256, 600,000 iterations), generates an RSA-2048 keypair, encrypts the private key with the master key, and uploads the ciphertext to the server.

Step 4: User copies and securely stores their 12-character emergency recovery key.

Step 5: Personal Vault is provisioned and ready for use.

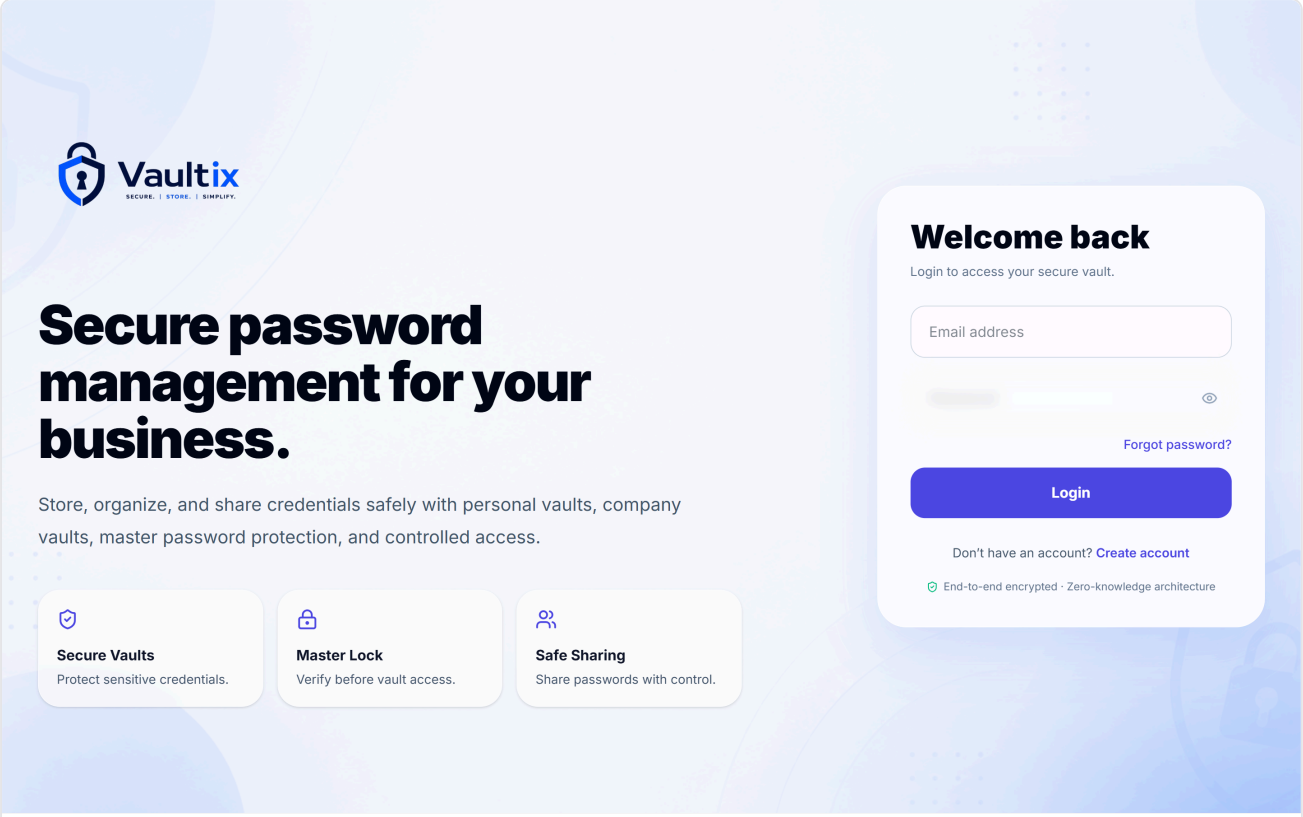

Figure 1: Account Login (`/login`). Primary authentication screen. Users log in with their registered email address and account password. The backend validates credentials using bcrypt and issues a session JWT.

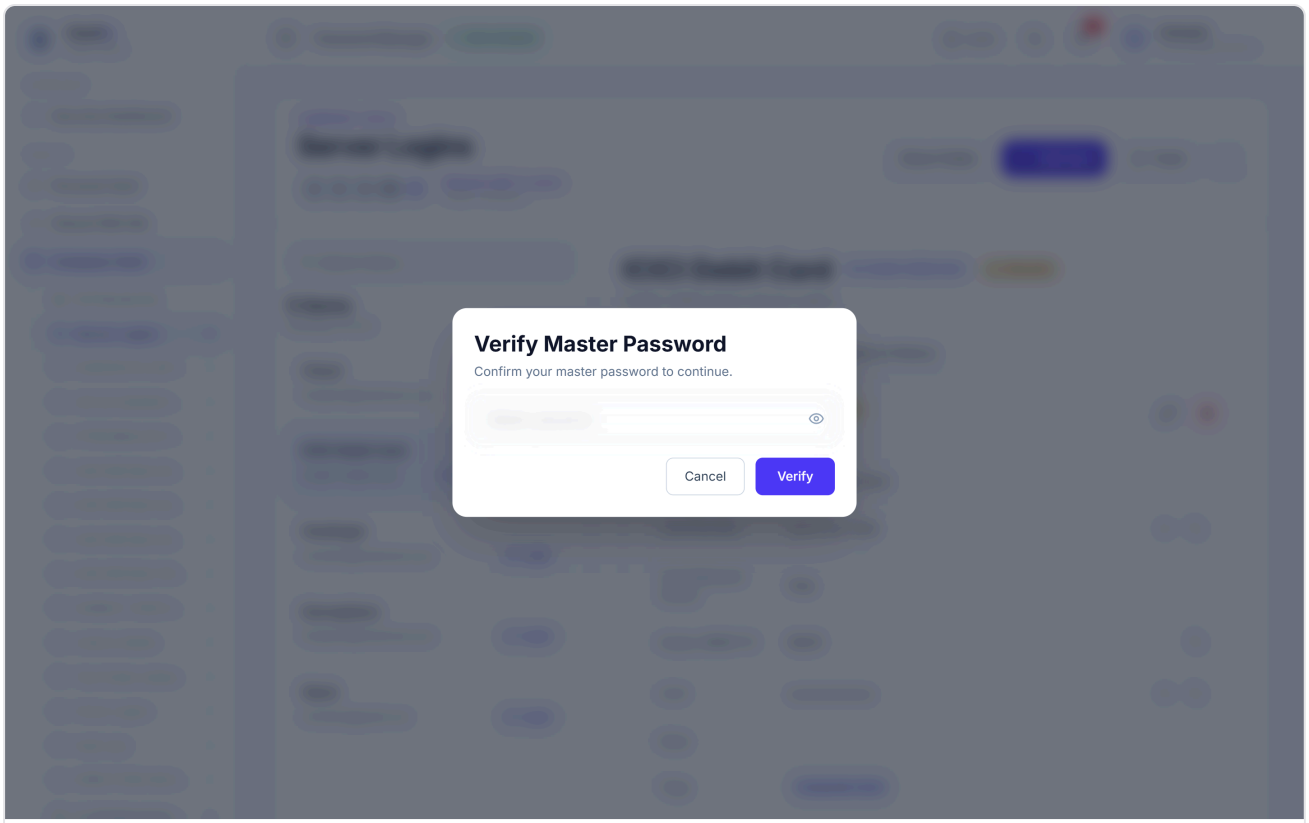

Figure 5: Master Password Verification & Vault Unlock (`Modal`). Vault unlock prompt. The Master Password derives a 256-bit AES key in browser memory via PBKDF2 (600,000 iterations) to decrypt the user's private RSA key.

Workflow 2: Adding Credentials with TOTP

Step 1: Navigate to Personal Vault or a Company Vault folder and click "+ Add Item".

Step 2: Select the item schema (Login, Card, Bank Account, Identity, etc.).

Step 3: Fill in Title, Username, Password (or click Generate), Website URL, and TOTP secret key.

Step 4: Add optional custom fields or color tags.

Step 5: Click Save. The browser generates a random AES key, encrypts all fields, wraps keys for authorized members, and sends ciphertext to the server.

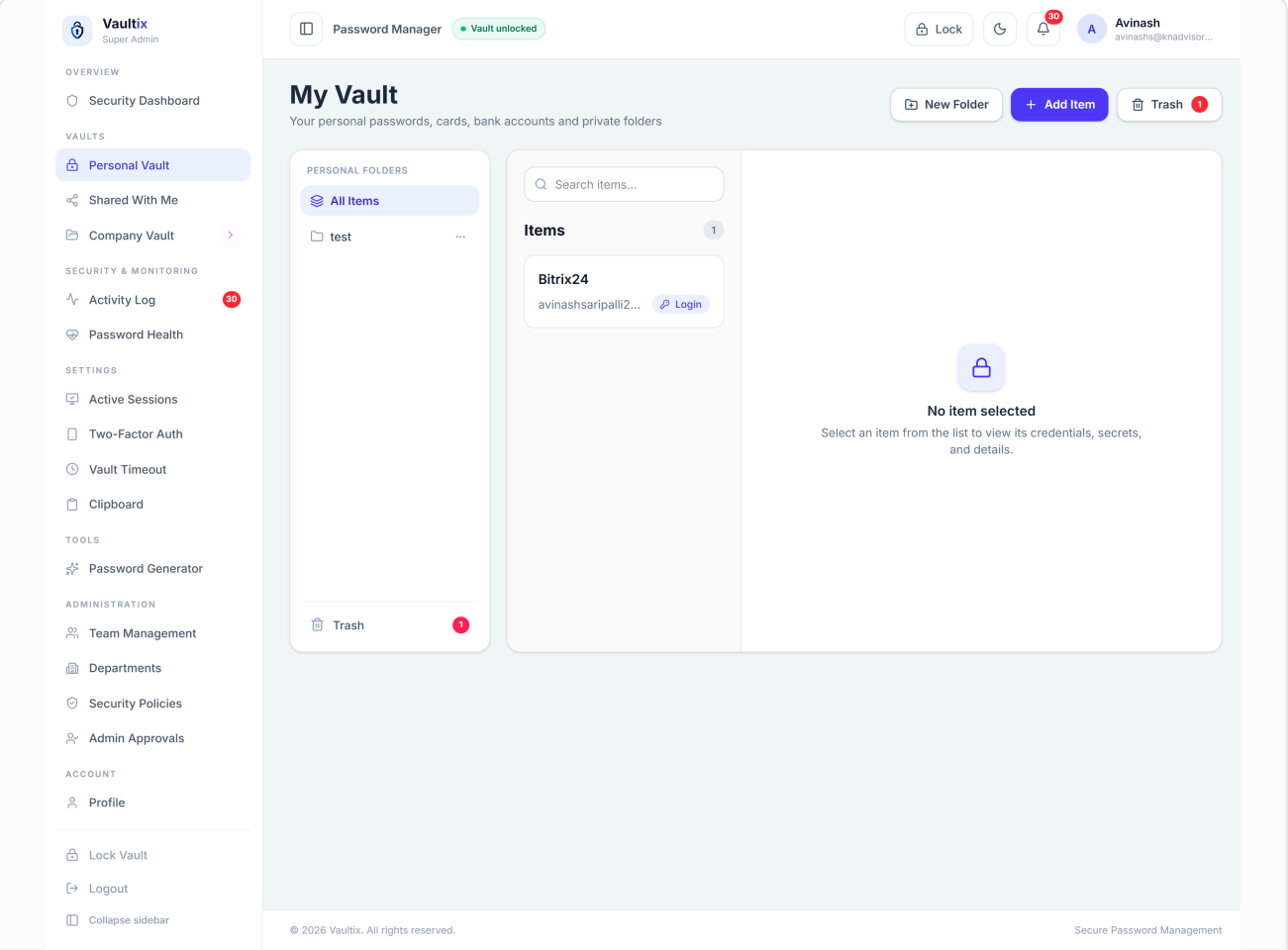

Figure 8: Personal Vault ("My Vault") (`/vault/personal`). Personal vault inventory. Contains private credentials, payment cards, bank accounts, and notes encrypted with the user's master key. Inaccessible to administrators.

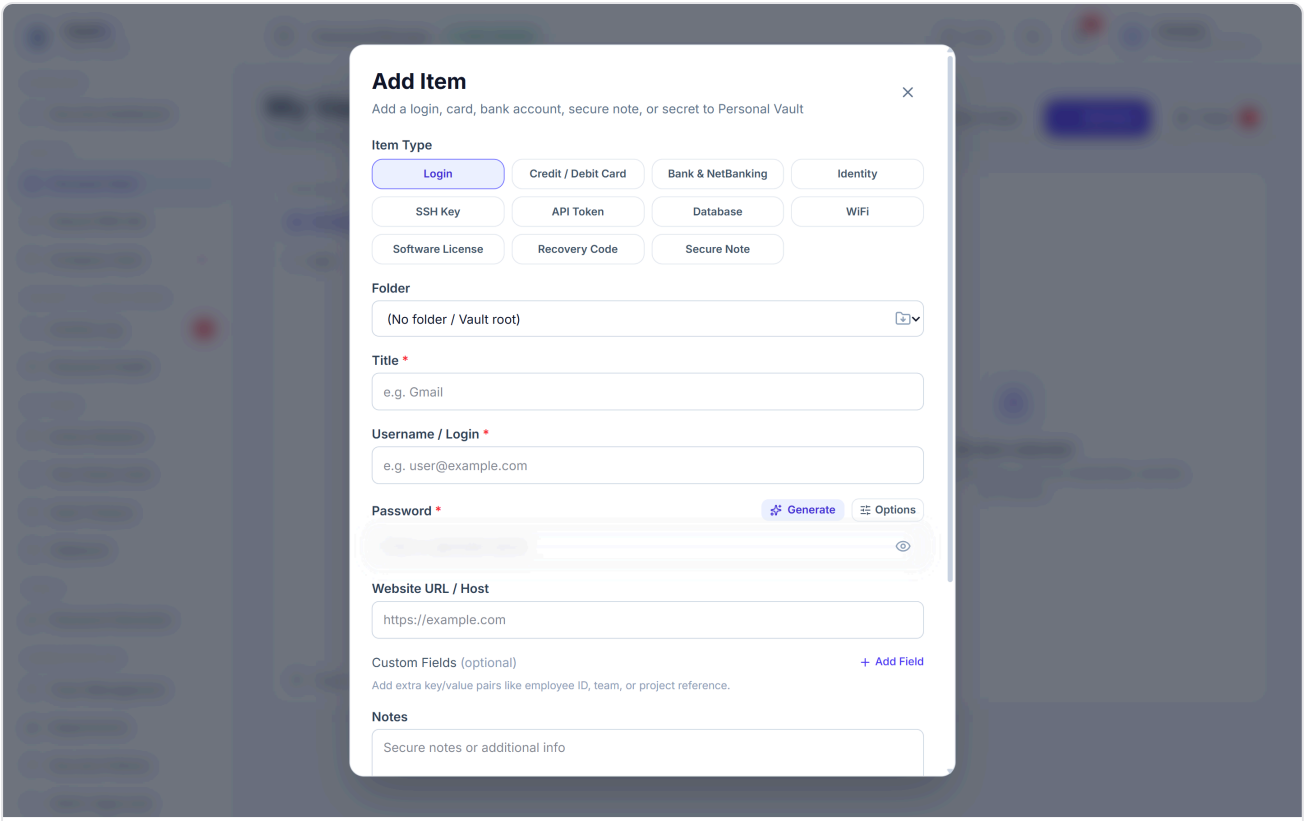

Figure 9: Add Credential Modal (`Modal`). Item creation modal supporting 11 schemas, custom fields (text, masked, hidden), built-in TOTP secret keys, and color tags.

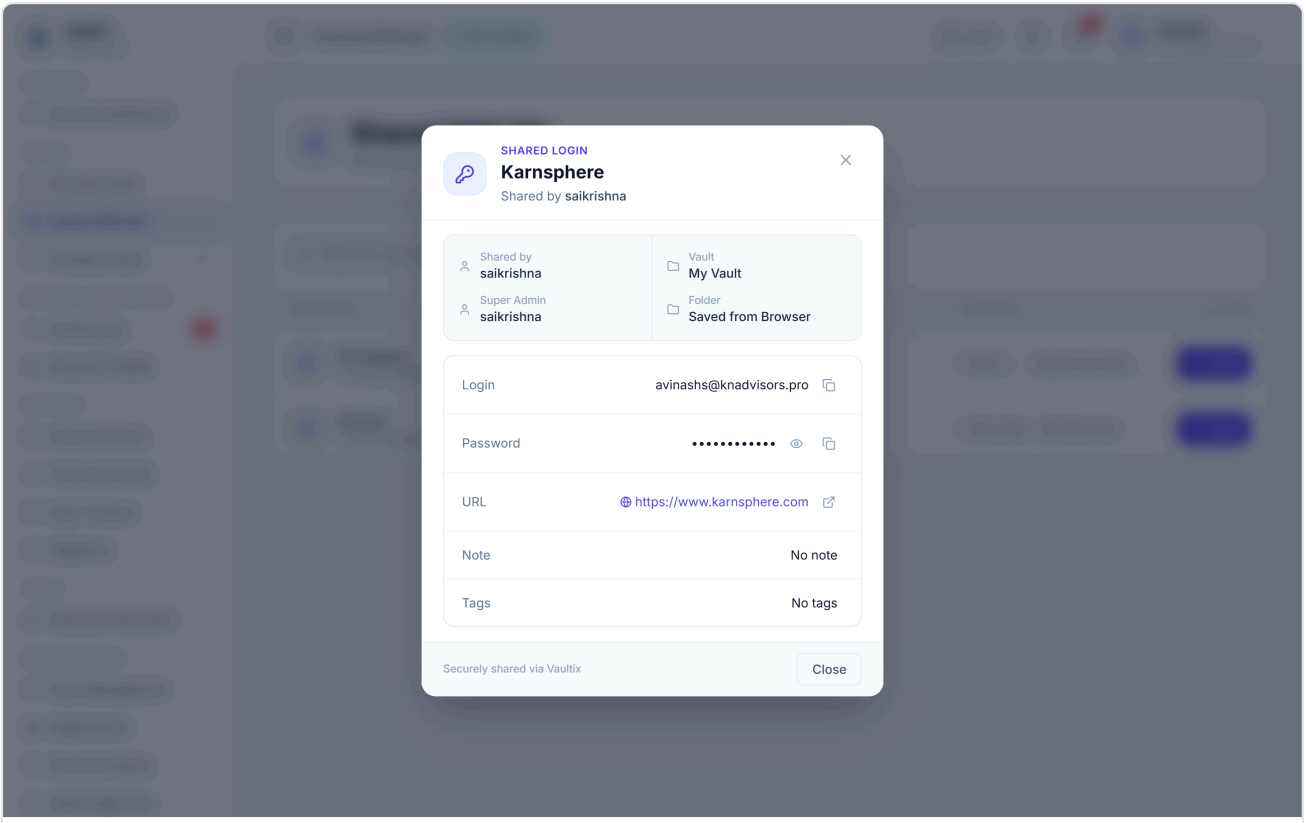

Figure 13: Credential Details & TOTP Modal (`Modal`). Decrypted item view displaying usernames, masked passwords, custom fields, and an active RFC 6238 TOTP circular countdown timer.

Workflow 3: Granting Shared Folder Permissions

Step 1: In Team Management, open the User Access modal for the target employee.

Step 2: Switch to the "+ Grant New Access" tab.

Step 3: Select the target Company Vault and Folder from the dropdown menus.

Step 4: Select the permission level: Full Access, Read & Write, or Read Only.

Step 5: Click Grant Access. The client retrieves the employee's public RSA key, wraps the folder key, and saves the record. The user can access the folder upon their next vault unlock.

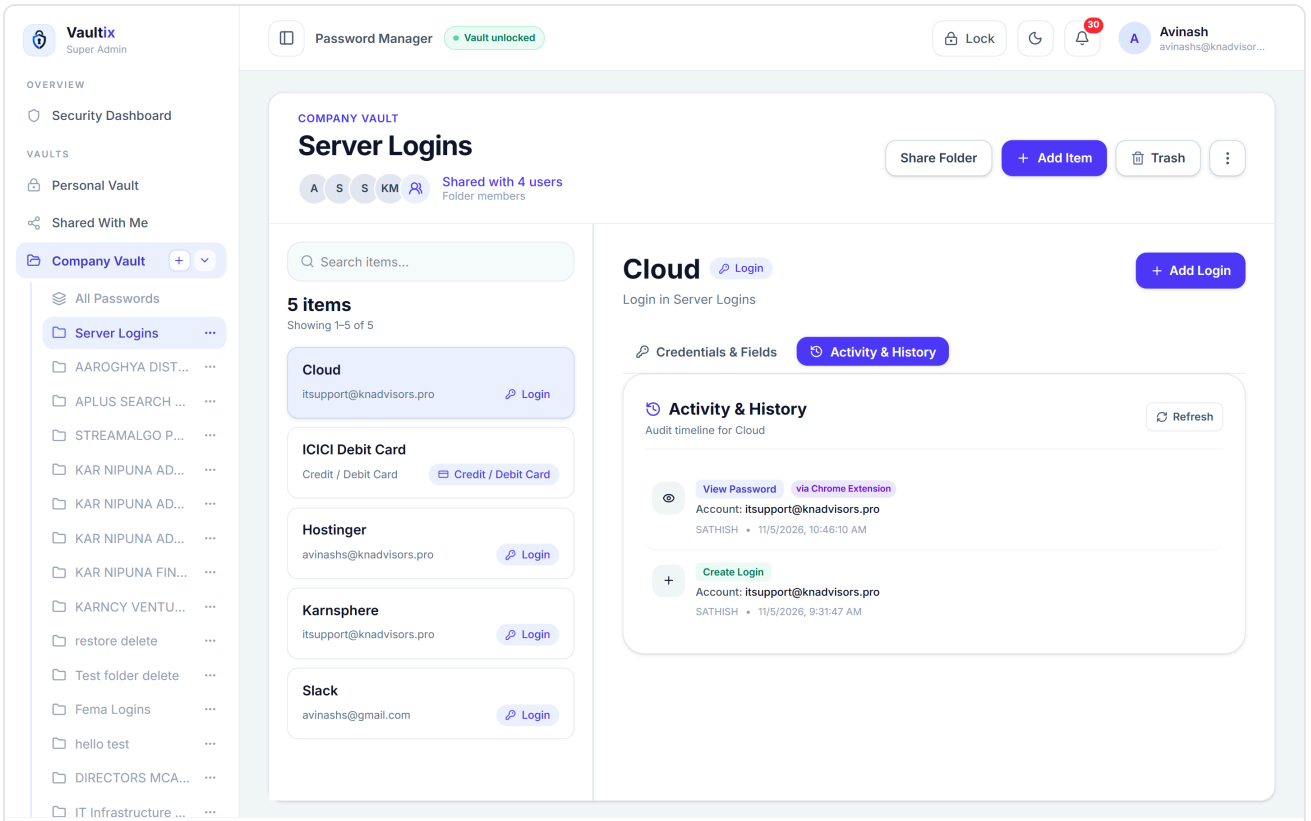

Figure 14: Company Vault Collaborative Workspace (`/vault/company`). Shared organizational vault. Holds shared team folders and credentials encrypted with hybrid RSA-OAEP / AES-256-GCM encryption.

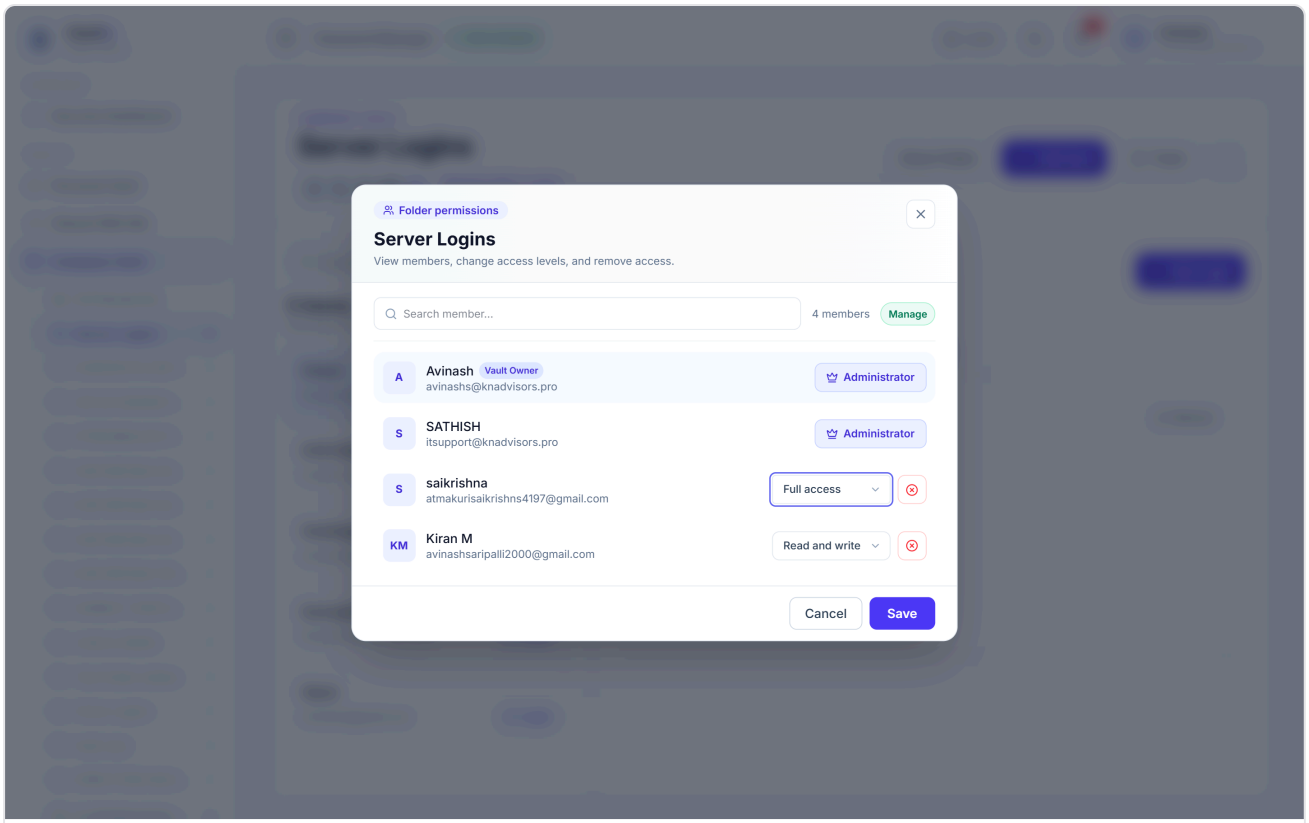

Figure 15: Folder Permissions Management (`Modal`). Permission assignment interface for granting direct user access (Full Access, Read/Write, Read Only) with client-side RSA key wrapping.

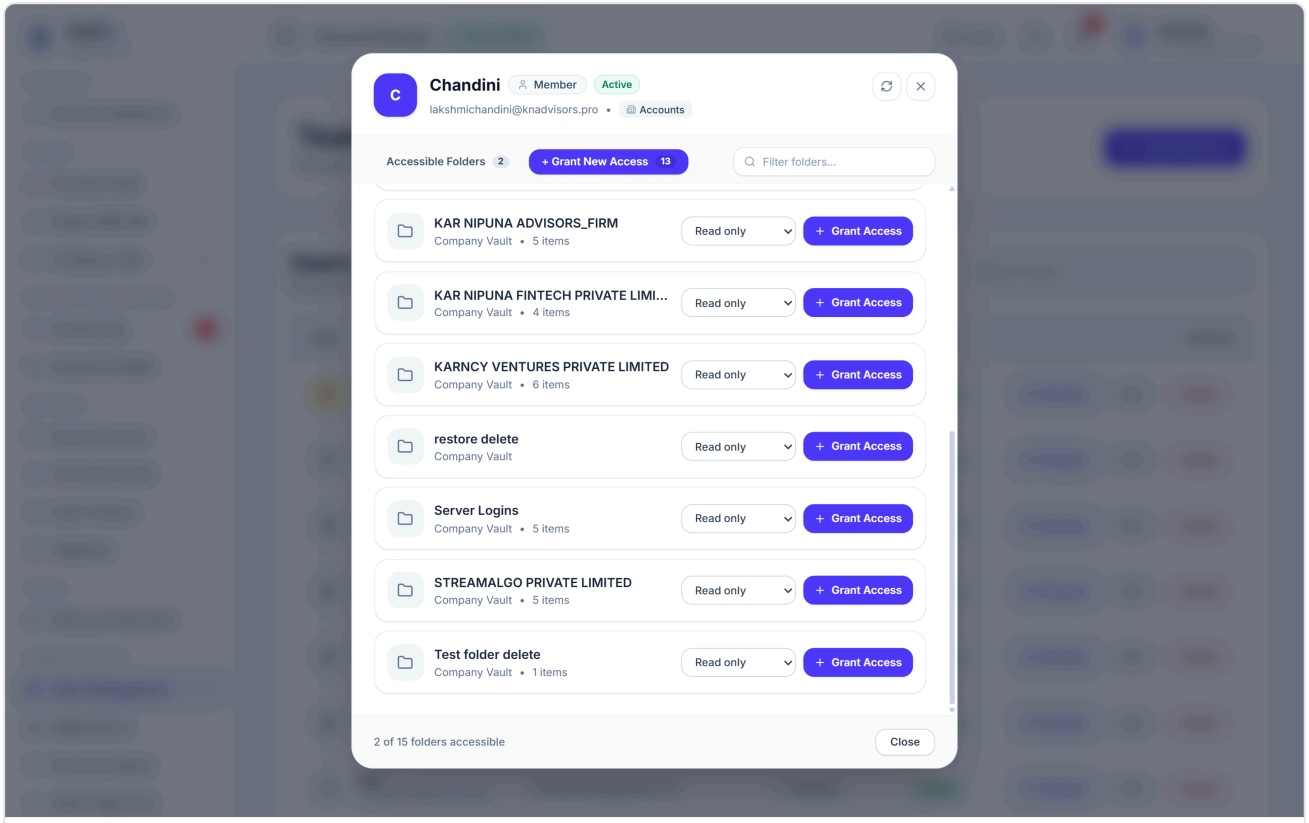

Figure 34: Grant Folder Access Modal (`Modal`). Interface for assigning shared folder permissions directly to a specific user with automated RSA key wrapping.

Workflow 4: Bulk Credential Migration

Step 1: Navigate to the target vault folder and click "Import Passwords".

Step 2: Upload an Excel (.xlsx) or CSV file (supports Dashlane, Bitwarden, KeePass, LastPass, or custom formats).

Step 3: The browser parses rows in local memory, maps external schema types, encrypts each item with AES-GCM, wraps keys for folder members, and batch-inserts the encrypted records.

Workflow 5: Requesting & Approving Sensitive Access

Step 1: User attempts to view a credential marked as Sensitive and clicks "Request Access".

Step 2: User inputs a business justification reason.

Step 3: Administrator reviews the pending request in the Admin Approvals portal.

Step 4: Administrator clicks "Approve". Access is granted with an immutable audit entry.

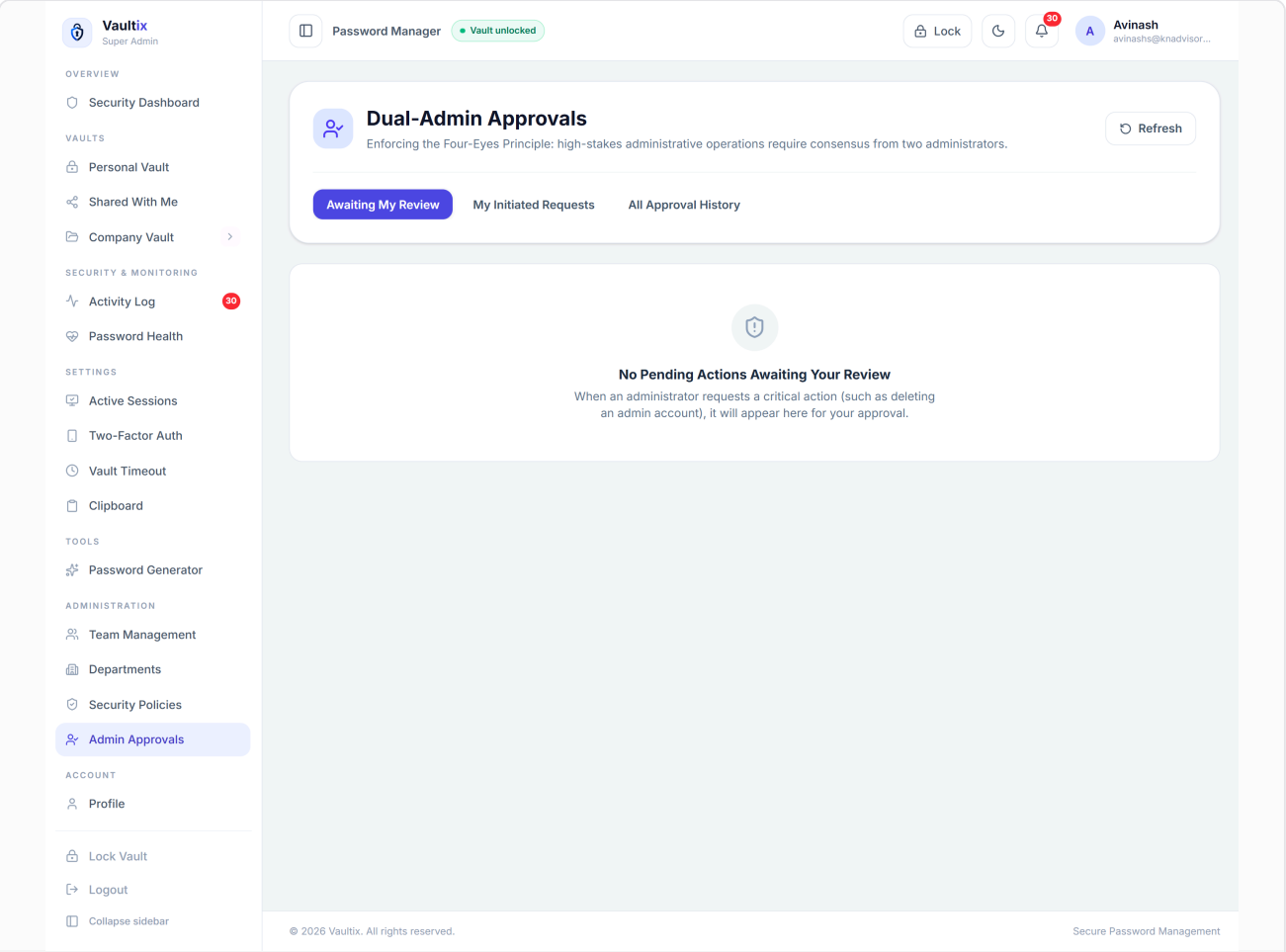

Figure 46: Pending Sensitive Access Approvals (`/approvals (Pending Tab)`). Approval queue where requests to view or copy sensitive credentials await administrator review.

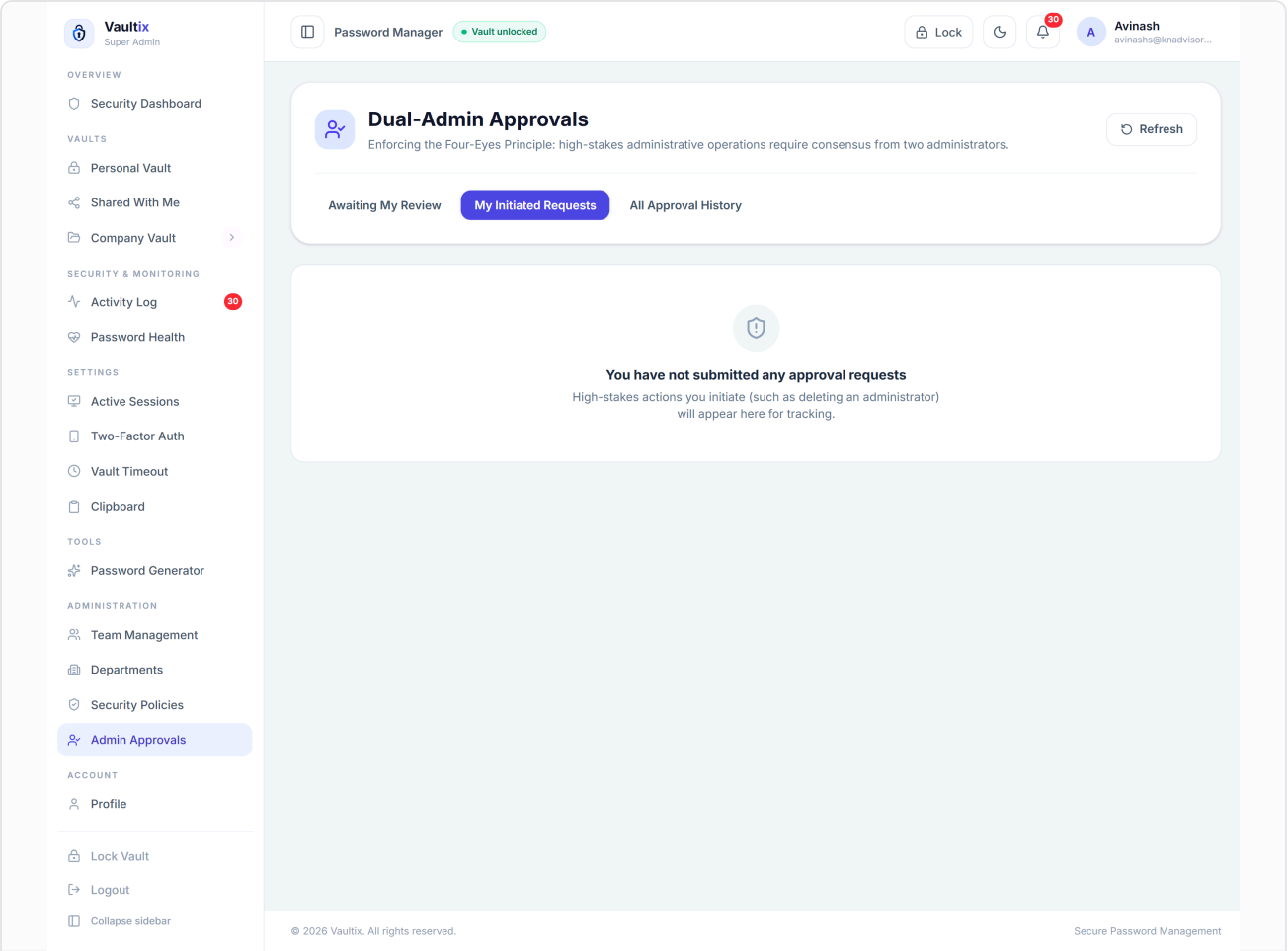

Figure 47: My Access Requests Tab (`/approvals (My Requests)`). Status view showing access requests submitted by the current user (Pending, Approved, Denied, Expired).

Workflow 6: Using the Browser Extension

Step 1: Install the extension and unlock your vault using your Master Password.

Step 2: Navigate to any login form (e.g. AWS, GitHub). The Vaultix icon appears inside the input fields.

Step 3: Click the icon to view matching credentials, and click an account to fill username and password.

Step 4: When registering on a new website, submit the form. Click "Save" on the slide-down banner to save the credential to Vaultix.

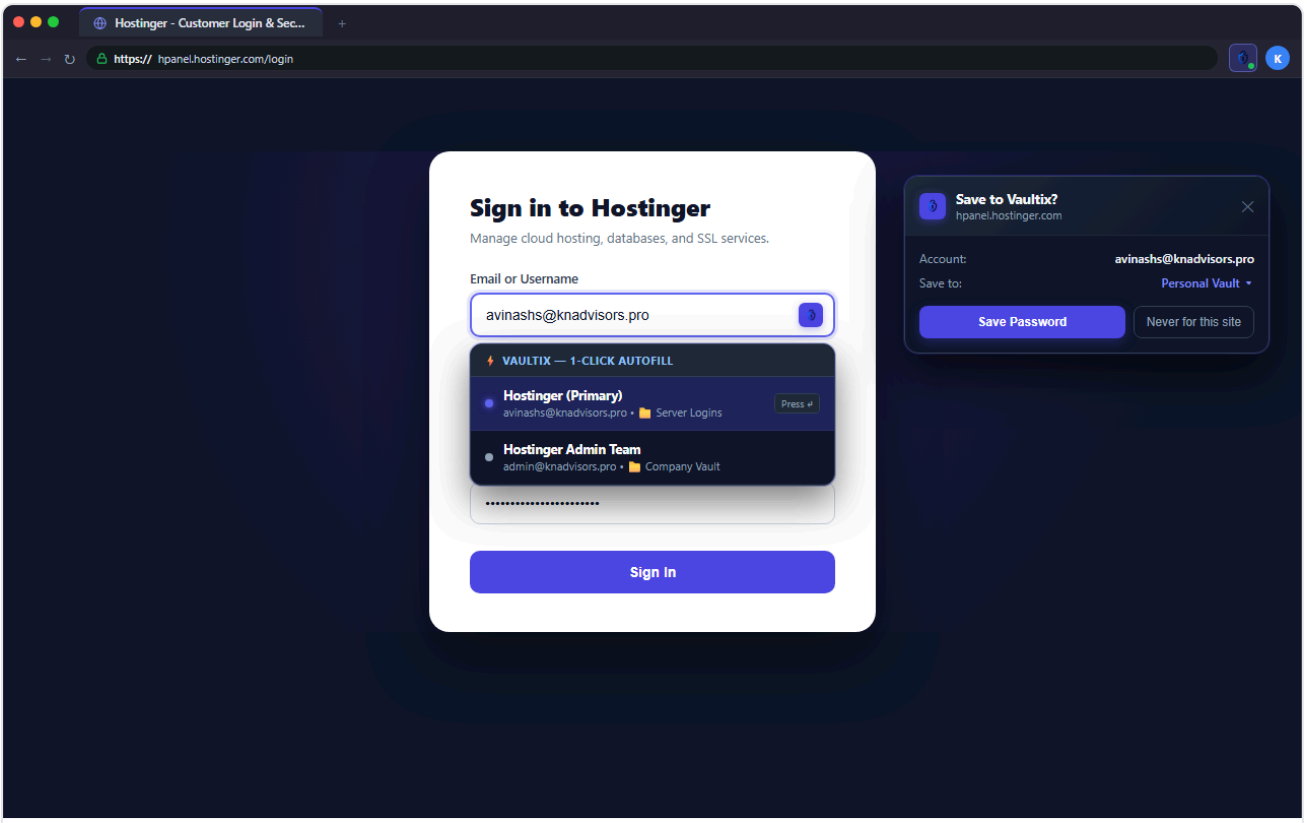

Figure 50: Browser Extension In-Page Autofill (`Extension Content Script`). Manifest V3 content script injecting an autofill dropdown into login fields using Shadow DOM isolation to prevent host page style conflicts.

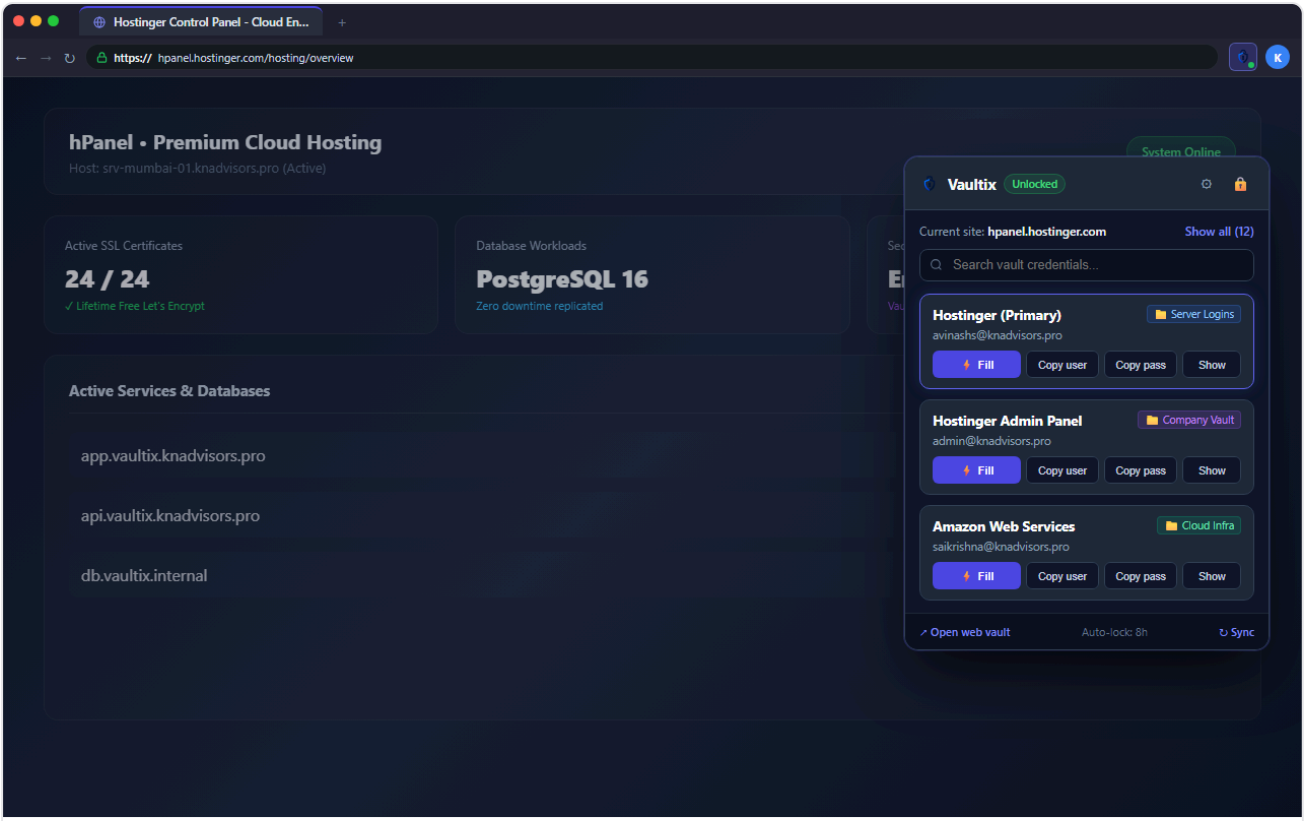

Figure 51: Browser Extension Toolbar Popup (`Extension Popup`). Extension popup providing quick vault search, one-click TOTP code copying, and inline password generation from the browser toolbar.

Workflow 7: Bitrix24 Sidepanel Access

Step 1: Log in to your Bitrix24 intranet portal.

Step 2: Click "Vaultix" in the left sidebar menu.

Step 3: The native Bitrix24 sidepanel opens with the embedded Vaultix interface.

Step 4: Unlock your vault with your Master Password to access and copy credentials without navigating away from active CRM tasks.

Visual Appendix: Screen & Action Index

Index of all captured screens, dialogs, and tools from the Vaultix production environment (`https://vaultix.knadvisors.pro`).

Figure	Screen Title	Route / UI Trigger	Purpose
Figure 1	Account Login	`/login`	Primary authentication screen. Users log in with their registered email address and account password. The backend validates credentials using bcrypt and issues a session JWT.
Figure 2	User Registration	`/register`	Account creation form for new team members. Registration initializes the user profile prior to administrator approval and role assignment.
Figure 3	Account Password Reset	`/forgot-password`	Password reset request screen. Resetting the account password updates server login credentials but does not affect the client-side master encryption key.
Figure 4	Master Password Recovery	`/forgot-master-password`	Master password recovery screen. Users input their 12-character emergency recovery key to re-derive their master key and regain access to their encrypted vault.
Figure 5	Master Password Verification & Vault Unlock	`Modal`	Vault unlock prompt. The Master Password derives a 256-bit AES key in browser memory via PBKDF2 (600,000 iterations) to decrypt the user's private RSA key.
Figure 6	Security Dashboard	`/dashboard`	Overview dashboard showing overall password health score, vulnerable/reused password counts, credential totals, and recent vault activity.
Figure 7	Password Generator	`/generator`	Cryptographic random password generator (crypto.getRandomValues). Supports configurable length (8–64 characters), character set toggles, and ambiguous character filtering.
Figure 8	Personal Vault ("My Vault")	`/vault/personal`	Personal vault inventory. Contains private credentials, payment cards, bank accounts, and notes encrypted with the user's master key. Inaccessible to administrators.
Figure 9	Add Credential Modal	`Modal`	Item creation modal supporting 11 schemas, custom fields (text, masked, hidden), built-in TOTP secret keys, and color tags.
Figure 10	Create Personal Folder Modal	`Modal`	Folder creation dialog for organizing personal vault credentials into hierarchical folders.

Figure	Screen Title	Route / UI Trigger	Purpose
Figure 11	Personal Vault Trash	`/vault/personal/trash`	Recycle bin for soft-deleted personal credentials. Items can be restored to their original folder or permanently purged.
Figure 12	Shared with Me Directory	`/vault/shared`	Directory displaying all folders and credentials that other team members or administrators have shared with the logged-in user.
Figure 13	Credential Details & TOTP Modal	`Modal`	Decrypted item view displaying usernames, masked passwords, custom fields, and an active RFC 6238 TOTP circular countdown timer.
Figure 14	Company Vault Collaborative Workspace	`/vault/company`	Shared organizational vault. Holds shared team folders and credentials encrypted with hybrid RSA-OAEP / AES-256-GCM encryption.
Figure 15	Folder Permissions Management	`Modal`	Permission assignment interface for granting direct user access (Full Access, Read/Write, Read Only) with client-side RSA key wrapping.
Figure 16	Folder Audit History	`Modal`	Change log recording credential updates, permission changes, additions, and deletions within the selected folder.
Figure 17	Vault Security Policy Settings	`Modal`	Folder-level policy configuration enforcing minimum password strength, maximum credential age, and allowed item schemas.
Figure 18	Folder Action Menu	`Context Menu`	Context menu for renaming folders, managing user access, reviewing audit history, configuring policies, and moving folders to trash.
Figure 19	Company Vault Trash	`/vault/company/trash`	Shared repository for soft-deleted company credentials. Items can be restored by folder managers or permanently purged by administrators.
Figure 20	System Activity Log	`/activity-log`	Audit log recording vault operations including credential views, copies, exports, permission changes, and user logins with timestamps and IP addresses.
Figure 21	Login Activity History	`/activity-log (Login History)`	Authentication audit trail recording login timestamps, client IP addresses, browser user agents, operating systems, and MFA verification outcomes.
Figure 22	Notifications Center	`/activity-log (Notifications)`	Central notification feed displaying shared folder invitations, security policy changes,

Figure	Screen Title	Route / UI Trigger	Purpose
			and pending access requests.
Figure 23	Header Notification Menu	`Header Widget`	Quick-access header dropdown displaying unread notifications, pending approval alerts, and system notices.
Figure 24	Password Health Audit	`/security/health`	Client-side password analysis tool evaluating password strength, detecting reused credentials, and flagging breach exposure using k-anonymity checks.
Figure 25	Weak Passwords Report	`/security/health (Weak Tab)`	Filtered list of credentials that do not meet length or complexity requirements.
Figure 26	Aging Passwords Report	`/security/health (Old Tab)`	Filtered list of credentials that have not been rotated in over 90 days.
Figure 27	Active Device Sessions	`/security/sessions`	Session manager showing active logins, client browsers, IP addresses, and last active times, with options to revoke individual or all other sessions.
Figure 28	Two-Factor Authentication Settings	`/security/2fa`	Two-factor configuration portal supporting authenticator applications (TOTP) and FIDO2/WebAuthn hardware passkeys.
Figure 29	Authenticator App Setup Modal	`Modal`	TOTP enrollment modal displaying an RFC 6238 QR code, manual base32 secret key, and code verification field.
Figure 30	Vault Inactivity Timeout Settings	`/security/timeout`	Inactivity auto-lock configuration for automatically clearing decrypted keys from browser memory after 1, 5, 15, or 60 minutes, or upon browser close.
Figure 31	Clipboard Auto-Clear Settings	`/security/clipboard`	Clipboard security settings for automatically clearing copied credentials from the operating system clipboard after 10, 30, 60, or 120 seconds.
Figure 32	Team Directory & User Roles	`/team`	Administrator console for managing team members, system roles (Super Admin, Admin, User), account statuses, and department assignments.
Figure 33	Edit User Modal	`Modal`	Dialog for updating team member details, changing system roles, and reassigning departments.
Figure 34	Grant Folder Access Modal	`Modal`	Interface for assigning shared folder permissions directly to a specific user with automated RSA key wrapping.
Figure 35	User Accessible Folders Modal	`Modal`	Audit view listing all shared folders accessible by a specific employee, with

Figure	Screen Title	Route / UI Trigger	Purpose
			quick-revoke controls.
Figure 36	Send Invitation Modal	`Modal`	Dialog for sending email onboarding invitations with pre-configured roles and department assignments.
Figure 37	Delete User Confirmation Dialog	`Dialog`	Confirmation prompt for deprovisioning user accounts and revoking their access to shared folder keys.
Figure 38	Member Invitation Page	`/team/invite`	Full-page form for inviting new employees, selecting their initial role, and assigning them to departments.
Figure 39	Departments Directory	`/departments`	Overview of organizational units showing department hierarchy, member counts, and assigned managers.
Figure 40	Department Members Drawer	`Drawer`	Slide-out panel displaying department leadership, member roster, and member roles (Manager vs Member).
Figure 41	Create Department Modal	`Modal`	Modal for creating top-level departments with custom descriptions and designated managers.
Figure 42	Edit Department Modal	`Modal`	Dialog for editing department details, updating descriptions, and reassigning managers.
Figure 43	Create Sub-Department Modal	`Modal`	Dialog for adding nested sub-departments within existing department structures.
Figure 44	Add Department Members Modal	`Modal`	Search and multi-select dialog for adding employees to a department.
Figure 45	Organization Security Policies	`/admin/policies`	Global policy controls enforcing minimum master password length, mandatory 2FA, session timeout limits, and clipboard auto-clear intervals.
Figure 46	Pending Sensitive Access Approvals	`/approvals (Pending Tab)`	Approval queue where requests to view or copy sensitive credentials await administrator review.
Figure 47	My Access Requests Tab	`/approvals (My Requests)`	Status view showing access requests submitted by the current user (Pending, Approved, Denied, Expired).
Figure 48	Access Request History Tab	`/approvals (History)`	Audit log of all sensitive credential access requests, showing reviewer decisions, timestamps, and justifications.
Figure 49	User Profile & Preferences	`/profile`	Profile settings page for managing personal information, rotating the Master

Figure	Screen Title	Route / UI Trigger	Purpose
			Password, and toggling dark/light mode.
Figure 50	Browser Extension In-Page Autofill	`Extension Content Script`	Manifest V3 content script injecting an autofill dropdown into login fields using Shadow DOM isolation to prevent host page style conflicts.
Figure 51	Browser Extension Toolbar Popup	`Extension Popup`	Extension popup providing quick vault search, one-click TOTP code copying, and inline password generation from the browser toolbar.